



DSU dsDNA Core Undergraduate Research Training Program

Using uafR for Reproducible Chemical and Mass Spectrometry Research

A 10-week applied research curriculum for undergraduate interns

Program Promise

Students finish this program with a reproducible research project, defensible chemical annotations, interpretable trait matrices, publication-style figures, and a clear plan for communicating results to scientific audiences.

DSU dsDNA Core

Version 1.0

May 27, 2026

Contents

Preface	ix
1 Program Architecture for a Full-Day Internship	1
1.1 Why This Is a 10-Week Full-Day Program	1
1.2 Daily Work Blocks	2
1.3 Weekday and Weekend Time Design	2
1.4 Product Ladder	2
1.5 Skill Progression	3
1.6 Research Integrity Rules	3
1.7 Program Decision Gates	4
1.8 Expected Weekly Time Budget	4
1.9 What Students Should Be Told Up Front	4
1.10 What the Program Should Prepare Before Week 1	5
1.11 How Work Accumulates	5
2 Complete 70-Day Training Spine	6
2.1 Week 1: Research Onboarding and Question Design	6
2.2 Week 2: R, RStudio, Git, and Reproducible Project Structure	11
2.3 Week 3: Chemical Data Literacy and Public Database Foundations	16
2.4 Week 4: uafR Core GC-MS Workflow	20
2.5 Week 5: Research-Grade <code>categorate()</code> Enrichment	25
2.6 Week 6: Research Variables and Trait Matrices	29
2.7 Week 7: Visualization and Exploratory Analysis	33
2.8 Week 8: Literature Integration and Evidence-Based Interpretation	37
2.9 Week 9: Manuscript-Style Research Product	41
2.10 Week 10: Presentation, Peer Review, Final Package, and Handoff	45

3	Research Onboarding and Question Design	50
3.1	Why This Matters	50
3.2	Concepts	50
3.3	uafR and R Skills	50
3.4	Guided Lab	51
3.5	Independent Extension	51
3.6	Deliverables	52
3.7	Quality Checklist	52
4	R, RStudio, Git, and Reproducible Project Structure	53
4.1	Why This Matters	53
4.2	Concepts	53
4.3	Recommended Project Structure	53
4.4	Guided Lab	54
4.5	Git Practice	54
4.6	Independent Extension	54
4.7	Deliverables	55
4.8	Quality Checklist	55
5	Chemical Data Literacy and Public Database Foundations	56
5.1	Why This Matters	56
5.2	Database Roles	56
5.3	Concepts	57
5.4	Guided Lab	57
5.5	Independent Extension	57
5.6	Deliverables	58
5.7	Quality Checklist	58
6	uafR Core GC-MS Workflow	59
6.1	Why This Matters	59
6.2	Required Input Columns	59
6.3	Guided Lab	59
6.4	Interpretation Practice	60
6.5	Independent Extension	60
6.6	Deliverables	60

6.7	Quality Checklist	61
7	Research-Grade categorate Enrichment	62
7.1	Why This Matters	62
7.2	Key Output Families	62
7.3	Guided Lab	62
7.4	Evidence Inspection	63
7.5	Independent Extension	63
7.6	Deliverables	64
7.7	Quality Checklist	64
8	Turning Chemical Metadata Into Research Variables	65
8.1	Why This Matters	65
8.2	Concepts	65
8.3	Guided Lab	65
8.4	Grouping Rules	66
8.5	Independent Extension	66
8.6	Deliverables	67
8.7	Quality Checklist	67
9	Visualization and Exploratory Analysis	68
9.1	Why This Matters	68
9.2	Figure Types	68
9.3	Guided Lab	68
9.4	Caption Standard	69
9.5	Independent Extension	69
9.6	Deliverables	69
9.7	Quality Checklist	69
10	Literature Integration and Evidence-Based Interpretation	71
10.1	Why This Matters	71
10.2	Evidence Ladder	71
10.3	Guided Lab	72
10.4	Finding Literature	72
10.5	Independent Extension	72
10.6	Deliverables	73

10.7 Quality Checklist	73
11 Manuscript-Style Research Product	74
11.1 Why This Matters	74
11.2 Report Structure	74
11.3 Methods Details To Include	74
11.4 Guided Lab	75
11.5 Independent Extension	75
11.6 Deliverables	75
11.7 Quality Checklist	75
12 Presentation, Peer Review, and Final Research Package	76
12.1 Why This Matters	76
12.2 Presentation Structure	76
12.3 Peer Review	76
12.4 Final Package Checklist	77
12.5 Independent Extension	77
12.6 Deliverables	78
12.7 Quality Checklist	78
13 Guided Lab and Exercise Compendium	79
13.1 Lab 1: Chemical Question Triage	79
13.2 Lab 2: Reproducible Project Build	80
13.3 Lab 3: Database Source Audit	81
13.4 Lab 4: Offline Core GC-MS Workflow	82
13.5 Lab 5: Research Enrichment Smoke Test	83
13.6 Lab 6: Evidence Court	84
13.7 Lab 7: Trait Matrix Design	85
13.8 Lab 8: Figure Clinic	86
13.9 Lab 9: Literature-to-Evidence Bridge	86
13.10Lab 10: Final Handoff Audit	87
13.11Lab 11: Identifier Ambiguity Drill	88
13.12Lab 12: Wrong-Column GC-MS Input Repair	89
13.13Lab 13: Source Coverage Missingness Audit	90
13.14Lab 14: Matrix Sparsity Sensitivity Analysis	90

13.15	Lab 15: Reviewer Response Exercise	91
13.16	Lab 16: Final Package Recovery Exercise	92
14	Supplementary Readings and Real-World Exercises	93
14.1	Core Reading List	93
14.2	Reading Annotation Template	94
14.3	Real-World Exercise Bank	94
14.3.1	Exercise A: The Unmatched Chemical	94
14.3.2	Exercise B: The Hazard Code Trap	94
14.3.3	Exercise C: Pathway Does Not Mean Presence	94
14.3.4	Exercise D: Natural Product Occurrence	94
14.3.5	Exercise E: Sensory Versus Biological Interpretation	95
14.3.6	Exercise F: Literature Conflict	95
14.3.7	Exercise G: Matrix Sparsity	95
14.3.8	Exercise H: The Reviewer Request	95
14.3.9	Exercise I: Public Data Ethics	95
14.3.10	Exercise J: Final Handoff Recovery	96
14.4	Scientific Claim Boundaries	96
14.4.1	Tentative Chemical Match	96
14.4.2	Database Presence	96
14.4.3	Source Annotation as Variable	96
14.4.4	Pathway Context	97
14.4.5	Hazard Screening	97
14.4.6	Natural Product Occurrence	97
14.4.7	Bioactivity and Mechanism	97
14.4.8	Sparse Trait Matrices	98
14.5	Discussion Prompts	98
14.6	Recommended Online References	98
15	Project Tracks for Student Interests	99
15.1	Track 1: Plant, Agriculture, and Natural Products	99
15.2	Track 2: Environmental Exposure and Toxicology	100
15.3	Track 3: Food, Flavor, Aroma, and Sensory Chemistry	100
15.4	Track 4: Biomedical, Pharmacology, and Public Health	101
15.5	Track 5: Methods, Data Science, and Tool Evaluation	102

15.6 Track 6: Chemistry Education and Communication	103
15.7 Capstone Options	103
15.8 Track Selection Worksheet	104
16 Program Completion Standard	105
16.1 Required Final Package	105
16.2 Claims That Require Explicit Boundaries	106
16.3 Final Acceptance Review	107
16.4 Minimum Passing Package	107
16.5 High-Impact Package	107
A Student Worksheets	108
A.1 Worksheet 1: Project Brief	108
A.2 Worksheet 2: Chemical Intake Table	109
A.3 Worksheet 3: Database Evidence Map	110
A.4 Worksheet 4: Source Coverage Report	110
A.5 Worksheet 5: GC-MS Input Column Audit	111
A.6 Worksheet 6: Query Settings Log	112
A.7 Worksheet 7: Trait Evidence Audit	112
A.8 Worksheet 8: Trait Grouping Dictionary	113
A.9 Worksheet 9: Matrix Sensitivity Log	113
A.10 Worksheet 10: Figure Planning Worksheet	114
A.11 Worksheet 11: Figure Caption Checklist	115
A.12 Worksheet 12: Literature and Evidence Synthesis	116
A.13 Worksheet 13: Claim-Evidence-Limitation Memo	116
A.14 Worksheet 14: Peer Review Form	117
A.15 Worksheet 15: Daily Research Log	118
A.16 Worksheet 16: Weekly Progress Review	118
A.17 Worksheet 17: Evidence Court	119
A.18 Worksheet 18: Final Handoff	120
A.19 Worksheet 19: Presentation Planning	121
A.20 Worksheet 20: Final Package Audit	122
B Assessment Rubrics	124
B.1 Performance Levels	124

B.2	Rubric 1: Weekly Deliverables	124
B.3	Rubric 2: Research Log	125
B.4	Rubric 3: Code and Reproducibility	125
B.5	Rubric 4: Chemical Evidence Traceability	126
B.6	Rubric 5: Trait Matrix and Grouping Dictionary	127
B.7	Rubric 6: Figures and Captions	127
B.8	Rubric 7: Literature Integration	128
B.9	Rubric 8: Final Report	128
B.10	Rubric 9: Presentation	129
B.11	Rubric 10: Final Package and Handoff	130
B.12	Recommended Overall Program Weight	131
C	Program Implementation Guide	132
C.1	Program Philosophy	132
C.2	Pre-Program Setup Checklist	132
C.3	Daily Check-In Script	132
C.4	Weekly Mentor Review Script	133
C.5	Decision-Gate Review Questions	133
C.6	Expected Student Answers by Program Stage	133
C.7	Common Coaching Moments	134
C.8	When To Slow Down	135
C.9	When To Accelerate	135
C.10	Running With Real Data, Case Data, or No Data	135
C.11	Protecting Unpublished and Private Data	135
C.12	Database Etiquette	135
C.13	Evaluating Final Packages	136
C.14	Archiving Projects	136
C.15	Recommended Grading Weight	136
D	dsDNA Core Review Notes and Answer Guidance	137
D.1	Facilitating Without Doing the Project for the Student	137
D.2	Daily Check-In Structure	137
D.3	Expected Answers by Program Stage	137
D.4	Common Program Responses	138
D.5	Review Notes for Real-World Exercises	138

D.5.1	Unmatched Chemical	138
D.5.2	Hazard Code Trap	138
D.5.3	Pathway Does Not Mean Presence	139
D.5.4	Natural Product Occurrence	139
D.5.5	Matrix Sparsity	139
D.6	When To Slow the Program Down	139
D.7	When To Accelerate	139
D.8	Final Review Questions	139
E	Installation and Setup Appendix	141
E.1	Recommended Student Installation	141
E.2	Install From the Student Bundle	141
E.3	Create a New Student Bundle	142
E.4	Confirm Bundle Integrity	142
E.5	Confirm the Installation Without Live Web Queries	142
E.6	Student Acceptance Test	143
E.7	Optional Live Smoke Test	143
E.8	Troubleshooting	144
F	uafR Table and Script Guide	145
F.1	High-Value categorate Tables	145
F.2	Training Scripts	146
F.3	Student Bundle Scripts	147
F.4	Choosing Detail Levels	147
F.5	Recommended Smoke Test Settings	147

Preface

This program trains undergraduate interns to move from raw chemical observations to reproducible, interpretable research products. It is built around uafR because the package connects GC-MS processing, public chemical databases, and analysis-ready chemical traits. The training is intentionally broader than one software tool: students also learn project organization, literature interpretation, figure design, peer review, and manuscript-style scientific communication.

This second-pass version is written as a full-day, 10-week research internship. It assumes students are present most weekdays and are expected to read, code, analyze, write, meet with the dsDNA Core team, revise, and produce tangible research artifacts. If the program is run part-time, the deliverables stay intact and the number of exercises is reduced rather than compressing every activity.

Who This Is For

The curriculum is designed for undergraduate interns from chemistry, biology, agriculture, environmental science, public health, data science, and related fields. No student needs to arrive with a finished project. The program helps them bring their own interests into a shared research workflow.

Audience and Prerequisites

The program is written for students who are ready to work carefully even if they are new to R, mass spectrometry, or chemical databases. Useful preparation includes basic spreadsheet literacy, willingness to keep a detailed research log, and enough chemistry background to learn terms such as compound, synonym, functional group, concentration, and tentative identification. Students with stronger R or chemistry preparation use the advanced paths; students with less preparation use the same quality standards with smaller chemical sets and more guided examples.

Required Student Commitments

- Work from a reproducible project folder rather than loose files.
- Keep a dated research log every calendar day of the program.
- Use cautious language for tentative chemical annotations and database evidence.
- Save intermediate outputs, validation summaries, and source coverage notes.
- Revise claims when evidence is weak, missing, or outside the project scope.

Program Outcomes

By the end of 10 weeks, each student should be able to:

- Define a research question that can be answered with chemical identities, chemical traits, and reproducible analysis.
- Organize an R project with clear file structure, version control, input data, scripts, outputs, and documentation.
- Use uafR to process tentative chemical identities, enrich query chemicals, and inspect database evidence.
- Convert chemical metadata into clean grouping variables and matrices for exploratory analysis.
- Build figures and tables that support a defensible scientific interpretation.
- Explain uncertainty, database provenance, and limits of tentative chemical annotation.
- Deliver a final research package that can support a poster, internal report, manuscript section, or future grant work.

What Students Produce

The program is product-driven. Students do not simply attend sessions; they build a research package piece by piece. Required products include a project brief, chemical intake table, reproducible R project, GC-MS input audit, saved uafR output objects, validation summaries, source coverage report, trait grouping dictionary, trait matrix, figure set, literature synthesis, manuscript-style report, presentation or poster, research log, final handoff audit, and continuation note.

What Mentors Review

Review focuses on artifacts rather than impressions. Weekly review checks the current product, the research log, evidence traceability, code reproducibility, claim strength, and readiness for the next gate. A strong review asks whether the work can be rerun, whether the evidence supports the wording, and whether a future student could continue the project.

How To Use This Manual

Each week has the same structure so students always know what is expected:

- **Why this matters:** the scientific reason for the week.
- **Concepts:** the vocabulary and logic students must understand.
- **uafR and R skills:** the specific computational skills taught.
- **Guided lab:** a dsDNA Core-supported activity.
- **Independent extension:** a student-owned application.

- **Deliverables:** concrete products that move the research project forward.
- **Quality checklist:** the standard students must meet before moving on.

How the Program Runs

Student work is reviewed weekly with the dsDNA Core team. The goal is not to demand perfect answers at the first pass. The goal is to train students to produce traceable work: every claim connects to data, code, a database field, or a cited source. When a result looks interesting, the review returns to one question: “What evidence supports that, and what could be wrong?”

Calendar-Day Participation

Weekdays are full internship days with concept work, guided technical labs, independent project work, evidence writing, and review. Saturdays and Sundays are lighter but still required asynchronous days. Weekend work usually includes reading annotation, reruns, research log updates, figure or text revision, peer review, or project maintenance. The calendar rhythm matters because research quality improves through repeated inspection and revision, not only through long coding sessions.

Core Principle

The student should never treat database text as a conclusion. Database records are evidence. The research conclusion comes from cleaned variables, source coverage checks, comparison across evidence types, and biological or chemical reasoning.

Recommended Weekly Rhythm

Meeting	Purpose
Monday kickoff	Introduce the week, confirm each student’s research question, assign readings, and define the deliverable.
Tuesday or Wednesday technical lab	Work through code, database interpretation, and analysis issues with dsDNA Core support.
Thursday writing and evidence clinic	Convert results into claims, captions, methods text, and documented limitations.
Friday review	Inspect deliverables, update the research log, complete peer review, and set next-week priorities.

One-Page 10-Week Arc

Week	Focus	Primary product	Decision gate
1	Question design and research integrity	Project brief and risk note	Question is feasible and scientifically cautious.
2	R, RStudio, Git, and project structure	Reproducible folder and README	Project can be understood from its root.
3	Chemical identities and database sources	Database evidence map	Sources are interpreted within limits.
4	Core GC-MS workflow	Offline workflow output and methods draft	Tentative annotation limits are documented.
5	Research-grade enrichment	Saved <code>categorate()</code> result and validation notes	Validation and source coverage are inspected before analysis.
6	Trait matrices and grouping variables	Grouping dictionary and final matrix	Variables are source-backed and analysis-ready.
7	Visualization	Publication-style figures and captions	Figures answer questions without hiding missingness.
8	Literature and interpretation	Claim-evidence-limitation memo	Claims survive evidence review.
9	Manuscript-style writing	Full draft report	Methods, results, and limitations align.
10	Presentation and handoff	Final package and continuation note	Work can be archived and continued.

Expected Final Package

Every student should finish with:

- A reproducible R project folder.
- A cleaned chemical query list and documented inclusion rules.
- uafR enrichment outputs and validation summaries.
- A trait matrix or grouped chemical table suitable for analysis.
- At least three publication-style figures or tables.
- A short manuscript-style report.
- A presentation or poster outline.
- A research log that documents decisions, errors, and interpretation changes.
- A continuation note that allows the next student to extend the project without starting over.

1 Program Architecture for a Full-Day Internship

1.1 Why This Is a 10-Week Full-Day Program

The program is not 10 weeks because students need 10 weeks to learn a few R functions. It is 10 weeks because the expected outcome is a research product with defensible evidence, reusable code, figures, interpretation, and a handoff package. A full-day internship gives students enough time to repeat the cycle that real research requires: learn, try, fail, diagnose, revise, document, and communicate.

Full-Day Assumption

Each week assumes roughly 30 to 35 focused hours. A typical day includes a short technical lesson, guided work, independent project time, dsDNA Core or peer feedback, and written documentation. Students who only attend part-time can still use the program; in that format the number of exercises is reduced rather than reducing the quality standard.

Weekend Asynchronous Expectation

The program uses all 70 calendar days. Weekend work is lighter than weekday work, but it is required. Weekend tasks are designed for one to three focused hours and usually include reading annotation, rerunning a small script, updating the research log, revising a worksheet, reviewing a peer artifact, or preparing the next week's files. Weekend work should never require live database access unless that activity has been explicitly planned and cached examples are available as a fallback.

1.2 Daily Work Blocks

Block	Purpose
Morning concept block	Introduce the scientific idea, database concept, coding pattern, or interpretation principle needed that day.
Guided technical lab	Run code, inspect outputs, troubleshoot errors, and learn the expected workflow with a small example.
Independent project block	Apply the day's method to the student's own project or assigned case study.
Evidence and writing block	Convert analysis into a research log entry, methods text, figure caption, or decision memo.
dsDNA Core check-point	Review a concrete artifact and decide whether the student can move forward or must revise.

1.3 Weekday and Weekend Time Design

Day type	Typical time	Expected work
Weekday concept and guided work	2 to 3 hours	Reading discussion, demonstration, code walk-through, or source interpretation exercise.
Weekday independent project work	2 to 3 hours	Student applies the day's method to their own project or a case dataset and saves an artifact.
Weekday evidence and writing	1 to 1.5 hours	Research log, method text, figure caption, worksheet, claim memo, or review response.
Weekday review check-point	20 to 45 minutes	Peer or dsDNA Core review of a specific artifact and next-step decision.
Weekend asynchronous work	1 to 3 hours	Reading annotation, rerun, project cleanup, source audit, reflection, peer review, or revision.

1.4 Product Ladder

The internship is organized as a product ladder. Each week creates an artifact that is needed later.

Week	Artifact	Why it matters
1	Project brief	Prevents a broad interest from becoming an untestable analysis.
2	Reproducible project	Gives the student a research workspace that can be audited and rerun.

Week	Artifact	Why it matters
3	Database evidence map	Separates identity, source coverage, biological context, and uncertainty.
4	Core GC-MS workflow	Establishes how tentative chemical evidence enters the project.
5	Enrichment result	Creates the main source-backed chemical evidence object.
6	Grouping dictionary	Converts source-backed fields into analysis-ready variables.
7	Figure set	Reveals patterns and missing data before interpretation hardens.
8	Evidence memo	Connects database evidence to peer-reviewed context.
9	Draft report	Forces the methods, results, figures, and limitations into a coherent argument.
10	Final package	Leaves behind work that the dsDNA Core team, a collaborator, or a future student can use.

1.5 Skill Progression

Students move through four levels:

1. **Operator:** can run a provided script and identify the output.
2. **Inspector:** can check whether an output is complete, duplicated, missing, or suspicious.
3. **Analyst:** can transform validated outputs into variables, figures, and tables.
4. **Researcher:** can defend interpretation with source evidence, literature, uncertainty, and next steps.

1.6 Research Integrity Rules

Rules Students Must Follow

- Never present a tentative chemical match as confirmed identity unless standards or equivalent confirmatory evidence are available.
- Never hide unmatched chemicals or missing source coverage.
- Never copy a database sentence into a result and treat it as an analyzed variable.
- Never store tokens, passwords, private data, or unpublished collaborator data in a repository.
- Never change raw data manually; all cleaning must be scripted or documented.
- Never make a figure that cannot be regenerated from code.

1.7 Program Decision Gates

Decision gates keep the program rigorous:

Gate	Student must show
Scope gate	A feasible question, reviewed chemical list, and documented risk note.
Reproducibility gate	A project folder with relative paths, scripts, logs, and no secrets.
Evidence gate	Source coverage, validation results, and evidence rows for important traits.
Variable gate	A grouping dictionary with source table, field, rule, allowed values, and caveat.
Interpretation gate	A claim supported by a figure, table, database field, and literature source.
Handoff gate	Final package can be understood by someone who was not present during the internship.

1.8 Expected Weekly Time Budget

Activity	Hours/week	Notes
Technical instruction and guided lab	6–8	Short lessons followed by hands-on work.
Independent project work	10–12	Student applies the workflow to their question or case study.
Reading and annotation	3–5	Database docs, methods papers, and project-specific literature.
Writing and documentation	4–6	Research log, methods, captions, interpretation, and handoff notes.
dsDNA Core and peer review	3–4	Weekly review, code walk-throughs, and figure critique.
Troubleshooting and revision	4–6	Expected time for errors, reruns, cleanup, and rethinking.

1.9 What Students Should Be Told Up Front

The internship is not a race to get a colorful heatmap. A student who finds that half their chemicals have weak source coverage has not failed. That is a useful result if it is documented clearly. A student who discovers that a project question is too broad has not failed. That is research judgment. The standard is not that every project produces a dramatic biological conclusion. The standard is that every project produces a trustworthy research package.

Student Briefing Before Week 1

- The program is a full-day research experience with required weekend asynchronous work.
- The final product is a reproducible research package, not only a presentation.

- uafR helps organize and enrich chemical evidence, but it does not confirm identities by itself.
- Public database evidence must be interpreted with source coverage, provenance, and caveats.
- Research logs are evaluated because they preserve decisions and make the project reusable.
- Students may bring their own interests, but each project must pass the same evidence and reproducibility gates.

1.10 What the Program Should Prepare Before Week 1

Preparation Checklist

- Current private uafR student bundle or verified installation route.
- Example chemical list and case dataset for students without immediate project data.
- Printed or digital worksheets for project brief, chemical intake, database evidence map, and research log.
- A clear data classification note for public, private, unpublished, and restricted materials.
- A shared location for program-safe examples, not for private raw collaborator data unless access is approved.
- A review calendar with Week 1 scope gate, Week 2 reproducibility gate, Week 5 validation gate, Week 6 variable gate, Week 8 interpretation gate, and Week 10 handoff gate.
- A backup plan for days when live public database services are unavailable.

1.11 How Work Accumulates

Each week reuses the previous week's artifact. The project brief controls the chemical intake table. The chemical intake table controls the enrichment query. The saved enrichment result controls source coverage and trait extraction. The grouping dictionary controls the matrix. The matrix controls figures. Figures control results paragraphs. Results paragraphs control the discussion and presentation. This chain prevents the final report from becoming a collection of disconnected outputs.

2 Complete 70-Day Training Spine

This chapter is the operating schedule for the internship. It is written for active use: students should know what to open, what to run, what to inspect, what to write, and what artifact closes the day. Weekdays are full internship days. Weekend days are lighter asynchronous work, but they are still required because research quality depends on steady reading, logging, reruns, revision, and handoff maintenance.

Daily Work Assumption

Weekdays normally use a full-day rhythm: 30 minutes for check-in and objectives, 60 to 90 minutes for concept or reading discussion, 90 to 120 minutes for guided technical work, 90 to 150 minutes for independent project work, 45 to 60 minutes for evidence and writing, and 30 minutes for peer or mentor checkpoint. Weekend days normally require 1 to 3 focused hours.

2.1 Week 1: Research Onboarding and Question Design

Week 1: Research onboarding and question design

Research milestone: A feasible, cautious chemical research question is selected.

Primary product: One-page project brief, risk note, chemical intake table v1, and research log setup.

Week 1 Operating Plan

Concepts: DSU dsDNA Core expectations, research conduct, chemical evidence types, tentative annotation, database scope, and project feasibility.

uafR/R skills: None required beyond opening RStudio and reading package examples. Students inspect names and tables before running analysis.

Required readings or references: Program preface, project tracks, PubChem and KEGG reference descriptions, and the evidence caution rules in this manual.

Required files/scripts/data: Project brief worksheet, chemical intake table template, evidence map worksheet, and research log template.

Weekly deliverables: Candidate question set, chemical intake table v1, evidence feasibility map, one-page project brief, risk note, and Week 2 readiness checklist.

Mentor review checkpoint: Day 5 scope gate. A project is approved only if the chemical set, comparison, evidence type, and likely limitation are explicit.

Common failure modes: Questions are too broad, students treat tentative hits as confirmed identities, or database annotations are treated as sample evidence.

Minimum viable path: Use a program-provided chemical list and write a narrow comparative question.

Advanced extension path: Bring a student-interest chemical set and justify each chemical with a source and ambiguity note.

Day 1: Orientation to DSU dsDNA Core Expectations

Purpose: Establish the research culture before analysis begins. Students learn that reproducibility, evidence discipline, and cautious claims matter as much as software output.

Learning objectives: Describe the program promise, explain what a research log is for, distinguish private data from shareable training data, and name at least three ways chemical evidence can be overclaimed.

Prerequisites and materials: This manual, project tracks chapter, daily research log worksheet, local code of conduct or lab expectations, and a notebook or digital log.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes orientation, 90 minutes project-domain discussion, 120 minutes interest inventory, 60 minutes log setup, 30 minutes review.

Morning concept block: Discuss responsible conduct, data integrity, file ownership, and how the dsDNA Core evaluates progress. Emphasize that a clean negative result or a well-supported limitation is valid research output.

Guided lab or reading discussion: Read the program outcomes and final package expectations. Annotate where the final report, reproducible folder, evidence trail, and continuation note will come from.

Independent project block: Write an interest inventory with systems, chemicals, methods, organisms, health questions, environmental questions, food or flavor questions, or data-science questions that could become projects.

Evidence and writing block: Draft a first research log entry that records today's orientation decisions and one scientific claim students already know they must avoid.

Research log prompt: What kind of chemical evidence would be exciting, and what would it not prove by itself?

End-of-day deliverable: Research log created and a one-page interest inventory.

Quality gate: The log must include a dated objective, at least one decision, one risk, and one question for review.

Common problems and fixes: If the project idea is a topic rather than a question, rewrite it as system plus chemical set plus possible comparison. If the log is vague, add concrete file names or decisions.

Optional advanced challenge: Find one published chemical study in a field of interest and identify its analytical method, chemical evidence, and main limitation.

Day 2: Turning Broad Interests Into Testable Chemical Questions

Purpose: Convert curiosity into a question that can be answered in 10 weeks with available data and source-backed chemical metadata.

Learning objectives: Write a question with system, chemical set, comparison, evidence source, and limitation; reject questions that require unavailable confirmation; and create three candidate

questions.

Prerequisites and materials: Day 1 interest inventory, project tracks chapter, project brief worksheet, and chemical intake table template.

Estimated time budget: Full internship day: 30 minutes check-in, 90 minutes question design, 90 minutes guided triage, 120 minutes independent drafting, 60 minutes project brief writing, 30 minutes review.

Morning concept block: Compare broad questions such as “What chemicals are important?” with feasible questions such as “Which tentative GC-MS hits in this list have source-backed sensory or plant-occurrence annotations?”

Guided lab or reading discussion: Complete Lab 1, Chemical Question Triage. Classify each candidate as feasible, feasible with narrower scope, or not feasible.

Independent project block: Draft three candidate questions and identify what data would be needed for each.

Evidence and writing block: For each question, write one sentence beginning “This analysis can support...” and one sentence beginning “This analysis cannot prove...”

Research log prompt: Which question is most feasible and what evidence would make it stronger?

End-of-day deliverable: Three candidate research questions with feasibility labels.

Quality gate: Each candidate must name a chemical set, source of chemical names, comparison or grouping idea, and major limitation.

Common problems and fixes: If a question depends on confirmed identity, rewrite it around tentative annotations or planned follow-up confirmation. If the comparison is unclear, define the groups before naming software.

Optional advanced challenge: Write a backup question that could still work if no student-generated data are available.

Day 3: Chemical Identity, Tentative Annotation, and Standards

Purpose: Build scientific caution around GC-MS annotation before students touch workflow outputs.

Learning objectives: Distinguish tentative library match, standard-confirmed identity, Pub-Chem record, synonym, CID, and analytical evidence; explain why a match factor is not proof.

Prerequisites and materials: Chemical intake table template, example GC-MS hit table description, Week 4 input column list, and database evidence map worksheet.

Estimated time budget: Full internship day: 30 minutes check-in, 90 minutes concept block, 90 minutes identifier examples, 120 minutes intake table work, 60 minutes uncertainty note, 30 minutes review.

Morning concept block: Explain that tentative GC-MS hits are hypotheses. Discuss standards, retention time, mass spectra, match factors, base peaks, and why names can be ambiguous.

Guided lab or reading discussion: Compare a simple compound, a synonym-heavy compound, and an ambiguous name. Record what makes each easy or risky.

Independent project block: Start the chemical intake table with candidate chemical names, source of the name, confidence, and reason for inclusion.

Evidence and writing block: Draft the first uncertainty paragraph for the project brief.

Research log prompt: Which chemical name in the current list is most ambiguous, and what would reduce that ambiguity?

End-of-day deliverable: Chemical intake table v1 and tentative identity caution paragraph.

Quality gate: No chemical should appear without a source and uncertainty note.

Common problems and fixes: If students copy names without source context, add the original file, paper, or database page. If they say “identified,” change to “tentatively annotated” unless standards confirm identity.

Optional advanced challenge: Add candidate CID or InChIKey fields for five chemicals and note conflicts.

Day 4: Evidence Source Landscape

Purpose: Teach students what public databases can and cannot support before enrichment outputs are interpreted.

Learning objectives: Describe roles for PubChem, KEGG, LOTUS, FEMA, FDA/SPL, MeSH, and literature; separate identity, property, pathway, occurrence, sensory, regulatory, and biomedical evidence.

Prerequisites and materials: Database evidence map worksheet, project chemical intake table, supplementary readings, and project tracks chapter.

Estimated time budget: Full internship day: 30 minutes check-in, 90 minutes evidence landscape, 90 minutes source comparison, 120 minutes evidence map work, 60 minutes claim caution writing, 30 minutes review.

Morning concept block: Explain evidence boundaries: database presence is not sample presence, pathway context is not pathway activity, and hazard screening is not risk assessment.

Guided lab or reading discussion: Compare what PubChem, KEGG, LOTUS, FEMA, FDA/SPL, and MeSH would contribute to the same chemical.

Independent project block: Build an evidence feasibility matrix for the three candidate questions.

Evidence and writing block: Write safe wording templates for the project’s most likely evidence types.

Research log prompt: Which source is likely to be most useful, and what claim would be unsafe from that source alone?

End-of-day deliverable: Evidence feasibility matrix.

Quality gate: Each source entry must say both what it can support and what it cannot support.

Common problems and fixes: If a source is described as “proves,” rewrite using “is annotated as,” “is associated with,” or “has a database record indicating.”

Optional advanced challenge: Add one reviewer challenge question for each evidence source.

Day 5: Scope Gate and Project Brief Review

Purpose: Choose a feasible project before students invest in setup and analysis.

Learning objectives: Defend a selected question, identify the minimum viable project, write a risk note, and define the Week 2 setup plan.

Prerequisites and materials: Candidate questions, chemical intake table v1, evidence feasibility matrix, project brief worksheet, and rubric.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes scope discussion, 90 minutes brief drafting, 120 minutes mentor review revisions, 60 minutes risk note, 30 minutes sign-off.

Morning concept block: Review the decision gate: the project must be answerable, reproducible, evidence-backed, and narrow enough for 10 weeks.

Guided lab or reading discussion: Students present candidate questions and receive structured critique on feasibility and evidence limits.

Independent project block: Revise the selected project brief and define the backup path if live databases, student data, or specific evidence sources fail.

Evidence and writing block: Write a risk note naming missing data, ambiguous identifiers, weak evidence sources, and overclaiming hazards.

Research log prompt: What was approved, what was rejected, and what evidence would trigger a later scope change?

End-of-day deliverable: Reviewed one-page project brief and risk note.

Quality gate: A reviewer can identify the question, chemical set, evidence sources, comparison, minimum viable product, and claim boundary in under five minutes.

Common problems and fixes: If the brief promises biological mechanism, rewrite to metadata-supported interpretation or follow-up hypothesis. If the chemical list is too large, cap it for the first analysis pass.

Optional advanced challenge: Draft a short alternative project brief for a methods or data-quality project.

Day 6: Weekend Reading Annotation and Intake Revision

Purpose: Keep the project moving with focused reading and cleanup while avoiding full weekday workload.

Learning objectives: Annotate one source, revise chemical intake fields, and identify one project assumption that needs support.

Prerequisites and materials: Project brief, chemical intake table, one approved reading or database reference, and research log.

Estimated time budget: Weekend asynchronous work, 1.5 to 2.5 hours.

Morning concept block: Not scheduled. Begin with the selected reading or source record.

Guided lab or reading discussion: Use the reading annotation template to capture system, method, chemical evidence, main claim, and limitation.

Independent project block: Revise the chemical intake table based on the reading.

Evidence and writing block: Add one source-backed sentence to the project brief or risk note.

Research log prompt: What did the reading change about the project scope or chemical list?

End-of-day deliverable: Reading annotation and revised intake table.

Quality gate: The annotation must include a limitation, not just a summary.

Common problems and fixes: If the reading is only background, label it as background and do not treat it as evidence for project results.

Optional advanced challenge: Add a second reading that disagrees with or complicates the first.

Day 7: Weekend Reflection and Week 2 Readiness

Purpose: Close Week 1 with a clean project direction and a setup checklist.

Learning objectives: Identify unresolved risks, prepare for R/RStudio setup, and make the project brief readable to someone else.

Prerequisites and materials: Project brief, risk note, chemical intake table, research log, and Week 2 installation checklist.

Estimated time budget: Weekend asynchronous work, 1 to 2 hours.

Morning concept block: Not scheduled. Focus on revision and readiness.

Guided lab or reading discussion: Reread the project brief as if it belonged to another student and mark unclear terms.

Independent project block: Clean the brief, update the risk note, and list files or accounts needed for Week 2.

Evidence and writing block: Write a Week 1 reflection naming the strongest project element and the biggest uncertainty.

Research log prompt: What must be true by the end of Week 2 for this project to remain feasible?

End-of-day deliverable: Week 2 readiness checklist.

Quality gate: The project brief can be understood without verbal explanation.

Common problems and fixes: If the brief still depends on undefined data, add a case-data fallback.

Optional advanced challenge: Make a one-slide project overview for Monday check-in.

2.2 Week 2: R, RStudio, Git, and Reproducible Project Structure

Week 2: R, RStudio, Git, and reproducible project structure

Research milestone: The project becomes a runnable folder rather than a collection of files.

Primary product: Installed uafR, project skeleton, README, first scripts, Git log, and reproducibility checkpoint.

Week 2 Operating Plan

Concepts: RStudio projects, relative paths, raw versus processed data, scripts as records, Git commits, ignored files, secrets, and clean-session testing.

uafR/R skills: Run student bundle checks, create project folders, read and write CSV files, inspect data frames, run `training/scripts/00_install_check.R` and `training/scripts/01_project_setup.R`.

Required readings or references: Installation appendix, quick reference, reproducible project structure notes, and Git safety guidance.

Required files/scripts/data: Student bundle scripts, `training/scripts/00_install_check.R`, `training/scripts/01_project_setup.R`, and project README template.

Weekly deliverables: Install verification, reproducible project folder, README, script order, session information, Git checkpoint, and peer handoff note.

Mentor review checkpoint: Day 12 clean-session rerun and folder review.

Common failure modes: Absolute paths, edited raw data, missing README instructions, committed private data, and scripts that only work interactively.

Minimum viable path: Build the standard project skeleton and document script order.

Advanced extension path: Add automated project checks and a reproducibility script that writes outputs to `results/`.

Day 8: Installation Check and Project Skeleton

Purpose: Confirm that the working environment is ready before analysis begins.

Learning objectives: Run bundle integrity checks, verify uafR installation, create a reproducible project folder, and record package versions.

Prerequisites and materials: Student bundle, `START_HERE.md`, `verify_bundle_integrity.R`, `preflight_check.R`, `install_uafR_from_bundle.R`, and `training/scripts/01_project_setup.R`.

Estimated time budget: Full internship day: 30 minutes check-in, 60 minutes installation concepts, 120 minutes setup lab, 120 minutes project skeleton work, 60 minutes documentation, 30 minutes review.

Morning concept block: Explain why installation logs and session information are part of research evidence.

Guided lab or reading discussion: Source `verify_bundle_integrity.R`, `preflight_check.R`, `install_uafR_from_bundle.R`, and `run_student_acceptance_test.R`. Review success messages and log files.

Independent project block: Source `training/scripts/01_project_setup.R` and run `create_training_project()` for the student project.

Evidence and writing block: Add the project question and data status to `README.md`; save `sessionInfo()` in `logs/`.

Research log prompt: What was installed, what version was installed, and what evidence proves the setup worked?

End-of-day deliverable: Project skeleton with `README`, logs folder, installation log, and session information.

Quality gate: Another person can open the folder and identify the project question, data folders, script folders, and setup evidence.

Common problems and fixes: If `library(uafR)` fails, restart RStudio and rerun verification. If paths point to a local desktop folder, replace them with project-relative paths.

Optional advanced challenge: Add a script named `scripts/00_check_setup.R` that reruns the installation check inside the student project.

Day 9: Reading, Inspecting, Cleaning, and Writing Tables

Purpose: Teach students how table operations become reproducible evidence.

Learning objectives: Read CSV files, inspect dimensions and column names, identify missing values, write processed outputs, and avoid editing raw files.

Prerequisites and materials: Project skeleton, chemical intake table template, evidence table template, and RStudio.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes table concepts, 120 minutes guided R practice, 120 minutes project table work, 60 minutes documentation, 30 minutes review.

Morning concept block: Explain raw data preservation, processed data, row counts, column names, and why cleaning choices must be scripted.

Guided lab or reading discussion: Practice `read.csv()`, `str()`, `names()`, `dim()`, `summary()`, and `write.csv()` on templates.

Independent project block: Create or revise a project chemical intake CSV and write it to `data_processed/` from a script.

Evidence and writing block: Write comments in the script explaining each cleaning decision.

Research log prompt: Which rows or columns changed today, and why?

End-of-day deliverable: Data import and cleaning script with a saved processed table.

Quality gate: Running the script from the project root recreates the processed table without manual steps.

Common problems and fixes: If a script depends on the working directory set by clicking, use an RStudio project root or relative paths. If raw files were edited, restore them and script the change.

Optional advanced challenge: Add checks that stop the script if required columns are missing.

Day 10: Git Basics and Safe Repository Habits

Purpose: Make version history a normal part of research practice.

Learning objectives: Explain commits, inspect status, avoid committing secrets or private data, and use `.gitignore`.

Prerequisites and materials: Project folder, Git-enabled RStudio or terminal, `.gitignore`, and README.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes Git concepts, 90 minutes guided commits, 120 minutes cleanup, 60 minutes README revision, 30 minutes review.

Morning concept block: Discuss what belongs in version control: scripts, documentation, templates, and small shareable derived outputs. Discuss what does not belong: tokens, private raw data, large temporary files, and local machine paths.

Guided lab or reading discussion: Run status checks, stage a small README change, commit it, then inspect history.

Independent project block: Add or revise `.gitignore`, remove scratch files from the candidate commit, and make a meaningful project checkpoint commit if appropriate.

Evidence and writing block: Write a short repository policy in the README explaining where raw data are stored and what is intentionally ignored.

Research log prompt: What did the commit preserve, and what files were intentionally excluded?

End-of-day deliverable: Clean repository status or documented reason for remaining untracked files.

Quality gate: No secrets, private data, local usernames, or absolute paths appear in committed text.

Common problems and fixes: If Git shows many generated files, update `.gitignore`. If private data were staged, unstage them and document the approved storage location.

Optional advanced challenge: Write a short commit message standard for the project.

Day 11: Numbered Workflows and README Structure

Purpose: Turn the project folder into a runnable research record.

Learning objectives: Define script order, separate raw and processed data, name outputs predictably, and explain how to rerun the project.

Prerequisites and materials: Project folder, Day 9 cleaning script, README, and final package checklist.

Estimated time budget: Full internship day: 30 minutes check-in, 60 minutes workflow design, 90 minutes guided folder review, 150 minutes independent script planning, 60 minutes README

writing, 30 minutes peer check.

Morning concept block: Explain numbered workflows: `00_check_setup.R`, `01_import_data.R`, `02_core_workflow.R`, and so on.

Guided lab or reading discussion: Review examples of clear and unclear README files. Identify what a future student needs to know first.

Independent project block: Rename or draft scripts in logical order and specify expected inputs and outputs for each script.

Evidence and writing block: Update README with research question, data description, script order, output locations, and current limitations.

Research log prompt: Which part of the project is now reproducible, and which part still depends on manual interpretation?

End-of-day deliverable: Numbered script plan and updated README.

Quality gate: A peer can identify the first script to run and where outputs should appear.

Common problems and fixes: If scripts do too many unrelated tasks, split them. If outputs overwrite each other, add descriptive file names.

Optional advanced challenge: Add an output manifest listing expected files after each script.

Day 12: Clean R Session Reproducibility Check

Purpose: Test the project as a user, not as the author.

Learning objectives: Restart R, rerun setup, document errors, and obtain mentor sign-off on folder structure.

Prerequisites and materials: Project folder, README, scripts, installation check script, and research log.

Estimated time budget: Full internship day: 30 minutes check-in, 60 minutes reproducibility expectations, 120 minutes clean-session rerun, 120 minutes fixes, 60 minutes documentation, 30 minutes sign-off.

Morning concept block: Explain that reproducibility is tested from a clean session because hidden objects in the workspace can mask broken scripts.

Guided lab or reading discussion: Restart RStudio, run the installation check, run each current script in order, and record failures.

Independent project block: Fix path errors, missing packages, unclear file names, and README gaps.

Evidence and writing block: Save session information and write a reproducibility checkpoint note.

Research log prompt: What failed from a clean session, what fixed it, and how can that failure be prevented?

End-of-day deliverable: Reproducibility checkpoint and mentor sign-off note.

Quality gate: The current project can be rerun from a clean R session through all scripts available so far.

Common problems and fixes: If objects are missing, the script is relying on interactive workspace state. Create objects inside the script or load them explicitly.

Optional advanced challenge: Ask a peer to rerun the project using README instructions only.

Day 13: Weekend Debugging Log and Command Recap

Purpose: Reinforce setup commands and package-version awareness.

Learning objectives: Record session information, summarize common commands, and maintain a debugging log.

Prerequisites and materials: Project folder, research log, session information, and quick reference.

Estimated time budget: Weekend asynchronous work, 1 to 2 hours.

Morning concept block: Not scheduled. Focus on command recall and documentation.

Guided lab or reading discussion: Review the commands used this week and annotate what each one does.

Independent project block: Add a command recap section to the README or research log.

Evidence and writing block: Save a fresh `sessionInfo()` note and list packages used so far.

Research log prompt: Which command is most important to remember, and what mistake would it prevent?

End-of-day deliverable: Debugging log and command recap.

Quality gate: The recap is specific enough for another student to follow.

Common problems and fixes: If entries say only “it worked,” add the exact command, output location, and package version.

Optional advanced challenge: Add a short troubleshooting table to the project README.

Day 14: Weekend Peer Handoff

Purpose: Test whether the project folder communicates without the author present.

Learning objectives: Review another project, identify missing instructions, and revise the README for handoff clarity.

Prerequisites and materials: Project folder, README, final package checklist, and peer review form.

Estimated time budget: Weekend asynchronous work, 1.5 to 3 hours.

Morning concept block: Not scheduled. Begin with folder review.

Guided lab or reading discussion: Exchange folders or screenshots and try to identify the project question, data status, and script order from README alone.

Independent project block: Revise README and folder organization based on reviewer confusion.

Evidence and writing block: Write a peer handoff note naming what was clear and what was fixed.

Research log prompt: What did the reviewer misunderstand, and how was the project documentation changed?

End-of-day deliverable: Peer handoff note and revised README.

Quality gate: A reviewer can understand the project without verbal explanation.

Common problems and fixes: If a reviewer cannot find data or scripts, add a top-level directory guide.

Optional advanced challenge: Create a one-paragraph “start here” section at the top of the README.

2.3 Week 3: Chemical Data Literacy and Public Database Foundations

Week 3: Chemical data literacy and public database foundations

Research milestone: Students understand identifiers, ambiguity, and source-specific evidence before enrichment.

Primary product: Identifier ambiguity note, database evidence map v1, source comparison table, and revised project scope.

Week 3 Operating Plan

Concepts: Names, synonyms, salts, mixtures, stereochemistry, CIDs, InChIKeys, PubChem records, KEGG biochemical context, LOTUS occurrence, FEMA/FDA/SPL/MeSH source roles, and literature context.

uafR/R skills: Inspect source-backed outputs conservatively. Optional use of `pubchemProfile()` and `keggProfile()` should be kept small and documented.

Required readings or references: PubChem, KEGG, LOTUS, reporting standards, and at least one approved field-specific source.

Required files/scripts/data: Database evidence map, chemical intake table, source comparison worksheet, and evidence table template.

Weekly deliverables: Identifier ambiguity note, PubChem evidence worksheet, KEGG evidence worksheet, source comparison table, evidence map v1, and scope adjustment.

Mentor review checkpoint: Day 19 evidence-source interpretation review.

Common failure modes: Treating synonyms as separate chemicals, assuming pathway activity, treating occurrence in one organism as occurrence in another, and ignoring mixture or salt ambiguity.

Minimum viable path: Audit five project chemicals and record which sources are useful.

Advanced extension path: Compare manual source records to uafR-assisted profiles for a small set.

Day 15: Chemical Names, Identifiers, and Ambiguity

Purpose: Prevent identifier errors from contaminating the whole project.

Learning objectives: Explain synonyms, salts, mixtures, stereochemistry, CIDs, and InChIKeys; identify ambiguous project chemicals; decide when a name is not safe enough for analysis.

Prerequisites and materials: Chemical intake table, identifier ambiguity drill, and source records for five chemicals.

Estimated time budget: Full internship day: 30 minutes check-in, 90 minutes identifier concept block, 90 minutes guided ambiguity drill, 120 minutes project chemical audit, 60 minutes evidence note, 30 minutes review.

Morning concept block: Demonstrate how one common name can map to multiple records and how salts, stereochemistry, and mixtures change interpretation.

Guided lab or reading discussion: Complete the Identifier Ambiguity Drill from the lab compendium.

Independent project block: Audit five chemicals from the project list and record preferred name, alternate names, CID if available, and ambiguity status.

Evidence and writing block: Write a rule for excluding or flagging ambiguous identifiers.

Research log prompt: Which identifier is most likely to mislead the analysis?

End-of-day deliverable: Identifier ambiguity note.

Quality gate: Every audited chemical has a preferred label, source, and ambiguity decision.

Common problems and fixes: If a name returns multiple plausible records, do not choose silently. Flag it and document the decision rule.

Optional advanced challenge: Add InChIKey or CID columns to the chemical intake table where defensible.

Day 16: PubChem Manual and uafR-Assisted Source Audit

Purpose: Learn PubChem as a source ecosystem rather than a single answer.

Learning objectives: Identify PubChem identity, property, classification, safety, literature, assay, and source-annotation records; distinguish manual source inspection from automated enrichment.

Prerequisites and materials: PubChem reading, chemical intake table, evidence map worksheet, and optional `pubchemProfile()` documentation.

Estimated time budget: Full internship day: 30 minutes check-in, 90 minutes PubChem concept block, 120 minutes guided source audit, 90 minutes project audit, 60 minutes writing, 30 minutes review.

Morning concept block: Explain PubChem CIDs, depositor-supplied records, annotations, and why source context matters.

Guided lab or reading discussion: Manually inspect one compound and identify which fields are identity, property, hazard, classification, literature, or assay evidence.

Independent project block: Audit two project chemicals and add PubChem-relevant evidence types to the evidence map.

Evidence and writing block: Write one strong and one weak PubChem-based claim.

Research log prompt: Which PubChem field appears useful, and what limitation should accompany it?

End-of-day deliverable: PubChem evidence worksheet.

Quality gate: The worksheet names the source field and evidence type, not just the compound page.

Common problems and fixes: If students cite PubChem broadly, require a table, heading, or source-specific field.

Optional advanced challenge: Run a small `pubchemProfile()` call and compare automated fields to manual inspection.

Day 17: KEGG Compounds, Reactions, Enzymes, Modules, and Pathways

Purpose: Teach biochemical context without overstating activity or presence.

Learning objectives: Explain KEGG compound records, reaction IDs, enzyme IDs, modules, pathways, and what each can support.

Prerequisites and materials: KEGG reading, evidence map worksheet, optional `keggProfile()` documentation, and project chemical list.

Estimated time budget: Full internship day: 30 minutes check-in, 90 minutes KEGG concept block, 120 minutes guided record review, 90 minutes project mapping, 60 minutes caution writing, 30 minutes review.

Morning concept block: Distinguish pathway membership or reaction participation from measured pathway activity.

Guided lab or reading discussion: Inspect a KEGG compound record when available and identify compound, pathway, reaction, enzyme, and module fields.

Independent project block: Record KEGG relevance for project chemicals that have plausible KEGG context.

Evidence and writing block: Write a safe pathway-context sentence and an unsafe pathway-activity sentence, then revise the unsafe sentence.

Research log prompt: What KEGG context is useful, and what would require experimental confirmation?

End-of-day deliverable: KEGG evidence worksheet.

Quality gate: No pathway claim should imply activity, abundance, or biological flux in the sample.

Common problems and fixes: If a pathway name is treated as a result, reframe it as context or hypothesis.

Optional advanced challenge: Compare KEGG context across two compounds and identify shared reactions or pathways if present.

Day 18: LOTUS, FEMA, FDA/SPL, MeSH, and Other Source Evidence

Purpose: Show that source families answer different kinds of questions.

Learning objectives: Separate natural-product occurrence, sensory/flavor context, regulatory or label context, biomedical terms, and literature evidence.

Prerequisites and materials: Source comparison worksheet, project chemical intake table, and readings on source-specific limitations.

Estimated time budget: Full internship day: 30 minutes check-in, 90 minutes source landscape, 120 minutes guided comparison, 90 minutes project source map, 60 minutes limitation writing, 30 minutes review.

Morning concept block: Discuss LOTUS occurrence evidence, FEMA flavor context, FDA/SPL labeling context, and MeSH biomedical indexing.

Guided lab or reading discussion: Compare source records for a familiar compound and label each as identity, occurrence, sensory, regulatory, biomedical, or literature context.

Independent project block: Add source roles to the evidence map for project chemicals.

Evidence and writing block: Draft source-specific caveats for each source that appears relevant.

Research log prompt: Which source is easiest to overinterpret and why?

End-of-day deliverable: Source comparison table.

Quality gate: Each source role has a matching “cannot support” note.

Common problems and fixes: If natural-product occurrence is treated as sample occurrence, add organism and source-context caveats.

Optional advanced challenge: Add a field-specific source approved by the program lead and define its evidence boundary.

Day 19: Database Evidence Map Review

Purpose: Freeze the first evidence plan before automated enrichment begins.

Learning objectives: Summarize source coverage expectations, revise project scope based on evidence, and identify missing evidence types.

Prerequisites and materials: Identifier note, PubChem worksheet, KEGG worksheet, source comparison table, evidence map.

Estimated time budget: Full internship day: 30 minutes check-in, 60 minutes synthesis, 120 minutes evidence map revision, 120 minutes scope adjustment, 60 minutes review memo, 30 minutes checkpoint.

Morning concept block: Discuss how weak source coverage can change a project question without making the project fail.

Guided lab or reading discussion: Review examples of evidence maps and identify unsupported cells.

Independent project block: Revise the evidence map and project scope based on actual source expectations.

Evidence and writing block: Write an evidence-source interpretation memo.

Research log prompt: What changed in the project because of database evidence limits?

End-of-day deliverable: Database evidence map v1 and scope adjustment note.

Quality gate: Each planned result has a plausible evidence source or is removed from scope.

Common problems and fixes: If the evidence map is mostly empty, narrow the question or switch to a methods/data-quality project.

Optional advanced challenge: Add confidence labels for each evidence source in the map.

Day 20: Weekend Reading Annotation on Public Databases

Purpose: Deepen source literacy with one focused reading.

Learning objectives: Annotate a source document or paper, extract evidence limits, and update the evidence map.

Prerequisites and materials: Approved PubChem, KEGG, LOTUS, metabolomics reporting, or field-specific reading.

Estimated time budget: Weekend asynchronous work, 1.5 to 3 hours.

Morning concept block: Not scheduled. Begin with reading annotation.

Guided lab or reading discussion: Complete the reading annotation template.

Independent project block: Add one note from the reading to the project evidence map.

Evidence and writing block: Write a limitation sentence that should appear later in the report.

Research log prompt: What did the reading clarify about source reliability?

End-of-day deliverable: Reading annotation and updated evidence map cell.

Quality gate: The annotation includes a method or source limitation.

Common problems and fixes: If the reading is not directly relevant, document it as background and choose a better source for evidence.

Optional advanced challenge: Compare the reading to a second source with a different database or method perspective.

Day 21: Weekend Evidence Map Revision

Purpose: Prepare for the GC-MS workflow by removing weak or unsupported project claims.

Learning objectives: Revise evidence map, update chemical intake table, and name Week 4 data needs.

Prerequisites and materials: Evidence map v1, chemical intake table, project brief, and research log.

Estimated time budget: Weekend asynchronous work, 1 to 2 hours.

Morning concept block: Not scheduled. Focus on revision.

Guided lab or reading discussion: Revisit each evidence map row and mark keep, revise, or remove.

Independent project block: Update the project brief and chemical intake table.

Evidence and writing block: Write a Week 3 summary paragraph for the research log.

Research log prompt: What evidence type will be most important in Week 4 and Week 5?

End-of-day deliverable: Revised evidence map and Week 4 readiness checklist.

Quality gate: The project can move into GC-MS workflow without unresolved identifier problems blocking every chemical.

Common problems and fixes: If more than half the list remains ambiguous, define a smaller verified subset.

Optional advanced challenge: Draft a source coverage hypothesis for the enrichment phase.

2.4 Week 4: uafR Core GC-MS Workflow

Week 4: uafR core GC-MS workflow

Research milestone: Students can run and explain the offline GC-MS workflow and its limits.

Primary product: Input column audit, offline workflow output, uncertainty note, methods paragraph, and processed-output cleanup.

Week 4 Operating Plan

Concepts: GC-MS hit-table anatomy, match factors, retention time, base peak, area, file names, duplicate peaks, suspicious values, and tentative identification limits.

uafR/R skills: Use `spreadOut()`, `mzExacto()`, and `exactoThese()` in a cautious workflow. Use `training/scripts/04_core_workflow.R` for offline examples.

Required readings or references: Function docs for `spreadOut()`, `mzExacto()`, and `exactoThese()`; Week 4 chapter; GC-MS input column audit worksheet.

Required files/scripts/data: `standard_data`, `standard_spread`, `standard_exacto`, `training/scripts/04_core_workflow.R`, and optional student or case dataset.

Weekly deliverables: Input audit, annotated offline workflow, duplicate/uncertainty note, processed output v1, and methods paragraph.

Mentor review checkpoint: Day 26 methods paragraph and data-preservation caveat.

Common failure modes: Wrong column names, duplicate peak keys, treating match factor as confirmation, and losing raw data provenance.

Minimum viable path: Use bundled offline outputs and explain them correctly.

Advanced extension path: Apply the workflow to a small project or case dataset after input audit passes.

Day 22: GC-MS Hit-Table Anatomy

Purpose: Ensure students understand the input format before running uafR.

Learning objectives: Identify required columns, explain match factor, retention time, base peak, component area, file name, and suspicious values.

Prerequisites and materials: Week 4 chapter, `spreadOut()` documentation, example `standard_data`, and GC-MS input column audit worksheet.

Estimated time budget: Full internship day: 30 minutes check-in, 90 minutes GC-MS table concept block, 90 minutes guided input audit, 120 minutes project or case input review, 60 minutes notes, 30 minutes review.

Morning concept block: Explain why `spreadOut()` requires `Component.RT`, `Component.Area`, `Base.Peak.MZ`, `File.Name`, `Compound.Name`, and `Match.Factor`.

Guided lab or reading discussion: Inspect `standard_data` and compare its columns to a malformed example.

Independent project block: Audit student or case data for required columns, missing values, duplicate peak keys, and suspicious match factors.

Evidence and writing block: Write an input-quality note for the methods folder.

Research log prompt: What input issue would most damage downstream interpretation?

End-of-day deliverable: GC-MS input column audit.

Quality gate: The audit states whether the file can be processed and what fixes are allowed.

Common problems and fixes: If a required column is absent, do not guess silently. Rename only when the source field is clearly equivalent and document it.

Optional advanced challenge: Write a small R check that stops when required columns are missing.

Day 23: Offline Core Workflow With Package Examples

Purpose: Run the core workflow without relying on live web services.

Learning objectives: Source the training workflow, inspect `standard_spread` and `standard_exacto`, explain where `spreadOut()`, `mzExacto()`, and `exactoThese()` fit.

Prerequisites and materials: Installed uafR, `training/scripts/04_core_workflow.R`, package example data, and research log.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes workflow overview, 120 minutes guided script run, 120 minutes output inspection, 60 minutes annotation, 30 minutes review.

Morning concept block: Explain that offline saved outputs are teaching artifacts; live lookup is a separate optional step.

Guided lab or reading discussion: Run `run_core_workflow(live_lookup = FALSE)` and inspect the returned list.

Independent project block: Annotate the workflow script with comments explaining inputs, outputs, and limitations.

Evidence and writing block: Draft a paragraph explaining what the exact-match output can and cannot show.

Research log prompt: Which object is raw input, which is reshaped output, and which is extracted result?

End-of-day deliverable: Annotated core workflow script and output inspection notes.

Quality gate: Students can explain why the offline workflow is reproducible and why live lookup may vary.

Common problems and fixes: If `uafR` is not found, rerun installation verification. If outputs are not understood, inspect structure before plotting or interpreting.

Optional advanced challenge: Use `exactoThese()` with a saved categorate result to create a targeted query list, noting legacy and research schema differences where relevant.

Day 24: Duplicate Hits, Ambiguous Peaks, and Tentative Limits

Purpose: Prevent students from treating workflow output as final truth.

Learning objectives: Identify duplicate or conflicting hits, explain why multiple candidates can share evidence, and write cautious interpretation.

Prerequisites and materials: Offline workflow output, input audit, duplicate troubleshooting exercise, and methods notes.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes uncertainty concept block, 120 minutes duplicate exercise, 120 minutes project/case review, 60 minutes writing, 30 minutes review.

Morning concept block: Discuss why GC-MS tentative annotation can produce similar or conflicting names and why decontamination or filtering must be documented.

Guided lab or reading discussion: Review a wrong-column or duplicate-key example and predict how it should fail.

Independent project block: Identify duplicate, ambiguous, or low-confidence features in project or case outputs.

Evidence and writing block: Write a duplicate and uncertainty note for the methods folder.

Research log prompt: Which hits should be treated with the greatest caution and why?

End-of-day deliverable: Duplicate and uncertainty note.

Quality gate: Each flagged issue has a decision: keep, remove, rerun, or mark as uncertain.

Common problems and fixes: If students delete rows manually, require a scripted filter and a reason.

Optional advanced challenge: Compare outputs under two documented filtering choices and record sensitivity.

Day 25: Applying the Workflow to Student or Case Data

Purpose: Move from example outputs to project-specific processing only when the input audit supports it.

Learning objectives: Run a small workflow subset, save output objects, and document all settings and caveats.

Prerequisites and materials: Passing input audit, project or case data, `training/scripts/04_core_workflow.R`, and output folders.

Estimated time budget: Full internship day: 30 minutes check-in, 60 minutes processing plan, 120 minutes guided run, 150 minutes independent troubleshooting, 45 minutes output saving, 30 minutes review.

Morning concept block: Explain why small test runs protect time and reduce confusion.

Guided lab or reading discussion: Plan the first project run using a limited chemical list or case dataset.

Independent project block: Run the workflow, save processed outputs, and capture errors.

Evidence and writing block: Write the processing settings and output locations in the research log.

Research log prompt: What ran successfully, what failed, and what should not be interpreted yet?

End-of-day deliverable: Processed output v1 or documented reason processing is not yet valid.

Quality gate: The output must be generated by a script and saved with clear file names.

Common problems and fixes: If live services are slow, return to saved offline outputs and document the live-query limitation.

Optional advanced challenge: Compare project output to bundled output structure and note differences.

Day 26: Methods Writing Clinic for GC-MS Processing

Purpose: Convert technical workflow steps into reproducible scientific methods.

Learning objectives: Write methods with input columns, package functions, settings, output objects, filtering rules, and tentative-identification caveats.

Prerequisites and materials: Input audit, workflow script, output object, duplicate note, and methods rubric.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes methods concept block, 90 minutes example critique, 120 minutes methods drafting, 75 minutes peer review, 30 minutes revision.

Morning concept block: Compare method text that is reproducible with text that only says “data were analyzed in R.”

Guided lab or reading discussion: Mark missing details in a weak methods paragraph.

Independent project block: Draft the GC-MS processing methods paragraph.

Evidence and writing block: Add a data-preservation caveat and tentative-identification caveat.

Research log prompt: What method detail would another student need to reproduce this step?

End-of-day deliverable: Methods paragraph draft.

Quality gate: The paragraph names functions, inputs, settings, output objects, and limitations without claiming confirmation.

Common problems and fixes: If the methods text lacks function names or settings, add them. If it says compounds were identified, revise to tentative language unless standards support confirmation.

Optional advanced challenge: Add a short workflow diagram in the project notes.

Day 27: Weekend Wrong-Column Troubleshooting Exercise

Purpose: Practice diagnosing input failures without damaging project data.

Learning objectives: Identify malformed input, explain expected error messages, and rerun from the project root.

Prerequisites and materials: Wrong-column exercise, input audit worksheet, project README, and research log.

Estimated time budget: Weekend asynchronous work, 1.5 to 2.5 hours.

Morning concept block: Not scheduled. Begin with the troubleshooting exercise.

Guided lab or reading discussion: Compare malformed columns to required columns and write the fix plan.

Independent project block: Rerun the valid workflow from the project root and confirm outputs still appear.

Evidence and writing block: Add one troubleshooting entry to the research log.

Research log prompt: What error would help diagnose a real input problem?

End-of-day deliverable: Wrong-column troubleshooting note.

Quality gate: The fix is described as a documented transformation, not a manual edit to raw data.

Common problems and fixes: If the exercise is solved by editing raw input, rewrite the solution as a script.

Optional advanced challenge: Add an input-validation function to the student's import script.

Day 28: Weekend Output Cleanup and Week 5 Readiness

Purpose: Prepare for enrichment by cleaning outputs and documenting the query list.

Learning objectives: Archive outputs, update research log, and define the chemical list for enrichment.

Prerequisites and materials: Processed output v1, chemical intake table, methods paragraph, and Week 5 checklist.

Estimated time budget: Weekend asynchronous work, 1 to 2 hours.

Morning concept block: Not scheduled. Focus on cleanup and readiness.

Guided lab or reading discussion: Review output folder and remove or archive confusing scratch outputs.

Independent project block: Create the Week 5 enrichment query list with inclusion rules.

Evidence and writing block: Update the research log with output locations and unresolved caveats.

Research log prompt: Which chemicals are ready for enrichment and which are excluded?

End-of-day deliverable: Week 5 readiness checklist and enrichment query list.

Quality gate: The enrichment list has an inclusion rule and excludes unresolved ambiguous identifiers.

Common problems and fixes: If the query list is too large, define a smoke-test subset and a batch plan.

Optional advanced challenge: Save the query list as a CSV with reason-for-inclusion and confidence columns.

2.5 Week 5: Research-Grade categorate() Enrichment

Week 5: Research-grade categorate enrichment

Research milestone: Students run small, validated enrichment before scaling.

Primary product: Validation outputs, source coverage report, saved enrichment object, query log, and Evidence Court notes.

Week 5 Operating Plan

Concepts: `categorate(detail = "research")`, caching, throttling, source coverage, validation-first interpretation, PubChem/KEGG/LOTUS/FEMA/FDA/SPL/MeSH output families, source diagnostics, and database etiquette.

uafR/R skills: Use `categorate()`, `validateCategorateResult()`, `chemicalDataDictionary()`, and saved RDS objects. Use `training/scripts/05_categorate_research.R`.

Required readings or references: `categorate()` documentation, table guide, installation appendix live-smoke guidance, and database etiquette note.

Required files/scripts/data: `training/scripts/05_categorate_research.R`, `library_data`, enrichment query list, query settings log, and evidence table template.

Weekly deliverables: Smoke-test result, validation summary, source coverage report, evidence audits, saved RDS enrichment object, and query log.

Mentor review checkpoint: Day 33 scaling plan and saved archive.

Common failure modes: Running too many compounds first, ignoring validation, overinterpreting empty tables, and treating source warnings as final conclusions.

Minimum viable path: Run or inspect a two-compound smoke test and validate outputs.

Advanced extension path: Batch a project list with caching and export selected tables.

Day 29: `categorate(detail = "research")` Smoke Test

Purpose: Start enrichment with a small reproducible test.

Learning objectives: Run a two-compound smoke test, use caching and throttling, save an RDS result, and avoid large first queries.

Prerequisites and materials: Installed `uafR`, `library_data`, `training/scripts/05_categorate_research.R`, and query settings log.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes enrichment overview, 120 minutes guided smoke test, 120 minutes output inspection, 60 minutes query log, 30 minutes review.

Morning concept block: Explain why live database services require small tests, caching, and respectful throttling.

Guided lab or reading discussion: Run `run_categorate_smoke()` or inspect a saved result if services are unavailable.

Independent project block: Create a project smoke-test list of two to five chemicals.

Evidence and writing block: Write query settings, date, package version, cache setting, and compound list in the query log.

Research log prompt: What was queried, what returned, and what should not be interpreted yet?

End-of-day deliverable: Smoke-test result and query settings log.

Quality gate: Students do not interpret traits until validation and source coverage are inspected.

Common problems and fixes: If the package says no library was detected, load `library_data` and pass `chemical_library = library_data`. If queries hang, reduce compound count and use cached results.

Optional advanced challenge: Compare `detail = "research"` to `detail = "standard"` for table count and purpose.

Day 30: Validation First

Purpose: Make validation the first analysis step after enrichment.

Learning objectives: Inspect `ValidationSummary`, `TableQuality`, `SourceCoverage`, `SourceDiagnostics`, and `DataDictionary`; decide which outputs are usable.

Prerequisites and materials: Smoke-test result, `validateCategorateResult()`, `chemicalDataDictionary()`, and source coverage worksheet.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes validation concept block, 120 minutes guided audit, 120 minutes source coverage report, 60 minutes interpretation limits, 30 minutes review.

Morning concept block: Explain row counts, missing required columns, duplicate analysis keys, source diagnostics, and source coverage. Note that saved `categorate()` results use `ValidationSummary`, while direct `validateCategorateResult()` audits use `Summary`.

Guided lab or reading discussion: Run `validateCategorateResult(result)` and inspect each validation table.

Independent project block: Build the first source coverage report for the smoke-test or project result.

Evidence and writing block: Write a short statement naming which tables are strong enough to use and which are weak or empty.

Research log prompt: Which table has the strongest coverage and which table should be ignored for now?

End-of-day deliverable: Validation summary and source coverage report.

Quality gate: Every planned interpretation is tied to a table that passed basic validation and coverage checks.

Common problems and fixes: If a table is empty, do not write that the chemical lacks the trait; write that the source returned no usable rows in this query.

Optional advanced challenge: Use `chemicalDataDictionary("ChemicalTraitReport")` to inspect expected columns.

Day 31: PubChem-Derived Enrichment Outputs

Purpose: Interpret PubChem-derived tables as source-backed evidence, not free text.

Learning objectives: Inspect identity, properties, hazards, classifications, literature, annotations, and measurement summaries conservatively.

Prerequisites and materials: Validated enrichment result, table guide, source coverage report, and evidence table template.

Estimated time budget: Full internship day: 30 minutes check-in, 90 minutes PubChem table families, 120 minutes guided table inspection, 90 minutes project audit, 60 minutes evidence writing, 30 minutes review.

Morning concept block: Explain PubChemIdentity, PubChemProperties, ChemicalHazards, ChemicalMeasurements, ChemicalMeasurementSummary, PubChemClassifications, and literature-related tables.

Guided lab or reading discussion: Trace one property and one hazard-related row to its source context.

Independent project block: Select three PubChem-derived outputs relevant to the project and document their evidence strength.

Evidence and writing block: Write safe templates for property, hazard, and literature claims.

Research log prompt: Which PubChem-derived value is analysis-ready, and which is narrative-only?

End-of-day deliverable: PubChem evidence audit.

Quality gate: Hazard annotations are described as screening/context evidence, not risk assessment.

Common problems and fixes: If a table has long descriptive text, look for discrete fields or route it to evidence notes rather than matrices.

Optional advanced challenge: Compare ChemicalMeasurements to ChemicalMeasurementSummary and decide which is better for figures.

Day 32: KEGG, LOTUS, FEMA, FDA/SPL, MeSH, and Related Outputs

Purpose: Inspect non-PubChem and source-specific outputs without overclaiming.

Learning objectives: Trace pathway, reaction, enzyme, module, taxonomy, occurrence, sensory, regulatory, and biomedical rows to source context.

Prerequisites and materials: Validated enrichment result, source coverage report, table guide, and evidence map.

Estimated time budget: Full internship day: 30 minutes check-in, 90 minutes output families, 120 minutes guided source tracing, 90 minutes project source audit, 60 minutes caveat writing, 30 minutes review.

Morning concept block: Explain KEGGPathways, KEGGReactions, KEGGEnzymes, KEGGModules, KEGGReactionParticipants, LOTUSProfile, FEMAProfile, FDA.SPL.Profile, MeSHProfile, ChemicalTaxonomy, and ChemicalOccurrences.

Guided lab or reading discussion: Trace one pathway or occurrence-like record and write what it can and cannot support.

Independent project block: Audit source-specific rows for project chemicals and mark them usable, context-only, or not relevant.

Evidence and writing block: Write three caveats: pathway context, organism occurrence, and regulatory or sensory context.

Research log prompt: Which non-PubChem source adds useful context, and what would be unsafe to claim from it?

End-of-day deliverable: KEGG/LOTUS/source-specific evidence audit.

Quality gate: No source-specific row is interpreted without naming source database and source field.

Common problems and fixes: If taxonomy fields are sparse or unexpected, inspect ChemicalTaxonomy and source rows before grouping.

Optional advanced challenge: Identify one KEGGReactionParticipants row and explain reactant/product role cautiously.

Day 33: Scaling, Batching, Caching, and Query Logs

Purpose: Scale enrichment responsibly after validation.

Learning objectives: Batch query lists, use cache settings, save RDS result objects, export selected tables, and maintain query logs.

Prerequisites and materials: Query list, smoke-test result, `training/scripts/05_categorate_research.R`, and query settings log.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes scaling strategy, 120 minutes guided batch plan, 120 minutes project run or saved-result archive, 60 minutes query log, 30 minutes review.

Morning concept block: Discuss database etiquette, service instability, and why cached RDS files are research artifacts.

Guided lab or reading discussion: Design batch sizes and naming conventions for saved results.

Independent project block: Run a defensible batch or archive saved enrichment outputs if live services are not used.

Evidence and writing block: Record all settings: compound list, date, detail level, cache, throttle, assay detail limit, and output file.

Research log prompt: What exact query produced the saved result object?

End-of-day deliverable: Saved enrichment object and query log.

Quality gate: The result object can be reloaded without repeating live queries.

Common problems and fixes: If a batch fails, reduce size and keep the failed query log rather than deleting it.

Optional advanced challenge: Use `exportCategorateWorkbook()` if optional spreadsheet dependencies are available, or export a CSV bundle of selected tables.

Day 34: Weekend Evidence Court

Purpose: Make students prove that important traits trace back to evidence.

Learning objectives: Trace five matrix or report signals to source evidence, rewrite unsupported claims, and document rejected claims.

Prerequisites and materials: Saved enrichment result, `ChemicalTraitMatrix`, `ChemicalTraitEvidence`, `ChemicalTraitReport`, and Evidence Court worksheet.

Estimated time budget: Weekend asynchronous work, 2 to 3 hours.

Morning concept block: Not scheduled. Begin with evidence tracing.

Guided lab or reading discussion: Use `chemicalTraitEvidence(result)` to inspect evidence for selected keys when available.

Independent project block: Trace five traits or report fields to source evidence and confidence metadata.

Evidence and writing block: Rewrite one weak claim into safer wording.

Research log prompt: Which trait was strongest, which was weakest, and why?

End-of-day deliverable: Evidence Court worksheet for five traits.

Quality gate: Each trait has source table, source field, evidence row or key, and caveat.

Common problems and fixes: If a trait cannot be traced, remove it from result claims and document why.

Optional advanced challenge: Compare evidence strength under medium versus high confidence thresholds.

Day 35: Weekend Source Coverage Report and Archive

Purpose: Close enrichment with a reproducible archive before matrix design begins.

Learning objectives: Save validation outputs, source coverage report, query log, and result object in predictable locations.

Prerequisites and materials: Saved enrichment result, validation output, source coverage report, query settings log, and project folder.

Estimated time budget: Weekend asynchronous work, 1 to 2 hours.

Morning concept block: Not scheduled. Focus on archiving.

Guided lab or reading discussion: Inspect the results folder and confirm file names match README.

Independent project block: Save the enrichment archive and update README script order.

Evidence and writing block: Write a Week 6 readiness note naming which tables can feed trait matrices.

Research log prompt: What saved object is the source of truth for Week 6?

End-of-day deliverable: Enrichment result archive and Week 6 readiness checklist.

Quality gate: A student can reload the saved RDS object and inspect validation without web queries.

Common problems and fixes: If the only output is printed in the Console, rerun and save RDS/CSV outputs.

Optional advanced challenge: Export selected validation tables to CSV for the final package.

2.6 Week 6: Research Variables and Trait Matrices

Week 6: Research variables and trait matrices

Research milestone: Students convert evidence into defensible analysis variables.

Primary product: Candidate variable list, trait matrices, grouping dictionary, sensitivity note, and variable gate decision.

Week 6 Operating Plan

Concepts: Discrete variables versus narrative text, matrix profiles, binary/count/confidence modes, confidence thresholds, grouping dictionaries, missingness, sparsity, and sensitivity.

uafR/R skills: Use `chemicalTraitMatrix()`, `chemicalTraitOntologyMatrix()`, `chemicalTraitEvidence()`, and `chemicalTraitReport()`. Use `training/scripts/06_trait_matrix_analysis.R`.

Required readings or references: Trait matrix chapter, table guide, evidence court notes, and matrix sparsity exercise.

Required files/scripts/data: Saved enrichment RDS, `training/scripts/06_trait_matrix_analysis.R`, grouping dictionary template, and trait evidence audit worksheet.

Weekly deliverables: Candidate variable list, core matrix, confidence matrix, ontology matrix, grouping dictionary, comparison exercise, and variable freeze note.

Mentor review checkpoint: Day 40 variable gate.

Common failure modes: Keeping too many sparse traits, using narrative text as a variable, ignoring confidence, and clustering uninterpretable matrices.

Minimum viable path: Use a core binary matrix with medium confidence or higher.
Advanced extension path: Compare matrix profiles and document sensitivity.

Day 36: Discrete Variables Versus Narrative Text

Purpose: Decide what is analysis-ready and what remains evidence context.

Learning objectives: Inspect `ChemicalTraits`, separate discrete values from descriptive text, and create a candidate variable list.

Prerequisites and materials: Saved enrichment result, `ChemicalTraits`, `ChemicalTraitReport`, and variable worksheet.

Estimated time budget: Full internship day: 30 minutes check-in, 90 minutes variable concept block, 120 minutes guided table inspection, 90 minutes project variable list, 60 minutes caveat writing, 30 minutes review.

Morning concept block: Explain that a useful variable has defined allowed values, source, confidence rule, and interpretation boundary.

Guided lab or reading discussion: Inspect `ChemicalTraits` and mark rows as matrix-ready, evidence-only, or not relevant.

Independent project block: Build a candidate variable list for the project question.

Evidence and writing block: Write a note explaining why some descriptive fields were excluded.

Research log prompt: Which attractive field is not analysis-ready, and why?

End-of-day deliverable: Candidate variable list.

Quality gate: Each candidate variable has source table, source field, allowed values, and confidence rule.

Common problems and fixes: If a variable is a sentence, either extract a discrete value with a documented rule or keep it as narrative evidence.

Optional advanced challenge: Add a column for expected figure use.

Day 37: Binary, Count, and Confidence-Weighted Matrices

Purpose: Build multiple matrix views and understand what each means.

Learning objectives: Use `chemicalTraitMatrix()` with binary, count, and confidence modes; compare profiles and thresholds.

Prerequisites and materials: Saved enrichment result, `training/scripts/06_trait_matrix_analysis.R`, and matrix sensitivity log.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes matrix concept block, 120 minutes guided matrix build, 120 minutes matrix comparison, 60 minutes methods note, 30 minutes review.

Morning concept block: Explain that binary matrices show presence, count matrices show row counts, and confidence matrices show evidence strength.

Guided lab or reading discussion: Run `build_trait_analysis()` or equivalent code to build core, confidence, and ontology matrices.

Independent project block: Compare matrix dimensions, sparsity, and top variables.

Evidence and writing block: Write a matrix settings note with profile, mode, confidence threshold, and max traits.

Research log prompt: Which matrix is most interpretable for the project question?

End-of-day deliverable: Trait matrix v1 and matrix settings note.

Quality gate: Matrix settings are documented and the matrix is not interpreted without coverage/sparsity review.

Common problems and fixes: If the matrix has too many columns, cap `max_traits` or use the `core` profile. If it is mostly zeros, revise the variable set.

Optional advanced challenge: Rebuild with high confidence only and compare which findings survive.

Day 38: Project-Specific Grouping Dictionary

Purpose: Freeze how variables will be named and interpreted.

Learning objectives: Define source table, source field, extraction rule, allowed values, confidence rule, caveat, and planned figure for each variable.

Prerequisites and materials: Candidate variable list, matrices, `ChemicalTraitEvidence`, grouping dictionary template.

Estimated time budget: Full internship day: 30 minutes check-in, 60 minutes dictionary concept block, 120 minutes guided examples, 150 minutes project dictionary work, 45 minutes review, 30 minutes revision.

Morning concept block: Explain that a grouping dictionary is the bridge between source evidence and analysis.

Guided lab or reading discussion: Review a strong dictionary row and a weak row.

Independent project block: Complete dictionary rows for all candidate variables.

Evidence and writing block: Add one caveat per variable or explain why the variable is robust.

Research log prompt: Which variable is most vulnerable to misinterpretation?

End-of-day deliverable: Grouping dictionary v1.

Quality gate: No variable is used in a figure or table unless it appears in the grouping dictionary.

Common problems and fixes: If the source field cannot be named, return to evidence tables before keeping the variable.

Optional advanced challenge: Add a reviewer challenge question for each variable.

Day 39: Shared Versus Distinct Trait Comparison

Purpose: Practice using matrices for a real comparison without overinterpreting.

Learning objectives: Compare two chemical sets or sample groups, identify shared/distinct traits, and write cautious interpretation.

Prerequisites and materials: Trait matrix v1, grouping dictionary, project comparison groups, and evidence map.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes comparison concept block, 120 minutes guided comparison, 120 minutes project analysis, 60 minutes interpretation note, 30 minutes review.

Morning concept block: Explain that matrix differences are patterns in source-backed metadata, not direct measurements of biological behavior.

Guided lab or reading discussion: Compare two example groups and identify traits that are shared, distinct, or missing.

Independent project block: Build a comparison table for the project.

Evidence and writing block: Write one cautious interpretation paragraph and one rejected overclaim.

Research log prompt: Which difference is interpretable, and which difference may reflect source coverage?

End-of-day deliverable: Comparison table and interpretation note.

Quality gate: Each interpreted difference has evidence and missingness context.

Common problems and fixes: If a group has fewer source records, do not treat absence of traits as biological absence.

Optional advanced challenge: Repeat the comparison under a stricter confidence threshold.

Day 40: Variable Gate

Purpose: Remove weak variables before visualization and report writing.

Learning objectives: Evaluate variables by relevance, coverage, confidence, interpretability, and caveat burden; freeze analysis inputs.

Prerequisites and materials: Candidate variable list, matrix v1, grouping dictionary, comparison table, and mentor review rubric.

Estimated time budget: Full internship day: 30 minutes check-in, 60 minutes gate criteria, 120 minutes variable review, 120 minutes revisions, 60 minutes methods note, 30 minutes sign-off.

Morning concept block: Discuss why removing weak variables improves the project.

Guided lab or reading discussion: Score variables as keep, revise, evidence-only, or remove.

Independent project block: Freeze final matrix and dictionary for Week 7.

Evidence and writing block: Write exclusion rules and missing-data note.

Research log prompt: Which variable was removed and what made that decision defensible?

End-of-day deliverable: Reviewed trait matrix and grouping dictionary.

Quality gate: Frozen inputs are saved, named, and documented.

Common problems and fixes: If students want to keep a weak variable because it is interesting, move it to Discussion or Future Work.

Optional advanced challenge: Add a matrix sensitivity summary comparing versions before and after exclusions.

Day 41: Weekend Matrix Sparsity and Sensitivity Challenge

Purpose: Test whether matrix conclusions survive reasonable filtering choices.

Learning objectives: Calculate or inspect sparsity, compare thresholds, and document sensitivity.

Prerequisites and materials: Frozen matrix, previous matrix versions, grouping dictionary, and sensitivity log.

Estimated time budget: Weekend asynchronous work, 2 to 3 hours.

Morning concept block: Not scheduled. Begin with matrix comparison.

Guided lab or reading discussion: Compare core versus full or medium versus high confidence matrices.

Independent project block: Record which findings remain stable and which disappear.

Evidence and writing block: Write a missing-data and sensitivity note.

Research log prompt: Which conclusion is robust to filtering, and which is not?

End-of-day deliverable: Matrix sparsity and sensitivity log.

Quality gate: At least one matrix limitation is explicitly documented.

Common problems and fixes: If sensitivity is ignored, add a paragraph explaining why the chosen matrix is acceptable.

Optional advanced challenge: Create a small table comparing matrix dimensions across settings.

Day 42: Weekend Variable Methods Paragraph

Purpose: Convert matrix design into methods text.

Learning objectives: Write the variable-selection method, confidence threshold, missing-data handling, and exclusion note.

Prerequisites and materials: Matrix settings note, grouping dictionary, sensitivity log, and methods draft.

Estimated time budget: Weekend asynchronous work, 1 to 2 hours.

Morning concept block: Not scheduled. Focus on writing.

Guided lab or reading discussion: Reread matrix settings and verify every setting appears in the methods paragraph.

Independent project block: Draft the variable methods paragraph.

Evidence and writing block: Add missing-data language and excluded-variable examples.

Research log prompt: What variable decision must future readers understand?

End-of-day deliverable: Variable-methods paragraph and missing-data note.

Quality gate: The paragraph names functions, profiles, modes, thresholds, and caveats.

Common problems and fixes: If the paragraph says “traits were analyzed” without settings, add exact function calls and matrix profile.

Optional advanced challenge: Add a reproducibility footnote with saved file names.

2.7 Week 7: Visualization and Exploratory Analysis

Week 7: Visualization and exploratory analysis

Research milestone: Students turn validated outputs into honest figures and captions.

Primary product: Figure plan, coverage heatmap, trait heatmap, summary figures/tables, captions, and visual narrative outline.

Week 7 Operating Plan

Concepts: Figure purpose, audience, missingness, visual honesty, caption evidence, exploratory analysis, and claim strength.

uafR/R skills: Use `training/scripts/07_visualization.R`, saved matrices, source coverage, and base R export settings.

Required readings or references: Figure planning worksheet, figure caption checklist, visualization script, and results rubric.

Required files/scripts/data: Saved enrichment result, frozen matrix, grouping dictionary, `training/scripts/07_visualization.R`, and figures folder.

Weekly deliverables: Figure plan, source coverage heatmap, trait matrix heatmap, at least two additional figures/tables, captions, and results paragraphs.

Mentor review checkpoint: Day 46 figure clinic.

Common failure modes: Figures display too many variables, captions overclaim, missingness is hidden, and axes are unreadable.

Minimum viable path: One source coverage figure, one trait matrix figure, and one focused summary table.

Advanced extension path: Add sensitivity figure or comparison figure with alternative matrix settings.

Day 43: Figure Purpose and Planning

Purpose: Plan figures before plotting so visuals answer a research question.

Learning objectives: Define figure question, audience, data source, rows, columns, grouping variable, expected pattern, and limitation.

Prerequisites and materials: Figure planning worksheet, frozen matrix, source coverage report, grouping dictionary.

Estimated time budget: Full internship day: 30 minutes check-in, 90 minutes figure concept block, 90 minutes guided critique, 120 minutes project planning, 60 minutes caption sketch, 30 minutes review.

Morning concept block: Explain why figures should not be decorative. Each figure should answer one question and expose uncertainty where necessary.

Guided lab or reading discussion: Critique example figures for axis labels, color choices, missingness, and claim strength.

Independent project block: Draft three figure plans.

Evidence and writing block: Write preliminary captions naming source table and limitation.

Research log prompt: Which figure is most central to the final argument?

End-of-day deliverable: Figure planning worksheet.

Quality gate: Each planned figure has a data source, question, and limitation.

Common problems and fixes: If a figure is only “interesting,” rewrite the question it answers or remove it.

Optional advanced challenge: Add a figure that shows missingness rather than hiding it.

Day 44: Source Coverage and Trait Matrix Heatmaps

Purpose: Visualize coverage and traits while keeping missingness visible.

Learning objectives: Generate heatmaps, inspect row/column labels, and explain what heatmap cells represent.

Prerequisites and materials: Saved enrichment result, `training/scripts/07_visualization.R`, figures folder, and caption checklist.

Estimated time budget: Full internship day: 30 minutes check-in, 60 minutes heatmap concepts, 120 minutes guided plotting, 120 minutes project figure revision, 60 minutes captions, 30 minutes review.

Morning concept block: Explain source coverage heatmaps versus trait matrix heatmaps.

Guided lab or reading discussion: Run `build_training_figures(result)` on a saved result.

Independent project block: Generate project heatmaps and revise labels for readability.

Evidence and writing block: Write captions that name source tables and missingness limitations.

Research log prompt: What does a blank or zero cell mean in this figure?

End-of-day deliverable: Coverage and trait heatmaps v1.

Quality gate: Labels are readable and captions do not imply unmeasured absence.

Common problems and fixes: If labels overlap, reduce variables, shorten labels, or create a focused subset.

Optional advanced challenge: Create a second heatmap using a stricter confidence threshold.

Day 45: Property Summaries, Bar Charts, Scatterplots, and Tables

Purpose: Build focused summaries that complement heatmaps.

Learning objectives: Choose appropriate summary figures or tables, connect them to the project question, and avoid overloading visuals.

Prerequisites and materials: Frozen matrix, `ChemicalMeasurementSummary`, `ChemicalTraitReport`, grouping dictionary, and figure plan.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes summary figure concepts, 120 minutes guided examples, 120 minutes project plotting/table building, 60 minutes captioning, 30 minutes review.

Morning concept block: Discuss when to use bar charts, scatterplots, and tables instead of heatmaps.

Guided lab or reading discussion: Build or inspect a trait-count bar chart and a focused summary table.

Independent project block: Create two additional figures or tables for the project.

Evidence and writing block: Draft captions with evidence source and limitation.

Research log prompt: Which figure or table best answers the research question?

End-of-day deliverable: Two additional figures or tables.

Quality gate: Every plotted value traces to a saved table or matrix.

Common problems and fixes: If a plot mixes incompatible evidence types, split it or turn it into a table.

Optional advanced challenge: Add a figure that compares results before and after a filtering choice.

Day 46: Figure Clinic

Purpose: Improve figures through structured critique before results writing.

Learning objectives: Review axes, legends, labels, captions, color choices, missingness, and claim strength.

Prerequisites and materials: Figure drafts, caption checklist, peer review form, and results rubric.

Estimated time budget: Full internship day: 30 minutes check-in, 60 minutes review criteria, 120 minutes peer critique, 120 minutes figure revision, 60 minutes caption revision, 30 minutes mentor checkpoint.

Morning concept block: Explain the difference between a figure that is visually polished and one that is scientifically honest.

Guided lab or reading discussion: Review one example as a group and rewrite its caption.

Independent project block: Revise all current figures based on peer and mentor feedback.

Evidence and writing block: Update captions and add figure source notes to the research log.

Research log prompt: What changed after critique, and how did that improve evidence discipline?

End-of-day deliverable: Revised figure set.

Quality gate: A reviewer can understand each figure without needing verbal explanation.

Common problems and fixes: If the caption makes a stronger claim than the data support, rewrite it using evidence-limited language.

Optional advanced challenge: Prepare one figure for poster-scale display and one for report-scale display.

Day 47: Results Paragraphs and Defensible Captions

Purpose: Connect figures to written results without overclaiming.

Learning objectives: Write results paragraphs that describe patterns, evidence source, and uncertainty; separate results from discussion.

Prerequisites and materials: Revised figures, captions, source coverage report, grouping dictionary, and results rubric.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes results-writing concept block, 90 minutes guided examples, 150 minutes project writing, 45 minutes peer review, 30 minutes revision.

Morning concept block: Explain that Results describes what the analysis shows; Discussion explains why it may matter.

Guided lab or reading discussion: Compare overclaimed and cautious figure captions.

Independent project block: Write one results paragraph per major figure or table.

Evidence and writing block: Add source table references and limitations to captions.

Research log prompt: Which sentence is most likely to be challenged by a reviewer?

End-of-day deliverable: Figure captions and results draft.

Quality gate: Every results paragraph points to a figure/table and avoids mechanism claims not supported by evidence.

Common problems and fixes: If the paragraph says “caused,” “proved,” or “confirmed,” check whether the evidence really supports that verb.

Optional advanced challenge: Write a one-sentence key result for a non-specialist audience and a technical audience.

Day 48: Weekend Figure Revision and Reproducibility Check

Purpose: Confirm that figures regenerate from scripts.

Learning objectives: Rerun figure scripts, verify output files, and update export settings.

Prerequisites and materials: Figure scripts, saved result object, figures folder, and research log.

Estimated time budget: Weekend asynchronous work, 1.5 to 3 hours.

Morning concept block: Not scheduled. Begin with rerun.

Guided lab or reading discussion: Restart R and rerun the figure workflow from the project root.

Independent project block: Fix missing inputs or output paths and save final-format images.

Evidence and writing block: Record figure file names and script names in the log.

Research log prompt: Which figures regenerate cleanly, and which need repair?

End-of-day deliverable: Figure reproducibility check.

Quality gate: Figures can be regenerated from scripts without manual editing.

Common problems and fixes: If a figure was manually edited, either script the edit or document it as presentation-only.

Optional advanced challenge: Add a small figure manifest with dimensions and file names.

Day 49: Weekend Visual Narrative Outline

Purpose: Organize figures into a report story.

Learning objectives: Choose figure order, identify central result, and connect each visual to a claim boundary.

Prerequisites and materials: Revised figures, results paragraphs, project question, and report outline template.

Estimated time budget: Weekend asynchronous work, 1 to 2 hours.

Morning concept block: Not scheduled. Focus on narrative.

Guided lab or reading discussion: Arrange figures in the order a reader should understand them.

Independent project block: Draft a visual narrative outline.

Evidence and writing block: Add one limitation under each figure.

Research log prompt: What is the central result and what evidence supports it?

End-of-day deliverable: Visual narrative outline for the final report.

Quality gate: The outline connects each figure to the research question and a limitation.

Common problems and fixes: If a figure does not fit the story, move it to supplementary material or remove it.

Optional advanced challenge: Draft the first version of a graphical abstract or poster flow.

2.8 Week 8: Literature Integration and Evidence-Based Interpretation

Week 8: Literature integration and evidence-based interpretation

Research milestone: Students connect uafR outputs to literature without inflating claims.

Primary product: Evidence ladder table, search log, annotated literature table, interpretation memo, and final report outline.

Week 8 Operating Plan

Concepts: Evidence ladder, literature search strings, reading papers efficiently, claim-evidence-limitation paragraphs, literature conflicts, and follow-up tests.

uafR/R skills: Link figures and matrix outputs back to `ChemicalTraitEvidence`, `ChemicalTraitReport`, validation outputs, and source coverage.

Required readings or references: At least three field-specific papers, reporting standards, and evidence ladder guidance.

Required files/scripts/data: Figures, evidence rows, literature table template, and claim-evidence-limitation memo.

Weekly deliverables: Evidence ladder table, search log, annotated literature table, interpretation memo, unsupported-claim removal note, and report outline.

Mentor review checkpoint: Day 54 interpretation gate.

Common failure modes: Citing papers that use different systems as direct proof, treating bioactivity annotations as mechanism, and ignoring conflicting literature.

Minimum viable path: Three strong sources and one cautious interpretation memo.

Advanced extension path: Add a literature conflict table and define follow-up tests.

Day 50: Evidence Ladder

Purpose: Rank project findings by strength of evidence.

Learning objectives: Place findings on a ladder from tentative match to confirmed standard and use ladder position to control claim strength.

Prerequisites and materials: Figures, source evidence, validation outputs, literature synthesis worksheet.

Estimated time budget: Full internship day: 30 minutes check-in, 90 minutes evidence ladder concept block, 90 minutes guided ranking, 120 minutes project ranking, 60 minutes writing, 30 minutes review.

Morning concept block: Compare tentative match, database descriptor, source annotation, occurrence/pathway context, peer-reviewed measurement, and confirmed standard.

Guided lab or reading discussion: Rank example claims and rewrite their strength.

Independent project block: Rank each major project finding.

Evidence and writing block: Write safe wording for the strongest and weakest findings.

Research log prompt: Which finding is most defensible and which needs follow-up confirmation?

End-of-day deliverable: Evidence ladder table.

Quality gate: Claim verbs match evidence strength.

Common problems and fixes: If a database descriptor is treated like measured sample evidence, move it down the ladder.

Optional advanced challenge: Add a follow-up experiment that would move one finding up the evidence ladder.

Day 51: Literature Search Strategy

Purpose: Find literature that clarifies, challenges, or contextualizes project results.

Learning objectives: Build search strings from chemical names, CIDs, traits, organisms, systems, methods, and database fields.

Prerequisites and materials: Evidence ladder table, chemical intake table, figures, and search log.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes search strategy, 120 minutes guided search, 120 minutes independent literature search, 60 minutes search log writing, 30 minutes review.

Morning concept block: Explain how search terms change when a project needs background, method support, interpretation, or follow-up context.

Guided lab or reading discussion: Build search strings for one compound and one trait.

Independent project block: Search for at least six candidate papers and select three to annotate.

Evidence and writing block: Record search terms, database or search engine, date, and reason for inclusion/exclusion.

Research log prompt: Which search string found the most relevant evidence?

End-of-day deliverable: Literature search log.

Quality gate: The search log includes failed searches, not only successful ones.

Common problems and fixes: If searches return too much, add system and method terms. If they return too little, use synonyms or broader trait terms.

Optional advanced challenge: Add one paper that complicates the expected interpretation.

Day 52: Reading Papers Efficiently

Purpose: Extract useful evidence without drowning in full-paper details.

Learning objectives: Identify system, method, chemical evidence, main claim, and limitation in each paper.

Prerequisites and materials: Three selected papers, literature table template, figures, and evidence ladder table.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes reading strategy, 120 minutes guided paper annotation, 120 minutes independent reading, 60 minutes synthesis, 30 minutes review.

Morning concept block: Discuss which paper details matter for chemical interpretation and which do not.

Guided lab or reading discussion: Annotate one paper as a group.

Independent project block: Annotate three papers in the literature table.

Evidence and writing block: Mark whether each paper supports background, method, interpretation, mechanism hypothesis, or follow-up only.

Research log prompt: Which paper is most directly relevant and why?

End-of-day deliverable: Annotated literature table.

Quality gate: Each paper entry includes a limitation and does not claim more than the paper measured.

Common problems and fixes: If a paper is about a different organism or method, label it as context unless it directly supports the method or interpretation.

Optional advanced challenge: Add one sentence comparing the project evidence to each paper's evidence level.

Day 53: Claim-Evidence-Limitation Paragraphs

Purpose: Build interpretation paragraphs that are both useful and cautious.

Learning objectives: Link uafR outputs, figures, source evidence, and literature into claim-evidence-limitation paragraphs.

Prerequisites and materials: Figures, evidence ladder, literature table, evidence rows, and memo template.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes paragraph structure, 90 minutes guided example revision, 150 minutes project writing, 45 minutes peer review, 30 minutes revision.

Morning concept block: Explain the paragraph pattern: claim, project evidence, source or literature support, limitation, and next-step implication.

Guided lab or reading discussion: Rewrite an overclaimed paragraph using evidence-limited language.

Independent project block: Draft two to four project interpretation paragraphs.

Evidence and writing block: Add citations or source table references to every paragraph.

Research log prompt: Which interpretation paragraph is most vulnerable to overclaiming?

End-of-day deliverable: Interpretation memo v1.

Quality gate: Each paragraph includes a limitation.

Common problems and fixes: If literature is used as proof of the student's sample, revise it as context or hypothesis support.

Optional advanced challenge: Add a reviewer rebuttal note for the strongest paragraph.

Day 54: Interpretation Gate

Purpose: Remove unsupported claims before report drafting.

Learning objectives: Challenge each claim, remove or weaken unsupported statements, and define follow-up tests.

Prerequisites and materials: Interpretation memo, figures, source evidence, literature table, and evidence ladder.

Estimated time budget: Full internship day: 30 minutes check-in, 60 minutes gate criteria, 120 minutes claim review, 120 minutes revisions, 60 minutes follow-up plan, 30 minutes mentor checkpoint.

Morning concept block: Discuss what not to claim: confirmed identity, biological mechanism, pathway activity, risk, organism occurrence, or clinical relevance unless evidence specifically supports it.

Guided lab or reading discussion: Review examples of statements that must be removed, weakened, or moved to Future Work.

Independent project block: Mark every interpretation sentence as keep, revise, remove, or future work.

Evidence and writing block: Write follow-up tests for claims that cannot be made now.

Research log prompt: What claim was removed today and why?

End-of-day deliverable: Reviewed interpretation memo and follow-up test list.

Quality gate: No major claim remains without figure/table evidence and source or literature support.

Common problems and fixes: If students resist removing an interesting claim, move it to a clearly labeled follow-up hypothesis.

Optional advanced challenge: Write a reviewer-style critique of the final remaining claims.

Day 55: Weekend Literature Conflict Exercise

Purpose: Practice handling conflicting or imperfect literature.

Learning objectives: Identify conflicts, explain differences in system or method, and revise interpretation.

Prerequisites and materials: Literature table, one conflicting or less-direct source, and interpretation memo.

Estimated time budget: Weekend asynchronous work, 1.5 to 3 hours.

Morning concept block: Not scheduled. Begin with literature comparison.

Guided lab or reading discussion: Compare two sources that differ in system, method, or conclusion.

Independent project block: Add conflict notes to the literature table.

Evidence and writing block: Revise one interpretation paragraph to acknowledge conflict.

Research log prompt: What conflict or limitation changed the interpretation?

End-of-day deliverable: Literature conflict note and annotated literature table update.

Quality gate: Conflict is not hidden; it is used to refine claim strength.

Common problems and fixes: If sources disagree, do not pick the convenient one without explaining method differences.

Optional advanced challenge: Add a decision tree for when literature is background, support, conflict, or not relevant.

Day 56: Weekend Interpretation Memo Revision

Purpose: Prepare the report outline with evidence-limited interpretation.

Learning objectives: Revise memo, define final report sections, and map each figure to a claim.

Prerequisites and materials: Reviewed interpretation memo, figures, report storyboard, and research log.

Estimated time budget: Weekend asynchronous work, 1 to 2 hours.

Morning concept block: Not scheduled. Focus on revision.

Guided lab or reading discussion: Reorder interpretation paragraphs to match the visual narrative.

Independent project block: Draft final report outline.

Evidence and writing block: Add citation or source-evidence notes for each outline subsection.

Research log prompt: What is the final report's central argument and its main limitation?

End-of-day deliverable: Revised interpretation memo and final report outline.

Quality gate: Every planned report claim has a source in the outline.

Common problems and fixes: If the outline follows chronology instead of argument, reorder around research question and figures.

Optional advanced challenge: Draft the abstract as a structured four-sentence summary.

2.9 Week 9: Manuscript-Style Research Product

Week 9: Manuscript-style research product

Research milestone: Students turn analysis into a complete scientific report.

Primary product: Report storyboard, methods, results, discussion, full draft, revision audit, and final-draft cleanup.

Week 9 Operating Plan

Concepts: Manuscript structure, reproducible methods, results tied to figures, limitations, references, and report revision.

uafR/R skills: Cite package version, functions, settings, saved result objects, validation outputs, and missing-data handling.

Required readings or references: Final report rubric, methods notes, figure captions, literature table, and evidence memo.

Required files/scripts/data: Report draft, figures, tables, saved RDS object, source coverage report, validation summaries, session information.

Weekly deliverables: Report outline, methods v1, results v1, discussion v1, complete draft, revision audit, and final-draft cleanup.

Mentor review checkpoint: Day 61 full draft review.

Common failure modes: Methods too vague, results overclaim, figures not cited, limitations minimized, and references inconsistent.

Minimum viable path: A complete report with cautious claims and reproducible methods.

Advanced extension path: Add supplement-style tables and a reviewer response memo.

Day 57: Report Storyboard

Purpose: Build the report argument before writing full prose.

Learning objectives: Define title, abstract structure, introduction logic, methods sections, results order, discussion themes, limitations, and reproducibility statement.

Prerequisites and materials: Report outline, figures, interpretation memo, final report rubric.

Estimated time budget: Full internship day: 30 minutes check-in, 90 minutes report structure, 90 minutes storyboard examples, 120 minutes project storyboard, 60 minutes abstract sketch, 30 minutes review.

Morning concept block: Explain how a manuscript guides readers from question to evidence to limitation.

Guided lab or reading discussion: Analyze a model report structure and identify where methods, results, and limitations appear.

Independent project block: Build a report storyboard with section-level claims and evidence.

Evidence and writing block: Draft a working title and abstract skeleton.

Research log prompt: What is the central result, and where will it appear?

End-of-day deliverable: Report storyboard.

Quality gate: The storyboard maps each section to specific figures, tables, scripts, or evidence rows.

Common problems and fixes: If the storyboard lists tasks rather than argument, rewrite it around research question and evidence.

Optional advanced challenge: Draft a graphical or poster version of the report flow.

Day 58: Methods Section From Scripts and Settings

Purpose: Make the report reproducible.

Learning objectives: Write methods from scripts, package versions, settings, filters, validation outputs, and missing-data handling.

Prerequisites and materials: Script folder, session information, query log, matrix settings note, validation outputs, and methods rubric.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes methods standards, 120 minutes guided methods extraction, 120 minutes project methods writing, 60 minutes peer review, 30 minutes revision.

Morning concept block: Explain why methods must name functions and settings, not just software.

Guided lab or reading discussion: Pull method details from scripts and output logs.

Independent project block: Write methods subsections for data source, GC-MS workflow, enrichment, validation, trait matrix, figures, and literature.

Evidence and writing block: Add package version and session information references.

Research log prompt: Which method detail was missing until today?

End-of-day deliverable: Methods section v1.

Quality gate: A reader can identify inputs, functions, settings, filters, and saved outputs.

Common problems and fixes: If methods say “default settings,” verify and name the defaults that matter.

Optional advanced challenge: Create a methods supplement table with script names and outputs.

Day 59: Results Writing Without Overclaiming

Purpose: Tie results to figures and tables while preserving uncertainty.

Learning objectives: Write results paragraphs, cite figure/table numbers, and avoid discussion or mechanism claims in Results.

Prerequisites and materials: Figures, captions, visual narrative outline, results draft, and evidence ladder.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes results standards, 90 minutes guided paragraph revision, 150 minutes project results writing, 45 minutes peer review, 30 minutes revision.

Morning concept block: Explain how results sentences should describe observed patterns and their source.

Guided lab or reading discussion: Revise example results that overstate source-backed metadata.

Independent project block: Write complete Results section v1.

Evidence and writing block: Check every paragraph against figure captions and evidence table.

Research log prompt: Which result is descriptive, and which interpretation belongs in Discussion instead?

End-of-day deliverable: Results section v1.

Quality gate: Every result refers to a figure or table and avoids unsupported verbs.

Common problems and fixes: If a result explains why a pattern occurs, move that sentence to Discussion unless directly supported.

Optional advanced challenge: Add a supplemental result that reports validation or source coverage.

Day 60: Discussion, Limitations, Next Steps, and What Not To Claim

Purpose: Interpret results with scientific restraint.

Learning objectives: Write discussion paragraphs, limitations, next steps, and explicit non-claims.

Prerequisites and materials: Results v1, interpretation memo, literature table, evidence ladder, and final completion standard.

Estimated time budget: Full internship day: 30 minutes check-in, 90 minutes discussion concept block, 90 minutes limitation examples, 120 minutes project discussion writing, 60 minutes next-step plan, 30 minutes review.

Morning concept block: Explain how limitations increase credibility when they are specific.

Guided lab or reading discussion: Rewrite an overclaimed discussion paragraph into cautious interpretation.

Independent project block: Draft Discussion, Limitations, and Next Steps.

Evidence and writing block: Add an explicit “what this project does not show” paragraph.

Research log prompt: What is the most important claim the project should not make?

End-of-day deliverable: Discussion section v1 and non-claim statement.

Quality gate: Discussion includes uncertainty, source limits, and realistic follow-up tests.

Common problems and fixes: If discussion claims causality, mechanism, risk, or clinical relevance, require evidence that specifically supports it or revise.

Optional advanced challenge: Write a reviewer response for a likely criticism.

Day 61: Full Draft Peer and Mentor Review

Purpose: Find problems while there is still time to fix them.

Learning objectives: Conduct structured review of logic, reproducibility, evidence traceability, figures, and claim strength.

Prerequisites and materials: Complete draft report, figures, methods, evidence table, peer review form, and rubrics.

Estimated time budget: Full internship day: 30 minutes check-in, 60 minutes review criteria, 150 minutes peer review, 120 minutes mentor review or revision planning, 60 minutes response memo, 30 minutes wrap-up.

Morning concept block: Explain that review is part of production, not a final punishment.

Guided lab or reading discussion: Review a sample paragraph and decide what feedback is actionable.

Independent project block: Exchange drafts and complete peer review.

Evidence and writing block: Write a revision plan that prioritizes reproducibility and evidence issues first.

Research log prompt: What feedback will most improve the final package?

End-of-day deliverable: Complete draft report and revision plan.

Quality gate: Review comments identify file, section, claim, figure, or evidence issue specifically.

Common problems and fixes: If feedback is vague, ask for evidence and a concrete revision target.

Optional advanced challenge: Draft a reviewer response table with issue, response, and file changed.

Day 62: Weekend Revision Sprint

Purpose: Fix the highest-impact report and reproducibility issues.

Learning objectives: Revise references, figure/table consistency, methods details, and reproducibility documentation.

Prerequisites and materials: Revision plan, draft report, project folder, references, and final package audit.

Estimated time budget: Weekend asynchronous work, 2 to 3 hours.

Morning concept block: Not scheduled. Begin with the highest-priority revision.

Guided lab or reading discussion: Sort revisions into must fix, should fix, and optional.

Independent project block: Complete must-fix revisions.

Evidence and writing block: Check every figure/table reference and methods setting.

Research log prompt: Which revision most improved reproducibility or evidence strength?

End-of-day deliverable: Revised draft and methods reproducibility audit.

Quality gate: Must-fix evidence and reproducibility issues are resolved or documented.

Common problems and fixes: If time is short, fix methods, figures, and unsupported claims before style.

Optional advanced challenge: Add a concise supplement with validation and source coverage outputs.

Day 63: Weekend Final-Draft Cleanup

Purpose: Prepare for presentation week with a coherent final draft.

Learning objectives: Clean formatting, references, captions, file names, and Week 10 readiness.

Prerequisites and materials: Revised draft, figures, references, package audit checklist, and research log.

Estimated time budget: Weekend asynchronous work, 1.5 to 2.5 hours.

Morning concept block: Not scheduled. Focus on cleanup.

Guided lab or reading discussion: Read the draft from title to limitations and mark unclear transitions.

Independent project block: Clean final draft and update package checklist.

Evidence and writing block: Write a Week 10 readiness note.

Research log prompt: What remains before presentation and final package submission?

End-of-day deliverable: Final-draft cleanup and Week 10 readiness checklist.

Quality gate: The report is complete enough for presentation planning.

Common problems and fixes: If figures and captions disagree, fix the caption or regenerate the figure before presentation work.

Optional advanced challenge: Draft slide titles from report section claims.

2.10 Week 10: Presentation, Peer Review, Final Package, and Handoff

Week 10: Presentation, peer review, final package, and handoff

Research milestone: Students communicate the project and leave a reusable research package.

Primary product: Final report, presentation/poster, reproducible folder, evidence trail, continuation note, and final checklist.

Week 10 Operating Plan

Concepts: Central result, audience, visual narrative, reproducibility audit, presentation practice, final package archive, continuation note, and future work.

uafR/R skills: Run `training/scripts/09_reproducibility_check.R`, reload saved result objects, verify figure regeneration, and save session information.

Required readings or references: Presentation rubric, final package rubric, handoff worksheet, final completion standard.

Required files/scripts/data: Final report, figures, saved RDS result, validation outputs, trait matrix, grouping dictionary, evidence trail, presentation, and README.

Weekly deliverables: Presentation outline, presentation draft, reproducibility audit, practice revision, final presentation, final report, project archive, continuation note, and exit reflection.

Mentor review checkpoint: Day 66 final reproducibility audit and Day 68 final submission.

Common failure modes: Slides too technical, final folder disorganized, figures not reproducible, missing continuation note, and unsupported claims in oral presentation.

Minimum viable path: A clear five-to-seven minute presentation and a complete auditable package.

Advanced extension path: Poster, short manuscript submission draft, or handoff project for the next intern.

Day 64: Presentation or Poster Outline

Purpose: Build a clear scientific story around one central result.

Learning objectives: Define audience, central claim, slide order, figure choices, and limitation language.

Prerequisites and materials: Final draft report, visual narrative outline, presentation rubric, figures.

Estimated time budget: Full internship day: 30 minutes check-in, 75 minutes presentation structure, 90 minutes guided outline review, 120 minutes project outline, 60 minutes talking points, 30 minutes review.

Morning concept block: Explain why a presentation cannot include every analysis detail.

Guided lab or reading discussion: Compare a report outline to a presentation outline and remove unnecessary detail.

Independent project block: Build a slide or poster outline around one central result.

Evidence and writing block: Draft talking points with source and limitation language.

Research log prompt: What is the one result the audience should remember?

End-of-day deliverable: Presentation or poster outline.

Quality gate: The outline includes question, method, central result, evidence, limitation, and next step.

Common problems and fixes: If there are too many results, choose the one most connected to the research question.

Optional advanced challenge: Create versions for scientific and general audiences.

Day 65: Visual Polishing and Story Flow

Purpose: Make presentation visuals readable and scientifically precise.

Learning objectives: Revise figure placement, captions, slide titles, story flow, and talking points.

Prerequisites and materials: Presentation outline, final figures, captions, presentation rubric.

Estimated time budget: Full internship day: 30 minutes check-in, 60 minutes visual standards, 120 minutes slide or poster build, 120 minutes peer critique, 60 minutes revision, 30 minutes review.

Morning concept block: Discuss readable labels, restrained color, slide titles as claims, and limitations in spoken explanations.

Guided lab or reading discussion: Review one slide for readability and claim strength.

Independent project block: Build presentation draft.

Evidence and writing block: Write speaker notes that state evidence source and limitation for each result.

Research log prompt: Which slide or poster panel needed the biggest change?

End-of-day deliverable: Presentation draft.

Quality gate: Every visual can be read at presentation size and every claim has evidence.
Common problems and fixes: If slides are text-heavy, convert details to speaker notes or backup slides.
Optional advanced challenge: Add one backup slide with validation or source coverage.

Day 66: Final Reproducibility and Handoff Audit

Purpose: Ensure the project can be reopened after the intern leaves.
Learning objectives: Run final project check, verify files, confirm saved outputs, and write package status.
Prerequisites and materials: Project folder, `training/scripts/09_reproducibility_check.R`, final package audit worksheet, and README.
Estimated time budget: Full internship day: 30 minutes check-in, 60 minutes audit criteria, 150 minutes clean-session rerun, 120 minutes package cleanup, 60 minutes handoff notes, 30 minutes checkpoint.
Morning concept block: Explain that final package quality is measured by whether another person can inspect or rerun the work.
Guided lab or reading discussion: Source the reproducibility check script and run available project checks.
Independent project block: Fix missing files, unclear names, broken script order, and incomplete README sections.
Evidence and writing block: Update final package audit and session information.
Research log prompt: What would block the next student from continuing this work?
End-of-day deliverable: Final handoff package v1 and audit results.
Quality gate: Saved result object, validation outputs, trait matrix, grouping dictionary, figures, report, and README are present and findable.
Common problems and fixes: If a figure cannot be regenerated, either fix the script or clearly label the figure as non-regenerable and explain why.
Optional advanced challenge: Have a peer run the handoff audit independently.

Day 67: Practice Presentation and Revision

Purpose: Improve oral explanation and prepare for questions.
Learning objectives: Deliver a practice presentation, answer evidence questions, revise weak slides, and prepare concise responses.
Prerequisites and materials: Presentation draft, final report, evidence trail, presentation rubric.
Estimated time budget: Full internship day: 30 minutes check-in, 60 minutes presentation expectations, 120 minutes practice talks, 120 minutes revision, 60 minutes question prep, 30 minutes review.
Morning concept block: Discuss how to answer questions without making unsupported claims.
Guided lab or reading discussion: Practice answering: What does this evidence support? What does it not support? What would confirm it?
Independent project block: Deliver practice talk and revise slides or poster.
Evidence and writing block: Write answers to five likely questions.
Research log prompt: Which question was hardest to answer, and what evidence is needed?
End-of-day deliverable: Revised final presentation and question-response notes.

Quality gate: Student can answer evidence, limitation, and reproducibility questions directly.
Common problems and fixes: If a spoken answer overclaims, script a safer response and practice it.

Optional advanced challenge: Prepare a one-minute version and a five-minute version of the talk.

Day 68: Final Presentation, Final Report, and Package Archive

Purpose: Submit and communicate the completed research package.

Learning objectives: Deliver final presentation, submit final report, archive project package, and record final status.

Prerequisites and materials: Final presentation, final report, final package folder, final package audit, and research log.

Estimated time budget: Full internship day: 30 minutes check-in, presentation block as scheduled, 90 minutes final fixes, 90 minutes archive preparation, 60 minutes final submission, 30 minutes closeout.

Morning concept block: Review final claim boundaries and the expectation that questions can be answered with evidence.

Guided lab or reading discussion: Final symposium or presentation session.

Independent project block: Apply last approved fixes to report and package.

Evidence and writing block: Submit final report and final package archive with final checklist.

Research log prompt: What was submitted, where is it stored, and what should not be claimed?

End-of-day deliverable: Final presentation, report, and package archive.

Quality gate: All required completion-standard items are present or explicitly marked not applicable with reason.

Common problems and fixes: If a required file is missing, submit a clear limitation note rather than hiding the gap.

Optional advanced challenge: Draft a short abstract for a campus research showcase.

Day 69: Weekend Continuation Note and Unresolved Issue Log

Purpose: Make future work realistic and transparent.

Learning objectives: Write continuation note, unresolved issue log, and next-student starting instructions.

Prerequisites and materials: Final package, research log, final report, and handoff worksheet.

Estimated time budget: Weekend asynchronous work, 1.5 to 2.5 hours.

Morning concept block: Not scheduled. Focus on handoff clarity.

Guided lab or reading discussion: Read the project as if starting it next semester.

Independent project block: Write continuation note with next steps, unresolved issues, file locations, and warnings.

Evidence and writing block: Add an explicit “do not claim” list.

Research log prompt: What should the next student do first?

End-of-day deliverable: Continuation note and unresolved issue log.

Quality gate: A future student can identify the next action and the biggest risk.

Common problems and fixes: If the continuation note is only aspirational, add exact files and specific next actions.

Optional advanced challenge: Write a short issue tracker with priority, owner, and evidence needed.

Day 70: Weekend Exit Reflection and Final Checklist

Purpose: Close the program with reflection, debrief prompts, and final accountability.

Learning objectives: Reflect on skill growth, name final project status, identify future work, and complete final checklist.

Prerequisites and materials: Final package, final checklist, continuation note, and research log.

Estimated time budget: Weekend asynchronous work, 1 to 2 hours.

Morning concept block: Not scheduled. Focus on reflection and final checklist.

Guided lab or reading discussion: Review the product ladder from Week 1 to Week 10 and mark evidence of growth.

Independent project block: Complete final package checklist one last time.

Evidence and writing block: Write exit reflection and mentor debrief prompts.

Research log prompt: What can this project support now, and what remains future work?

End-of-day deliverable: Exit reflection, future-work plan, mentor debrief prompts, and final checklist.

Quality gate: The final reflection names skills gained, evidence limits, and next steps without inflating the project.

Common problems and fixes: If reflection becomes generic, require references to exact outputs, scripts, figures, or decisions.

Optional advanced challenge: Write a recommendation letter-style summary of the student's technical contributions for program records.

3 Research Onboarding and Question Design

Week 1: Research Onboarding and Question Design

Research milestone: Turn a broad interest into a testable chemical research question.

Primary product: One-page project brief with question, system, candidate chemicals, expected evidence, and risks.

3.1 Why This Matters

Strong computational research starts before any code is run. Students need a clear scientific question, a defensible reason for studying a chemical set, and a plan for evaluating evidence. Without this foundation, database enrichment becomes a list of facts rather than a research workflow.

3.2 Concepts

- **Research system:** the organism, environment, treatment, material, food, exposure, sample type, or process being studied.
- **Chemical evidence:** tentative identifications, mass spectral matches, database annotations, literature support, functional groups, biological activities, hazard terms, occurrence records, pathways, reactions, and measured properties.
- **Analysis-ready variable:** a clean field that can be grouped, counted, filtered, modeled, or visualized.
- **Provenance:** the source and retrieval context for a piece of evidence.
- **Uncertainty:** the gap between a database hit and a confirmed scientific conclusion.

3.3 uafR and R Skills

Students do not need to run the full workflow in Week 1. They should learn what the tool can produce:

- Core GC-MS workflow: `spreadOut()`, `mzExacto()`, and `exactoThese()`.

- Chemical enrichment: `categorate(detail = "research")`.
- Research trait tables: `ChemicalTraits`, `ChemicalTraitOntology`, and `ChemicalTraitMatrix`.
- Hazard summary table: `ChemicalHazards`.
- Occurrence summary table: `ChemicalOccurrences`.
- Measurement summary table: `ChemicalMeasurementSummary`.
- KEGG-specific interpretation table: `KEGGReactionParticipants`.
- Quality checks: `validateCategorateResult()` and `TableQuality`.

3.4 Guided Lab

1. Choose one research domain: plant metabolites, environmental exposure, food chemistry, microbial products, pharmacology, materials, forensic chemistry, toxicology, or another reviewed topic.
2. Write three possible research questions. Each question must name the system, chemicals or chemical class, comparison, and expected evidence.
3. Identify a starting chemical list. This can come from GC-MS results, literature, standards, known metabolites, or a small exploratory list.
4. Mark each chemical as known, tentative, uncertain, or exploratory.
5. Write a risk note: what could make the analysis misleading?

Example Research Questions

- Do volatile compounds detected in two plant treatments differ in sensory trait profiles, natural product occurrence, or pathway roles?
- Which detected compounds have safety or exposure signals that require additional review before biological interpretation?
- Do compounds associated with a sample group cluster by functional group, pathway, or literature-backed biological activity?

3.5 Independent Extension

Students interview the program lead, the dsDNA Core mentor, or a project lead for 20 minutes and capture:

- The research context.
- Why the chemical list matters.
- What decisions the analysis should support.
- What evidence would be convincing.
- What evidence would be weak or unacceptable.

3.6 Deliverables

Deliverable	Evidence of quality	Submission format
Project brief	Includes question, system, comparison, candidate chemical list, and expected output.	PDF or Markdown
Chemical intake table	Has one row per chemical and columns for source, confidence, project relevance, and notes.	CSV
Risk note	Names at least three ways the analysis could be wrong or overinterpreted.	Research log

3.7 Quality Checklist

Before Week 2

- The question is narrow enough to analyze in 10 weeks.
- The chemical list has a clear source and inclusion rule.
- The student can explain why chemical traits, database evidence, or GC-MS patterns matter for the question.
- The project scope has passed dsDNA Core review.

4 R, RStudio, Git, and Reproducible Project Structure

Week 2: R, RStudio, Git, and Reproducible Project Structure

Research milestone: Build a project that another researcher can open, run, and audit.

Primary product: Working R project with folders, install check, research log, and first commit.

4.1 Why This Matters

Reproducibility is a research skill, not an afterthought. A student who can organize files, name outputs clearly, and rerun an analysis has already solved many of the problems that slow down publication.

4.2 Concepts

- **Project root:** the top-level folder that contains scripts, data, results, and documentation.
- **Relative path:** a file path that works from the project root on any machine.
- **Raw data:** input data that should not be manually edited.
- **Derived data:** tables created by scripts.
- **Version control:** a record of changes to files and analysis decisions.
- **Research log:** a dated record of decisions, errors, fixes, and interpretation changes.

4.3 Recommended Project Structure

```
student-project/  
  README.md  
  analysis/  
  data_raw/  
  data_processed/  
  figures/  
  results/
```

```
scripts/  
reports/  
logs/
```

4.4 Guided Lab

Run the install and environment check in `training/scripts/00_install_check.R`. Students should save the output in their research log.

```
source("training/scripts/00_install_check.R")  
check_training_install()
```

Then create a project folder and initialize the file structure:

```
source("training/scripts/01_project_setup.R")  
create_training_project("student-project")
```

4.5 Git Practice

Students should make small, readable commits. A good commit message says what changed and why it matters:

```
Add chemical intake table for plant volatile project  
Document first uafR install check  
Add Week 1 project brief
```

4.6 Independent Extension

Students customize their project README with:

- Project title.
- Research question.
- Data source.
- Main scripts.
- Contact and dsDNA Core support.
- Current status.

4.7 Deliverables

Deliverable	Evidence of quality	Submission format
Project folder	Uses the required folder structure and relative paths.	Local folder or repository
Install check	Shows R version, uafR version, key dependencies, and package data availability.	Research log
First commit	Commit includes Week 1 and Week 2 materials without private tokens or sensitive data.	Git history

4.8 Quality Checklist

Before Week 3

- No script uses an absolute path to another person's computer.
- No token, password, or private key appears in any file.
- The project can be opened from the project root and scripts can find their inputs.
- The student can explain the difference between raw and processed data.

5 Chemical Data Literacy and Public Database Foundations

Week 3: Chemical Data Literacy and Public Database Foundations

Research milestone: Learn what public chemical databases can and cannot prove.

Primary product: Database evidence map for the student's chemical list.

5.1 Why This Matters

Public databases are powerful, but they are not interchangeable. PubChem, KEGG, LOTUS, FEMA, FDA/SPL, MeSH, and literature records answer different questions. Students need to know which source supports which type of claim.

5.2 Database Roles

Source	Useful evidence	Important caution
PubChem	Identifiers, synonyms, properties, annotations, hazards, classifications, assays, literature links.	Records can aggregate broad text from many sources; verify source-specific context.
KEGG	Compound IDs, reactions, pathways, enzymes, modules, biochemical context.	Coverage is strongest for curated biochemical systems; absence is not evidence of no role.
LOTUS	Natural product occurrence and taxonomic context.	Occurrence records can be sparse or uneven across organisms.
FEMA	Flavor-use context and sensory relevance.	Flavor presence does not prove abundance or biological role in a sample.
FDA/SPL	Drug product, route, dosage form, and regulatory text.	Regulatory annotation may describe a formulation context rather than the chemical alone.
MeSH	Pharmacologic actions and biomedical terms.	Terms may reflect literature indexing rather than direct mechanism.

5.3 Concepts

- **Identifier ambiguity:** the same name can map to multiple structures or salts.
- **Synonym noise:** synonyms include trade names, old names, registry names, abbreviations, and mixture names.
- **Annotation source:** a database field should be interpreted through its source and section.
- **Ground-truth extraction:** clean values must come from traceable source fields, not guesses.
- **Negative space:** missing data can mean no record, no query match, no source coverage, or a real absence.

5.4 Guided Lab

Students inspect three chemicals from different contexts. For each chemical, they answer:

- What are the stable identifiers?
- What synonyms might confuse a search?
- What database evidence is relevant to the project?
- Which fields are clean enough for grouping?
- Which fields are only useful as narrative evidence?

```
library(uafR)
data("library_data", package = "uafR")

compounds <- c("aspirin", "caffeine", "methyl salicylate")

profile <- pubchemProfile(
  compounds = compounds,
  profile = "minimal",
  cache = TRUE,
  throttle = 0.1
)

profile$identity
profile$properties
head(profile$synonyms)
```

5.5 Independent Extension

Students create a source evidence map:

- Rows are chemicals.
- Columns are evidence types: identity, properties, hazard, occurrence, pathway, reaction, sensory, biomedical, literature, assay, and regulatory.
- Cells contain yes, no, uncertain, or not relevant.

5.6 Deliverables

Deliverable	Evidence of quality	Submission format
Evidence map	Shows source coverage by chemical and evidence type.	CSV or spreadsheet
Ambiguity note	Identifies naming or identifier problems for at least three chemicals.	Research log
Database interpretation paragraph	Explains which sources are most relevant to the project and why.	Report draft

5.7 Quality Checklist

Before Week 4

- The student can explain why a PubChem annotation, KEGG pathway, and LOTUS occurrence do not mean the same thing.
- The chemical intake table has stable names or identifiers whenever available.
- The project has a plan for handling unmatched or ambiguous chemicals.

6 uafR Core GC-MS Workflow

Week 4: uafR Core GC-MS Workflow

Research milestone: Convert GC-MS-style hit tables into a form that supports targeted extraction.

Primary product: Processed example workflow using `spreadOut()`, `mzExacto()`, and `exactoThese()`.

6.1 Why This Matters

Before enrichment, students need to understand the chemical identity table coming from a mass spectrometry workflow. Tentative compound names, retention time, base peak, match factor, file name, and peak area must be reshaped carefully so chemical evidence can be compared across samples.

6.2 Required Input Columns

The core workflow expects a GC-MS-style input table with these columns:

- `Component.RT`
- `Component.Area`
- `Base.Peak.MZ`
- `File.Name`
- `Compound.Name`
- `Match.Factor`

6.3 Guided Lab

Use the package example data first. Students should not begin with their own messy file until they can run and explain the example. The support script uses bundled exact-match output by default so the classroom workflow does not depend on live services.

```
library(uafR)
source("training/scripts/04_core_workflow.R")
```

```
core <- run_core_workflow(live_lookup = FALSE)

names(core$spread)
head(core$exact)
```

When services are available and live lookup behavior needs to be tested, run:

```
live_core <- run_core_workflow(live_lookup = TRUE)
head(live_core$exact)
```

Then subset an existing categorized object:

```
data("standard_categorized", package = "uafR")
subset_reactives <- exactoThese(
  standard_categorized,
  subsetBy = "Database",
  subsetArgs = "reactives"
)
subset_reactives
```

6.4 Interpretation Practice

Students should describe:

- Which compounds were retained and why.
- Whether match factors are high enough for the intended claim.
- Whether retention time or base peak patterns suggest duplicated or ambiguous hits.
- What should be considered tentative until confirmed by standards.

6.5 Independent Extension

Students adapt the workflow to their own data or a program-provided example. They must document every cleaning step and preserve the original input file unchanged.

6.6 Deliverables

Deliverable	Evidence of quality	Submission format
Core workflow script	Runs from project root and produces spread and exact outputs.	R script
Input column audit	Confirms required columns, missing values, duplicates, and unusual match factors.	CSV or log
Interpretation note	Explains what can and cannot be concluded from the processed hits.	Research log

6.7 Quality Checklist

Before Week 5

- The script preserves raw data and writes processed outputs separately.
- The student can explain why tentative library matches are not structural confirmation.
- Duplicate or suspicious hit keys are documented.
- The workflow uses clear object names and comments only where they clarify decisions.

7 Research-Grade categorate Enrichment

Week 5: Research-Grade categorate Enrichment

Research milestone: Use public database enrichment to convert chemical names into traceable research tables.

Primary product: Validated `categorate(detail = "research")` result and source coverage report.

7.1 Why This Matters

The upgraded `categorate()` workflow gives students much more than a simple database lookup. It returns source-specific tables, normalized chemical traits, ontology-style terms, measurement summaries, hazard fields, occurrence evidence, KEGG roles, quality diagnostics, and data dictionaries. This is where the chemical list becomes a research dataset.

7.2 Key Output Families

- **Identity and properties:** `PubChemIdentity`, `PubChemProperties`, `PubChemIdentifiers`.
- **Source profiles:** `SafetyProfile`, `FEMAPProfile`, `FDA_SPL_Profile`, `LOTUSProfile`, `MeSHProfile`, `LiteratureProfile`.
- **Discrete trait data:** `ChemicalTerms`, `ChemicalTraits`, `ChemicalTraitOntology`.
- **Analysis matrices:** `ChemicalTraitMatrix`, `ChemicalTraitOntologyMatrix`.
- **Research summaries:** `ChemicalTraitReport`, `ChemicalMeasurementSummary`, `SourceCoverage`, `DerivedGroups`.
- **Database quality:** `DataDictionary`, `TableQuality`, `SourceDiagnostics`, `ValidationSummary`.

7.3 Guided Lab

Run a small enrichment first. Use caching for routine work and set a gentle throttle when live services are queried.

```
library(uafR)
data("library_data", package = "uafR")

compounds <- c("aspirin", "caffeine", "methyl salicylate")

result <- categorate(
  compounds = compounds,
  chemical_library = library_data,
  input_format = "wide",
  detail = "research",
  cache = TRUE,
  throttle = 0.2,
  assay_detail_limit = 0,
  trait_matrix_profile = "core",
  trait_matrix_min_confidence = "medium",
  trait_matrix_max_traits = 100
)

validateCategorateResult(result)$Summary
result$SourceCoverage
result$ChemicalTraitReport
result$ChemicalTraitMatrix
```

7.4 Evidence Inspection

Students should not trust a matrix until they inspect the supporting evidence.

```
trait_key <- names(result$ChemicalTraitMatrix)[2]

evidence <- chemicalTraitEvidence(result)
evidence[evidence$MatrixKey == trait_key, ]
```

7.5 Independent Extension

Students run `categorate()` on their reviewed chemical list. If the list is long, they should begin with 5 to 10 representative chemicals, inspect output quality, then scale up.

7.6 Deliverables

Deliverable	Evidence of quality	Submission format
Enrichment object	Saved result contains validation tables, source coverage, and trait matrices.	RDS
Coverage report	Summarizes which sources returned useful data by chemical.	CSV or table
Evidence audit	Traces five matrix traits back to source evidence.	Research log

7.7 Quality Checklist

Before Week 6

- `ValidationSummary` has been inspected.
- Missing source data are documented rather than silently ignored.
- At least five high-value traits have been traced to evidence rows.
- The student can identify which output tables are raw-ish source evidence and which are normalized analysis tables.

8 Turning Chemical Metadata Into Research Variables

Week 6: Turning Chemical Metadata Into Research Variables

Research milestone: Convert enrichment outputs into clean variables for grouping, filtering, and comparison.

Primary product: Analysis-ready trait table and project-specific grouping dictionary.

8.1 Why This Matters

Full sentences are difficult to analyze. Researchers need discrete values: hazard codes, pathway IDs, organism names, property bins, pharmacologic actions, measurement units, reaction roles, and presence or absence of curated traits. Week 6 teaches students to build those variables without inventing unsupported labels.

8.2 Concepts

- **Trait type:** a broad evidence domain such as safety, sensory, pathway, taxonomy, bioactivity, or physicochemical behavior.
- **Trait value:** the specific clean value, such as a GHS code, KEGG pathway, organism, or functional term.
- **Matrix key:** a stable column name used in a wide analysis matrix.
- **Confidence:** a source-aware estimate of how reliable or direct an extracted trait is.
- **Project grouping dictionary:** the student's reviewed mapping from traits to analysis groups.

8.3 Guided Lab

Start with the normalized tables:

```
traits <- result$ChemicalTraits
ontology <- result$ChemicalTraitOntology
matrix <- result$ChemicalTraitMatrix
```

```
table(traits$TraitType, useNA = "ifany")
table(ontology$OntologyDomain, useNA = "ifany")

head(traits[order(traits$Query, traits$TraitType), ])
```

Build a focused matrix:

```
core_matrix <- chemicalTraitMatrix(
  result,
  profile = "core",
  mode = "binary",
  min_confidence = "medium",
  max_traits = 100
)

confidence_matrix <- chemicalTraitMatrix(
  result,
  profile = "full",
  mode = "confidence",
  min_confidence = "medium",
  max_traits = 200
)
```

8.4 Grouping Rules

Every grouping variable must pass four tests:

1. It is useful for the research question.
2. It is extracted from a traceable source field.
3. It has a reproducible rule.
4. It does not overstate what the database evidence proves.

Good Versus Weak Grouping Variables

Good: A binary variable marking chemicals with source-backed GHS hazard codes. The evidence row stores the code and source.

Weak: A manually assigned variable called “dangerous” based on reading broad safety sentences.

8.5 Independent Extension

Students design a project-specific grouping dictionary with 8 to 20 variables. Each row should include:

- Variable name.
- Source table.
- Source field.

- Extraction rule.
- Allowed values.
- Confidence rule.
- Reason the variable matters.

8.6 Deliverables

Deliverable	Evidence of quality	Submission format
Trait table	Cleaned long table with selected traits and confidence fields.	CSV
Trait matrix	Wide matrix ready for visualization or modeling.	CSV
Grouping dictionary	Documents how each project variable was derived.	CSV or spreadsheet

8.7 Quality Checklist

Before Week 7

- No grouping variable is based only on unsupported intuition.
- The student can trace each selected variable to a source table and evidence row.
- Matrix columns are named clearly enough to interpret later.
- Missing values are handled deliberately.

9 Visualization and Exploratory Analysis

Week 7: Visualization and Exploratory Analysis

Research milestone: Use figures to reveal patterns, gaps, and hypotheses without overclaiming.

Primary product: At least three publication-style exploratory figures with captions.

9.1 Why This Matters

Figures are where students often discover whether the research question is working. A good exploratory figure makes patterns visible, but also shows uncertainty, missing data, and source coverage. Week 7 trains students to make figures that can survive dsDNA Core review.

9.2 Figure Types

Figure	Use
Source coverage heatmap	Shows which databases contributed evidence for each chemical.
Trait matrix heatmap	Groups chemicals by shared discrete traits.
Property scatterplot	Compares measured or computed properties such as exact mass, molecular weight, XLogP, or TPSA when available.
Bar chart	Summarizes trait counts, hazard code counts, source hits, or pathway roles.
Network sketch	Shows chemical-trait or chemical-pathway relationships for a small set.

9.3 Guided Lab

Use the script in `training/scripts/07_visualization.R`. It creates example figures from a saved `categorate()` result.

```
source("training/scripts/07_visualization.R")

plots <- build_training_figures(result, output_dir = "figures")
```

```
names(plots)
```

9.4 Caption Standard

Each figure caption should include:

- What the figure shows.
- Which table or matrix was used.
- What counts as evidence.
- What limitation matters most.
- One interpretation that can be defended.

9.5 Independent Extension

Students produce a figure set that answers one of these:

- Which chemicals share the most traits?
- Which source contributed the most useful evidence?
- Which chemical classes or traits separate sample groups?
- Which compounds need additional confirmation before interpretation?
- Which chemical traits connect most directly to the project question?

9.6 Deliverables

Deliverable	Evidence of quality	Submission format
Figure set	At least three figures, each with a caption and source table.	PNG or PDF
Figure script	Recreates all figures from processed data.	R script
Exploratory interpretation	Names patterns, caveats, and next analysis decisions.	Report draft

9.7 Quality Checklist

Before Week 8

- Every figure can be regenerated from a script.
- Axes, labels, colors, and legends are readable.

- Captions do not claim more than the data support.
- Missing source coverage is visible or discussed.

10 Literature Integration and Evidence-Based Interpretation

Week 8: Literature Integration and Evidence-Based Interpretation

Research milestone: Connect database evidence to peer-reviewed context.

Primary product: Annotated evidence table and interpretation memo.

10.1 Why This Matters

Database enrichment helps students find leads, but publication-quality interpretation requires literature context. A student must distinguish between a chemical being listed in a database, a chemical being measured in a system, and a chemical having a demonstrated role in a biological or environmental process.

10.2 Evidence Ladder

Evidence level	Interpretation strength
Tentative GC-MS match	Suggests possible presence; requires caution and preferably standards.
Database identifier or property	Supports identity, descriptors, and grouping.
Source annotation	Provides context from curated or aggregated source text.
Pathway, reaction, or occurrence evidence	Suggests biochemical, ecological, or taxonomic context.
Peer-reviewed measurement in similar system	Stronger support for relevance to the student's project.
Confirmed standard or targeted validation	Strongest support for chemical presence in the specific sample.

10.3 Guided Lab

Students select three high-priority findings from their Week 7 figures. For each finding, they fill an evidence table:

- Chemical.
- Trait or pattern.
- uafR table and field.
- Source database.
- Literature citation.
- Interpretation.
- Limitation.
- Follow-up test.

10.4 Finding Literature

Use database records to find search terms, not conclusions. Good searches combine:

- Chemical name or CID.
- System, organism, tissue, material, food, environment, disease, or exposure.
- Trait term, pathway, reaction, hazard code, sensory term, or measured property.
- Analytical method such as GC-MS, LC-MS, headspace, metabolomics, or targeted assay.

10.5 Independent Extension

Students write a two-page interpretation memo with:

- One central result.
- Two supporting figures or tables.
- Three literature sources.
- One paragraph on uncertainty.
- One proposed follow-up experiment or analysis.

10.6 Deliverables

Deliverable	Evidence of quality	Submission format
Annotated evidence table	Links database fields, literature, interpretation, and limits.	CSV or spreadsheet
Interpretation memo	Explains one result with evidence and caveats.	PDF
Reference library	Contains cleaned citations for final report.	BibTeX, Zotero, or plain bibliography

10.7 Quality Checklist

Before Week 9

- Claims are tied to a table, figure, or citation.
- The memo distinguishes tentative identity from confirmed presence.
- Literature sources are relevant to the system, not just the compound.
- Uncertainty is written clearly instead of hidden.

11 Manuscript-Style Research Product

Week 9: Manuscript-Style Research Product

Research milestone: Convert the analysis into a concise scientific report.

Primary product: Draft report with abstract, methods, results, figures, and limitations.

11.1 Why This Matters

Writing is part of analysis. A report exposes weak logic, missing methods, unclear figures, and overclaims. Week 9 turns the student's project into a product that can become a poster, internal report, thesis chapter, manuscript section, or grant preliminary result.

11.2 Report Structure

- **Title:** specific, not inflated.
- **Abstract:** question, approach, main result, limitation, and significance.
- **Introduction:** system background and why chemical traits matter.
- **Methods:** data source, uafR workflow, enrichment settings, filtering, visualization, and literature review.
- **Results:** figures and tables with restrained interpretation.
- **Discussion:** what the results suggest, what remains uncertain, and what should be done next.
- **Reproducibility statement:** how to rerun the analysis and where outputs are stored.

11.3 Methods Details To Include

- uafR version.
- R version and operating system.
- Chemical query list source.
- `categorate()` detail level, cache setting, throttle setting, and assay detail limit.

- Trait matrix profile, confidence threshold, and maximum traits.
- Any manual filtering or grouping rules.
- How missing data and unmatched chemicals were handled.

11.4 Guided Lab

Run the reproducibility check:

```
source("training/scripts/09_reproducibility_check.R")

check_training_project(".")
```

Then draft the methods paragraph from script settings and validation summaries.

11.5 Independent Extension

Students complete a 4 to 6 page report. The goal is not length. The goal is a defensible chain from question to data to evidence to interpretation.

11.6 Deliverables

Deliverable	Evidence of quality	Submission format
Draft report	Includes all required sections and at least three figures or tables.	PDF or Word export
Reproducibility check	Confirms project files, scripts, results, and session information.	Log file
Peer review notes	Student gives and receives structured feedback.	Worksheet

11.7 Quality Checklist

Before Week 10

- The methods section is specific enough to rerun the analysis.
- Results are separated from speculation.
- Each figure is cited in the text.
- The discussion includes limitations and next steps.

12 Presentation, Peer Review, and Final Research Package

Week 10: Presentation, Peer Review, and Final Research Package

Research milestone: Communicate the project clearly and leave behind a reusable research package.

Primary product: Final report, presentation, clean project folder, and continuation note.

12.1 Why This Matters

The final week is about making the work useful to others. A strong undergraduate project should be understandable to a future student, the dsDNA Core team, a collaborator, or a reviewer. The final package should preserve what was done, why it was done, and how to continue.

12.2 Presentation Structure

1. Research question and system.
2. Why the chemical evidence matters.
3. Data source and uafR workflow.
4. Main figure or result.
5. Evidence supporting interpretation.
6. Limitations.
7. Next experiment or analysis.

12.3 Peer Review

Students review each other's final products using the rubric appendix. Reviewers should focus on:

- Clarity of research question.
- Traceability of chemical evidence.

- Quality of cleaned variables and matrices.
- Figure readability.
- Strength of interpretation.
- Reproducibility.

12.4 Final Package Checklist

Required Handoff Contents

- README.md explains the project and how to rerun the analysis.
- data_raw/ contains unchanged input files or instructions for secure access.
- data_processed/ contains derived tables.
- scripts/ contains numbered scripts.
- figures/ contains final figures.
- results/ contains saved enrichment objects and summary tables.
- reports/ contains final report and presentation.
- logs/ contains research log and session information.

12.5 Independent Extension

Students prepare a continuation note:

- What worked.
- What failed.
- Which chemicals or traits deserve follow-up.
- Which database fields were most useful.
- Which results should not be overinterpreted.
- What the next student should do first.

12.6 Deliverables

Deliverable	Evidence of quality	Submission format
Final report	Revised after dsDNA Core and peer feedback.	PDF
Presentation	8 to 12 minute talk or poster-style slide deck.	PDF or slides
Research package	Clean folder with scripts, data, figures, results, and logs.	Project folder
Handoff note	Explains next steps for the project.	Markdown or PDF

12.7 Quality Checklist

Program Completion Standard

- The dsDNA Core team can rerun or inspect the analysis without guessing file locations.
- The final interpretation is evidence-based and appropriately cautious.
- The student can answer where each major result came from.
- The project creates a foundation for continued research.

13 Guided Lab and Exercise Compendium

The daily schedule names the work students complete each day. This compendium gives the reusable lab bank behind that schedule. Each lab has a concrete product, an inspection step, a research log requirement, and a review standard. Labs may be run with real student data, a dsDNA Core case dataset, or bundled uafR teaching data. Live database access is optional unless a program lead deliberately schedules it; offline examples and saved result objects are preferred for core instruction.

Lab Use Standard

Every lab produces a file, table, figure, memo, worksheet, or code object that remains in the project folder. A lab is not complete because a command ran. It is complete when the student can name the input, inspect the output, describe what the output supports, and document what it does not support.

13.1 Lab 1: Chemical Question Triage

Objective: Convert broad interests into feasible chemical research questions that can be answered with available evidence.

Required inputs: Three candidate interests, the Project Brief worksheet, a starter chemical list or example dataset, and one relevant background source.

Setup: Open a new research log entry. Create a project folder or use the Week 1 planning folder. Keep all draft questions in `planning/question_triage.md`.

Procedure:

1. Write each candidate interest as a question with a system, chemical set, comparison, evidence type, and expected output.
2. Mark the evidence needed for each question: tentative GC-MS hits, public database annotation, trait matrix, literature support, or confirmed standard.
3. Classify each question as feasible in 10 weeks, feasible after narrowing, or not feasible with available data.
4. Rewrite weak questions so they avoid proof language and name the decision the analysis can support.

5. Select one primary question and one backup question. Record the main scientific risk for both.

Expected outputs: Completed Project Brief worksheet, ranked candidate questions, and a one-paragraph scope note.

Interpretation prompts: What would count as evidence for the question? What result would be inconclusive? Which claim would require standards, new experiments, or literature beyond the current project?

Quality gate: The selected question must identify the chemical set, comparison, source of evidence, and final product. It must not depend on proving biological activity, therapeutic value, or sample occurrence from database presence alone.

Troubleshooting: If every question is too broad, reduce the chemical set, source domain, or comparison. If no comparison exists, use a descriptive evidence atlas question. If the question requires data that are not available, convert it to a pilot or methods question.

Research log entry: Record the original broad interest, the narrowed question, why alternatives were rejected, and the strongest remaining limitation.

Review guidance: Strong answers separate what uafR can organize and enrich from what the student's experiment has actually measured. Weak answers use verbs such as "prove," "cause," or "confirm" without confirmatory evidence.

Minimum viable version: One feasible question, 8 to 15 chemicals, one comparison or evidence theme, and one expected figure.

Advanced version: Two linked questions: one chemical evidence question and one methods or source-coverage question.

Rubric checklist: Feasible scope; named data source; clear limitation; no overclaiming; usable Week 2 project brief.

13.2 Lab 2: Reproducible Project Build

Objective: Build a project folder that another student can open, understand, and rerun.

Required inputs: RStudio, Git if available, uafR installation or local package source, `training/scripts/01_project_setup.R`, and the Reproducible Project checklist.

Setup: Start from a clean R session. Work from a project root, not from the Downloads folder or a desktop scratch folder.

Procedure:

1. Source `training/scripts/01_project_setup.R`.
2. Run `create_training_project()` if available, or create the same folders manually: `data_raw/`, `data_processed/`, `scripts/`, `outputs/`, `figures/`, `logs/`, and `docs/`.
3. Edit `README.md` to include the research question, folder map, script order, and contact/ownership notes.
4. Add `scripts/00_check_setup.R`. It should load packages, print `sessionInfo()`, and check that expected folders exist.

5. Save `logs/session_info.txt` from a clean R session.
6. If Git is used, make an initial commit after confirming that private data, tokens, and local machine paths are not present.

Expected outputs: Project skeleton, README, setup script, session information file, and first research log entry.

Interpretation prompts: Could a reviewer find the raw data, processed data, scripts, figures, and report without explanation? Which files are human-authored and which are generated?

Quality gate: The setup script runs from the project root and the README names the first script to run. No absolute local paths or secrets are stored.

Troubleshooting: If a script only works from one computer, replace absolute paths with project-relative paths. If package loading fails, record the version and installation error before editing analysis scripts.

Research log entry: Record the project root, script order, package versions, and any setup problem that had to be solved.

Review guidance: Accept projects that are plain and orderly. Do not reward elaborate folder structures that hide the script order or make reruns harder.

Minimum viable version: README, `scripts/00_check_setup.R`, `raw/processed` data folders, `outputs` folder, and session information.

Advanced version: Add a reproducibility badge table in the README showing which scripts run cleanly and which outputs they generate.

Rubric checklist: Runnable setup; clear script order; no secrets; no edited raw data; documented package state.

13.3 Lab 3: Database Source Audit

Objective: Teach students to treat public databases as evidence sources with scope, strengths, and limits.

Required inputs: Three project chemicals, PubChem pages or uafR outputs, KEGG records when available, LOTUS/FEMA/FDA/SPL/MeSH evidence when available, and the Database Evidence Map worksheet.

Setup: Use a small chemical set. The goal is careful interpretation, not broad querying.

Procedure:

1. Record each chemical name, synonym, CID if available, and known ambiguity such as salt form, mixture, stereochemistry, or generic name.
2. For each source, mark evidence as identity, property, hazard, occurrence, sensory, biomedical, regulatory, pathway, reaction, or literature.
3. Classify fields as analysis-ready, evidence-only narrative, or not relevant to the question.
4. Write one safe sentence for each source: what it can support and what it cannot support.
5. Identify the highest-value source and weakest source for the project.

Expected outputs: Database Evidence Map worksheet and a source suitability paragraph.

Interpretation prompts: Does the source support a chemical property, a database annotation, a biological occurrence, or an experimental measurement? Does it apply to the student's sample?

Quality gate: The audit must distinguish database presence from sample presence and source annotation from analyzed variable.

Troubleshooting: If a chemical has multiple identities, prefer stable identifiers and record ambiguity. If KEGG or LOTUS has no record, keep the missingness instead of replacing it with a weak synonym.

Research log entry: Record which sources were inspected, which were useful, and which source gaps may affect the project.

Review guidance: Strong audits explain absence of evidence. Weak audits silently remove sources that did not return results.

Minimum viable version: Three chemicals and at least three evidence-source columns.

Advanced version: Add source URLs, date accessed, and one reviewer-style challenge question per evidence type.

Rubric checklist: Stable identifiers; source roles; missingness retained; limitations stated; no unsupported biological claims.

13.4 Lab 4: Offline Core GC-MS Workflow

Objective: Run the core uafR GC-MS workflow without relying on live web services.

Required inputs: `training/scripts/04_core_workflow.R`, bundled example data such as `standard_data`, `standard_spread`, and `standard_exacto`, and the GC-MS Input Column Audit worksheet.

Setup: Start from a clean R session. Load uafR from the installed package. When working from the package source, use `devtools::load_all(".")`.

Procedure:

1. Source `training/scripts/04_core_workflow.R`.
2. Run `run_core_workflow(live_lookup = FALSE)`.
3. Inspect the resulting list object. Confirm that spread and exact-match outputs exist.
4. Compare required input columns to the GC-MS Input Column Audit worksheet.
5. Write a methods paragraph naming the offline teaching data and noting that no live lookup was required.
6. Optional: when live services are appropriate, rerun a small live lookup and compare output structure only.

Expected outputs: Saved offline workflow object, input column audit, and draft methods paragraph.

Interpretation prompts: Which outputs summarize peak organization? Which outputs depend on chemical identity matching? Which object should be preserved for later reruns?

Quality gate: Students can explain what `spreadOut()`, `mzExacto()`, and `exactoThese()` contribute and where tentative annotation limits begin.

Troubleshooting: Missing required columns usually means the input table is not ready for uafR. Duplicate peak keys should be documented and resolved by rule, not ignored.

Research log entry: Record command used, object names created, row counts, column names, and any mismatch between example data and project data.

Review guidance: Strong work inspects object structure before interpretation. Weak work jumps from a successful script run to a biological conclusion.

Minimum viable version: Offline workflow runs and methods paragraph names inputs and outputs.

Advanced version: Compare offline bundled output with one guarded live lookup and document any differences.

Rubric checklist: Required columns checked; object structure inspected; tentative status preserved; saved outputs present.

13.5 Lab 5: Research Enrichment Smoke Test

Objective: Run a small `categorate(detail = "research")` test and inspect validation before interpreting enriched data.

Required inputs: `training/scripts/05_categorate_research.R`, two or three common compounds, `library_data` when a chemical library is required, and the Query Settings Log worksheet.

Setup: Use caching and throttling. Set `assay_detail_limit = 0` for smoke tests unless assay-level bioactivity is the explicit lesson.

Procedure:

1. Source `training/scripts/05_categorate_research.R`.
2. Run `run_categorate_smoke()` if available, or run `categorate()` on two compounds with `detail = "research"`, caching, and a small throttle.
3. Immediately inspect `ValidationSummary`, `TableQuality`, `SourceCoverage`, `SourceDiagnostics`, and `DataDictionary`.
4. Save the full result as an RDS file.
5. Identify three tables that are useful for the project and three tables that are absent, sparse, or not relevant.

Expected outputs: Saved enrichment RDS, query settings log, validation notes, and source coverage summary.

Interpretation prompts: Which sources returned data? Which tables are empty? Which columns are discrete enough for grouping? Which are evidence text only?

Quality gate: No trait matrix or figure is interpreted before validation and source coverage are inspected.

Troubleshooting: If uafR reports that no library was detected, load `library_data` and pass it as `chemical_library`. If multiple compounds appear to hang, reduce the query size, use cached results,

and test one compound before scaling.

Research log entry: Record compounds queried, cache setting, throttle, detail level, assay limit, date, and any failed source.

Review guidance: Strong smoke tests are small, documented, and saved. Weak smoke tests repeatedly query live services without preserving settings.

Minimum viable version: Two compounds, saved result, validation inspected.

Advanced version: Compare `detail = "standard"` and `detail = "research"` for table families and research value.

Rubric checklist: Small query; validation first; cache/throttle recorded; result saved; source gaps documented.

13.6 Lab 6: Evidence Court

Objective: Decide whether a draft claim deserves to remain in the project by tracing it back to source evidence.

Required inputs: `ChemicalTraitMatrix`, `ChemicalTraitEvidence`, `ChemicalTraitReport`, one draft claim, and the Evidence Court worksheet.

Setup: Choose a claim that is tempting but not yet fully defended. Claims about pathway activity, safety, bioactivity, and natural occurrence are especially useful.

Procedure:

1. Identify the figure, table, matrix column, or result sentence behind the claim.
2. Pull evidence rows for the relevant trait key with `chemicalTraitEvidence()` where possible.
3. Record source database, source field, evidence text, source URL or identifier, confidence, and extraction rule.
4. Decide whether the evidence supports a result, a discussion hypothesis, a limitation note, or removal.
5. Rewrite the claim using cautious wording.

Expected outputs: Completed Evidence Court worksheet and a revised claim-evidence-limitation paragraph.

Interpretation prompts: Does evidence come from the student's data, a public database, a literature source, or an inference across sources? Which link is weakest?

Quality gate: Accepted claims trace to evidence. Rejected claims are documented so they do not reappear later.

Troubleshooting: If evidence rows are missing, inspect source tables directly or withdraw the claim. If evidence text is narrative, extract only defensible discrete variables.

Research log entry: Record the original claim, evidence reviewed, decision, and revised wording.

Review guidance: Strong claims survive reviewer-style challenge questions. Weak claims rely on one broad database sentence.

Minimum viable version: One claim fully traced and rewritten.

Advanced version: Run Evidence Court for the three strongest final report claims and include one removed claim.

Rubric checklist: Source table named; evidence row recorded; confidence considered; claim strength corrected; limitation explicit.

13.7 Lab 7: Trait Matrix Design

Objective: Build analysis-ready matrices that answer the research question rather than maximize column count.

Required inputs:

- Saved enrichment result and `ChemicalTraits`.
- `chemicalTraitMatrix()`, `chemicalTraitOntologyMatrix()`, and `chemicalTraitEvidence()`.
- Trait Grouping Dictionary worksheet.

Setup: Freeze the chemical set before comparing matrix settings. Do not change compounds and matrix settings at the same time.

Procedure:

1. Build a core binary matrix.
2. Build a confidence-weighted matrix.
3. Build or inspect ontology-derived variables when appropriate.
4. Compare matrix dimensions, missingness, sparsity, and interpretability.
5. Create a grouping dictionary with source table, source field, extraction rule, allowed values, confidence rule, and caveat.
6. Select the final matrix and document excluded variables.

Expected outputs: Matrix files, grouping dictionary, sensitivity note, and exclusion list.

Interpretation prompts: Are variables biologically or chemically meaningful? Are columns too sparse to interpret? Would changing the confidence threshold change the conclusion?

Quality gate: Every final variable has a source and an extraction rule. Narrative text is not treated as a categorical variable until it has been cleaned into discrete values.

Troubleshooting: If the matrix has too many columns, use `max_traits`, a narrower profile, or a higher confidence threshold. If the matrix is mostly zeros, rethink the question or trait profile.

Research log entry: Record function calls, profile, mode, thresholds, dimensions, removed columns, and justification.

Review guidance: Strong matrices are smaller and better defended than the widest possible output.

Minimum viable version: One final matrix and one grouping dictionary.

Advanced version: Compare binary, count, and confidence modes and show which conclusions are stable.

Rubric checklist: Reproducible function calls; documented thresholds; source-backed variables; sparsity evaluated; exclusions justified.

13.8 Lab 8: Figure Clinic

Objective: Convert validated outputs into figures that answer explicit research questions.

Required inputs: `training/scripts/07_visualization.R`, saved enrichment result, final or draft matrix, source coverage table, and Figure Planning worksheet.

Setup: Sketch the figure purpose before plotting. Identify the result sentence the figure is supposed to support.

Procedure:

1. Generate or inspect a source coverage heatmap.
2. Generate or inspect a trait matrix heatmap.
3. Build one project-specific figure such as a property summary, trait count bar chart, or focused evidence table.
4. Check axis labels, legends, color choices, missing-value treatment, caption, and export size.
5. Revise the caption so it states source, scope, and limitation.

Expected outputs: At least two scripted figures, one figure plan, and one peer review note.

Interpretation prompts: What question does the figure answer? What is missing? Would a table be clearer? Does the caption overclaim?

Quality gate: A figure must be regenerated from code and must have a caption that names the source table or matrix.

Troubleshooting: If labels overlap, simplify variables or export at a larger size. If a figure is visually striking but scientifically vague, rewrite the result question or remove the figure.

Research log entry: Record figure file names, source objects, script name, and what changed after critique.

Review guidance: Strong figures reduce cognitive load and clarify evidence. Weak figures show everything available.

Minimum viable version: Source coverage figure, one trait figure, and captions.

Advanced version: Add a sensitivity panel showing how a threshold choice changes or does not change interpretation.

Rubric checklist: Scripted export; readable labels; source named; limitation stated; figure tied to a result sentence.

13.9 Lab 9: Literature-to-Evidence Bridge

Objective: Link database-derived patterns to peer-reviewed literature without inflating claims.

Required inputs: One figure or table, one source evidence row, three papers, and the Literature and Evidence Synthesis worksheet.

Setup: Search by chemical name, stable identifier, trait, organism or system, method, and project context. Avoid relying only on broad review articles.

Procedure:

1. For each paper, record system, method, chemical evidence, main claim, and limitation.
2. Link each paper to a specific project result or background need.
3. Mark the paper's role: background, interpretation, method comparison, conflict, or follow-up.
4. Write a claim-evidence-limitation paragraph that includes the uafR evidence and literature role.
5. Identify one claim that should be weakened or removed after reading.

Expected outputs: Annotated literature table and one revised interpretation paragraph.

Interpretation prompts: Does the paper measure the same chemical, system, organism, analytical method, or outcome? Is it direct support or contextual support?

Quality gate: Literature must sharpen uncertainty and next steps. It must not be used to imply confirmed identity, pathway activity, mechanism, or clinical relevance without direct evidence.

Troubleshooting: If papers are too general, add method and organism terms. If no direct literature exists, document the gap and treat it as a future-work point.

Research log entry: Record search terms, databases searched, papers retained, papers rejected, and why.

Review guidance: Strong synthesis connects specific evidence to a modest interpretation. Weak synthesis adds impressive citations without changing the claim.

Minimum viable version: Three annotated papers and one paragraph.

Advanced version: Add a conflict table showing where literature and database evidence disagree.

Rubric checklist: Search terms recorded; paper roles assigned; limitations included; claim revised; citation quality adequate.

13.10 Lab 10: Final Handoff Audit

Objective: Ensure the project can be understood, rerun, and extended after the student leaves.

Required inputs: Final project folder, `training/scripts/09_reproducibility_check.R`, README, saved results, final report, figures, and Final Handoff worksheet.

Setup: Run from a clean R session. If possible, have a different student or mentor perform the audit.

Procedure:

1. Run `check_training_project()` when available.
2. Confirm that scripts run from the project root in the documented order.
3. Confirm that each final figure can be regenerated.

4. Confirm that the final report points to scripts, tables, source evidence, and citations.
5. Move obsolete scratch files to an archive folder or delete them only after review.
6. Write a continuation note naming the next best step, unresolved issues, and claims not to make.

Expected outputs: Final package audit, continuation note, clean README, and reproducibility status.

Interpretation prompts: What would a future student need first? Which result is ready to extend? Which result needs confirmation before publication?

Quality gate: A reviewer who did not work on the project can identify the question, rerun path, main outputs, limitations, and next steps.

Troubleshooting: If the project cannot rerun, preserve the failure message and fix the earliest failing script first. If a figure is missing its source object, rebuild the output chain rather than manually recreating the figure.

Research log entry: Record audit date, reviewer, failures, fixes, and final archive location.

Review guidance: Strong handoffs are boring in the best way: files are named, scripts are ordered, and decisions are documented.

Minimum viable version: README, saved result, scripts, figures, report, session info, and continuation note.

Advanced version: A second reviewer reruns the full package and signs the audit worksheet.

Rubric checklist: Clean session rerun; source evidence present; final products linked; unresolved issues named; continuation note actionable.

13.11 Lab 11: Identifier Ambiguity Drill

Objective: Practice resolving chemical-name ambiguity before database enrichment.

Required inputs: A list containing synonyms, salts, mixtures, stereochemical variants, and common names; PubChem or saved identity outputs; Chemical Intake Table worksheet.

Setup: Use chemicals where ambiguity is visible. Include at least one common name, one salt form, and one stereochemistry example.

Procedure:

1. Record the submitted name exactly as received.
2. Identify synonyms and candidate stable identifiers.
3. Flag ambiguity type: synonym, salt, mixture, stereochemistry, isomer, generic class, or misspelling.
4. Choose an inclusion rule for the project.
5. Document excluded names and why they were excluded.

Expected outputs: Cleaned chemical intake table and identifier ambiguity note.

Interpretation prompts: Would two students choose the same identifier from the rule? Is the rule appropriate for GC-MS tentative annotation?

Quality gate: No ambiguous identifier enters enrichment without a documented inclusion decision.

Troubleshooting: If the source name is a mixture or class rather than a compound, remove it or split it only with defensible source support.

Research log entry: Record identifier rule, ambiguous entries, and final query name.

Review guidance: Strong entries preserve the original name and the cleaned query name.

Minimum viable version: Five chemicals with ambiguity flags.

Advanced version: Add InChIKey or CID checks where available and compare results across synonyms.

Rubric checklist: Original names preserved; ambiguity type assigned; inclusion rule documented; exclusions justified.

13.12 Lab 12: Wrong-Column GC-MS Input Repair

Objective: Diagnose and repair malformed GC-MS hit tables before running uafR workflows.

Required inputs: A malformed CSV, required column list, `spreadOut()` documentation, and GC-MS Input Column Audit worksheet.

Setup: Work on a copy of the input file. Preserve the original raw export unchanged.

Procedure:

1. Read the CSV and print column names.
2. Compare names to expected fields such as `Component.RT`, `Component.Area`, `Base.Peak.MZ`, `File.Name`, `Compound.Name`, and `Match.Factor`.
3. Identify missing, renamed, duplicated, or nonnumeric columns.
4. Write a cleaning script that renames columns and coerces types with warnings.
5. Save a processed copy and rerun the input audit.

Expected outputs: Column audit, cleaning script, processed input file, and rerun note.

Interpretation prompts: Which changes are mechanical? Which changes require scientific judgment? Were any rows removed?

Quality gate: The raw file is preserved and every repair is scripted.

Troubleshooting: If duplicate peak keys exist, create a documented rule before processing. If match factor or area columns are nonnumeric, inspect examples before coercion.

Research log entry: Record original columns, repairs, row counts before and after, and remaining warnings.

Review guidance: Strong repair scripts are explicit and reversible.

Minimum viable version: Rename and audit one malformed file.

Advanced version: Write a reusable audit function for future exports from the same instrument.

Rubric checklist: Raw preserved; column map created; type checks documented; duplicates handled; processed file reproducible.

13.13 Lab 13: Source Coverage Missingness Audit

Objective: Determine whether missing database evidence changes the interpretation.

Required inputs:

- Saved `categorate()` result.
- `SourceCoverage`, `TableQuality`, and `SourceDiagnostics`.
- Source Coverage Report worksheet.

Setup: Use the same saved result object that will support figures. Do not rerun live queries just to fill gaps until the missingness is understood.

Procedure:

1. Count source coverage by chemical and source family.
2. Mark missingness as expected, concerning, or project-limiting.
3. Identify whether missingness affects one chemical, one source, or one evidence type.
4. Write a limitation statement for the final report.
5. Decide whether to query an alternate identifier, reduce scope, or keep the missingness.

Expected outputs: Source coverage report, missingness note, and any identifier follow-up list.

Interpretation prompts: Are missing values random, source-specific, or tied to chemical class? Does a missing field mean no evidence was found or that the source was not queried successfully?

Quality gate: Missing source coverage is visible in methods, figures, or limitations.

Troubleshooting: If diagnostics show failed requests, repeat a small cached smoke test. If the source truly has no data, keep that result.

Research log entry: Record source coverage pattern, decision, and any rerun settings.

Review guidance: Strong audits prevent students from overinterpreting chemicals with richer records.

Minimum viable version: One source coverage table and one paragraph.

Advanced version: Compare conclusions with and without low-coverage chemicals.

Rubric checklist: Coverage summarized; diagnostics checked; missingness classified; interpretation adjusted.

13.14 Lab 14: Matrix Sparsity Sensitivity Analysis

Objective: Test whether matrix-based conclusions are stable under reasonable trait filtering choices.

Required inputs: Saved trait outputs, `chemicalTraitMatrix()`, Matrix Sensitivity Log worksheet, and final grouping dictionary draft.

Setup: Freeze the chemical set. Change one matrix setting at a time.

Procedure:

1. Build a baseline core binary matrix.
2. Build a higher-confidence matrix.
3. Build a narrower profile or lower maximum trait count.
4. Record dimensions, percent nonzero cells, dropped traits, and changed patterns.
5. Decide which matrix is final and which sensitivity result belongs in the report or appendix.

Expected outputs: Sensitivity table, final matrix choice, and missing-data note.

Interpretation prompts: Are the same chemicals grouped together? Are conclusions driven by one broad or low-confidence trait? Are sparse columns useful or misleading?

Quality gate: The final matrix choice is justified by interpretability and evidence quality, not by the most visually appealing result.

Troubleshooting: If all matrices are sparse, narrow the question or treat the project as a source-coverage analysis.

Research log entry: Record settings, matrix dimensions, selected matrix, and reason.

Review guidance: Strong sensitivity analysis shows what changed and what stayed stable.

Minimum viable version: Two matrix settings compared.

Advanced version: Add a similarity comparison using `chemicalTraitSimilarity()` when appropriate.

Rubric checklist: One setting changed at a time; sparsity measured; conclusion stability assessed; final settings documented.

13.15 Lab 15: Reviewer Response Exercise

Objective: Practice responding to critique with evidence and revision rather than defensiveness.

Required inputs: Draft report section, peer review form, rubrics, and source evidence.

Setup: Choose three comments: one scientific, one reproducibility, and one writing or figure comment.

Procedure:

1. Classify each comment as accept, partially accept, clarify, or reject with reason.
2. Locate the file, figure, table, or paragraph affected.
3. Make the revision or write the evidence-based reason for not changing it.
4. Create a response table with reviewer comment, action taken, file location, and remaining concern.
5. Update the research log and final report.

Expected outputs: Reviewer response table and revised report or figure.

Interpretation prompts: Did the revision improve evidence traceability? Did it weaken an overclaim? Did it make the project easier to rerun?

Quality gate: Every accepted comment produces a concrete change and every declined comment has a defensible reason.

Troubleshooting: If feedback conflicts, prioritize scientific accuracy and reproducibility over visual preference.

Research log entry: Record feedback received, decisions, and files changed.

Review guidance: Strong responses are specific and professional.

Minimum viable version: Three comments resolved.

Advanced version: Add a before-and-after claim table showing how interpretation improved.

Rubric checklist: Comments classified; actions specific; evidence cited; files revised; unresolved issues named.

13.16 Lab 16: Final Package Recovery Exercise

Objective: Test whether the final package is robust when a file is missing, a session is clean, or a new reviewer starts from the README.

Required inputs: Near-final project folder, final package audit, README, scripts, and saved outputs.

Setup: Work on a copy of the project folder. Do not damage the authoritative final folder.

Procedure:

1. Ask a reviewer to start from the README and identify the rerun sequence.
2. Temporarily move one generated output to an archive folder and rerun the script that should recreate it.
3. Start a clean R session and rerun the setup and final figure scripts.
4. Record any failure caused by missing data, unclear paths, package versions, or manual steps.
5. Fix the project so the same failure is unlikely for the next user.

Expected outputs: Recovery audit, fixed README or scripts, and final archive note.

Interpretation prompts: Which files are essential? Which outputs are regenerateable? Which files are historical records that should be preserved but not rerun?

Quality gate: The final package can recover generated outputs from scripts and can explain any file that cannot be regenerated.

Troubleshooting: If recovery fails because raw data are private, document access rules and provide a case-data substitute when possible.

Research log entry: Record recovery test, failure points, fixes, and final status.

Review guidance: Strong final packages are resilient to clean sessions and new reviewers.

Minimum viable version: One generated output is regenerated successfully.

Advanced version: A reviewer on another machine completes the recovery test from the student bundle.

Rubric checklist: README usable; outputs regenerate; private data protected; recovery failures fixed or documented; archive note complete.

14 Supplementary Readings and Real-World Exercises

Readings are assigned to support the work, not to fill time. Students should annotate each reading with three items: what the source teaches, how it changes their project decisions, and what limitation or caution it raises.

14.1 Core Reading List

Week	Reading	Reason for assigning it
1	Wilson et al., Good enough practices in scientific computing [10].	Gives students a practical standard for files, scripts, documentation, and collaboration.
2	R Markdown documentation [1].	Helps students see how code, results, and text can live in one reproducible report.
3	PubChem update paper [4] and PUG-REST update [5].	Explains why PubChem is more than a name lookup and why source/provenance matters.
3	PubChem PUG-REST documentation [6].	Shows the programmatic access model behind many chemical data workflows.
3	KEGG API documentation [3].	Teaches students KEGG REST operations and reinforces the three-calls-per-second academic-use limit.
3	KEGG reference resource paper [2].	Gives biochemical context for compounds, pathways, enzymes, reactions, and annotation.
3	LOTUS natural products occurrence database [7].	Helps students interpret natural product occurrence and taxonomy evidence.
4	Metabolomics Standards Initiative chemical analysis guidance [8].	Establishes why chemical identification rigor and reporting detail matter.
8	Viant et al. metabolomics reporting and best practices [9].	Provides a mature example of how metabolomics evidence is used in high-stakes toxicology contexts.

14.2 Reading Annotation Template

For every assigned reading, students should write:

- **Main idea:** one or two sentences.
- **Project consequence:** what the reading changes about the student's data, code, interpretation, or reporting.
- **Caution:** what the reading says or implies that should prevent overclaiming.
- **Useful phrase:** one technical phrase the student should be able to define.
- **Question for discussion:** one specific question raised by the reading.

14.3 Real-World Exercise Bank

14.3.1 Exercise A: The Unmatched Chemical

Scenario: A student has 20 tentative chemical hits. Four do not return strong source coverage.

Task: Build a missingness table. For each unmatched chemical, decide whether to rename, remove, retain as unmatched, search by synonym, or request follow-up validation.

Expected learning: Missing data are not clutter. They are evidence about the limits of the workflow.

14.3.2 Exercise B: The Hazard Code Trap

Scenario: A chemical has safety annotations and GHS-style terms, but the project is ecological.

Task: Decide which safety terms are relevant to the research question and which belong only in a cautionary appendix.

Expected learning: A hazard code can support screening or caution, but it does not automatically explain biological behavior in a sample.

14.3.3 Exercise C: Pathway Does Not Mean Presence

Scenario: KEGG links a compound to a pathway. A student wants to claim that the pathway is active in their sample.

Task: Rewrite the claim at three evidence levels: source-backed context, hypothesis, and confirmed pathway activity.

Expected learning: Pathway annotation is context, not direct measurement of pathway flux or activity.

14.3.4 Exercise D: Natural Product Occurrence

Scenario: LOTUS-derived evidence suggests a compound occurs in several organisms, but the taxonomic context differs from the student's organism.

Task: Separate direct occurrence evidence, related-organism evidence, and broad natural product evidence.

Expected learning: Occurrence records can guide hypotheses while still requiring taxonomic caution.

14.3.5 Exercise E: Sensory Versus Biological Interpretation

Scenario: FEMA or annotation text suggests flavor or odor relevance.

Task: Decide whether the sensory trait belongs in the main result, a descriptive table, or background motivation.

Expected learning: Sensory descriptors are useful grouping variables only when the research question is sensory, food, aroma, behavior, or exposure related.

14.3.6 Exercise F: Literature Conflict

Scenario: One paper reports a chemical as biologically active, but another paper treats it as a contaminant or background compound.

Task: Build a conflict table with system, dose or abundance, method, evidence strength, and relevance to the student's project.

Expected learning: Conflicting literature is handled by context, not by choosing the more exciting citation.

14.3.7 Exercise G: Matrix Sparsity

Scenario: A full trait matrix has hundreds of columns, but most are present in only one compound.

Task: Compare a full matrix, a core matrix, and a medium-confidence capped matrix. Choose one for the report and justify the choice.

Expected learning: More variables can make analysis weaker when they are sparse, redundant, or poorly supported.

14.3.8 Exercise H: The Reviewer Request

Scenario: A reviewer asks, "How do you know these compounds are meaningfully grouped?"

Task: Prepare a response with matrix construction rules, evidence fields, source coverage, and a limitation statement.

Expected learning: A grouping claim needs source-backed variables, not only visual clustering.

14.3.9 Exercise I: Public Data Ethics

Scenario: A collaborator wants to upload unpublished sample metadata to a public repository before approval.

Task: Write a data handling decision memo: what can be shared, what must be withheld, and what metadata can be generalized.

Expected learning: Reproducibility and confidentiality must be balanced deliberately.

14.3.10 Exercise J: Final Handoff Recovery

Scenario: A student inherits a project folder with no README, ambiguous scripts, and several saved result objects.

Task: Reconstruct the workflow order, identify missing context, and write a recovery plan.

Expected learning: The value of documentation becomes obvious when students experience the cost of missing documentation.

14.4 Scientific Claim Boundaries

The safest writing pattern is to name the evidence type before making the claim. Students should practice replacing broad claims with source-backed claims that state uncertainty directly.

14.4.1 Tentative Chemical Match

Weak claim	This sample contains methyl salicylate.
Safer wording	A tentative GC-MS annotation matched methyl salicylate; confirmation would require standards or other confirmatory evidence.
Reviewer challenge	What evidence distinguishes a library match from confirmed identity?

14.4.2 Database Presence

Weak claim	The chemical occurs in this plant because it appears in a public database.
Safer wording	Public database records report occurrence or annotation evidence for this chemical in specified sources; the current sample requires its own evidence.
Reviewer challenge	Is the database record about the same organism, sample, and analytical method?

14.4.3 Source Annotation as Variable

Weak claim	This full database sentence proves the compound is anti-inflammatory.
Safer wording	The source annotation contains biomedical language that can be recorded as evidence; an analysis-ready trait requires a documented extraction rule and caveat.
Reviewer challenge	Which exact field was extracted, and what values are allowed in the variable?

14.4.4 Pathway Context

Weak claim	The pathway is active in the sample.
Safer wording	KEGG provides pathway or reaction context for the compound; pathway activity was not measured in this project.
Reviewer challenge	Was pathway flux, expression, enzyme activity, or metabolite abundance directly tested?

14.4.5 Hazard Screening

Weak claim	The sample is dangerous.
Safer wording	One or more chemicals have hazard annotations that justify screening or follow-up; risk assessment requires exposure, concentration, route, organism, and context.
Reviewer challenge	What dose or exposure information is available?

14.4.6 Natural Product Occurrence

Weak claim	LOTUS proves this compound is in the student's organism.
Safer wording	LOTUS-derived occurrence or taxonomy evidence supports natural product context; occurrence in the student's organism requires matching source evidence or direct measurement.
Reviewer challenge	Does the source record name the same organism or only a related taxonomic group?

14.4.7 Bioactivity and Mechanism

Weak claim	The compound has a therapeutic mechanism in this system.
Safer wording	Bioactivity, target, potency, or pharmacology annotations can support hypothesis generation; mechanism or clinical relevance requires direct experimental evidence.
Reviewer challenge	What assay system, target, potency, and biological context produced the annotation?

14.4.8 Sparse Trait Matrices

Weak claim	The heatmap shows clear chemical groups.
Safer wording	The trait matrix suggests exploratory grouping under the selected profile and confidence threshold; sparse columns and source coverage must be checked before stronger interpretation.
Reviewer challenge	Do the groups persist after low-confidence or sparse variables are removed?

14.5 Discussion Prompts

- When is a database result strong enough for a Results section, and when does it belong in Discussion?
- What is the difference between a chemical being present in PubChem and a chemical being present in a biological sample?
- How should a student report a potentially interesting result with weak source coverage?
- What makes a trait matrix scientifically meaningful rather than merely computationally convenient?
- How should a researcher discuss compounds that are tentatively identified but biologically plausible?

14.6 Recommended Online References

- PubChem PUG-REST documentation: <https://pubchem.ncbi.nlm.nih.gov/docs/pug-rest>
- KEGG API documentation: <https://www.genome.jp/kegg/rest/>
- KEGG API manual: <https://www.genome.jp/kegg/rest/keggapi.html>
- PubChem LOTUS data source page: <https://pubchem.ncbi.nlm.nih.gov/source/25132>
- R Markdown documentation: <https://rmarkdown.rstudio.com/>

15 Project Tracks for Student Interests

Students should be able to bring their own interests to the program while using a common quality standard. The tracks below define feasible questions, outputs, caveats, and capstone products for different research domains. A track is a starting point, not a restriction. The same project may combine a biological question with a methods question, or a food chemistry question with a communication product.

15.1 Track 1: Plant, Agriculture, and Natural Products

Good research questions:

- Which tentative GC-MS hits in two plant treatments have source-backed natural product occurrence or chemical class annotations?
- Do cultivar, tissue, or developmental-stage groups differ in trait profiles derived from validated chemical evidence?
- Which compounds have pathway or occurrence context that supports follow-up targeted analysis?

Bad or overbroad questions:

- Which plant is medicinal?
- Did this treatment activate a pathway based only on KEGG context?
- Are these compounds present in the plant because LOTUS reports them in another organism?

Recommended uafR tables and outputs:

- `ChemicalOccurrences`, `ChemicalTaxonomy`, `LOTUSProfile`, and `ChemicalClasses`.
- `ChemicalTraits`, `ChemicalPathwayRoles`, `KEGGPathways`, and `KEGGReactions`.
- `SourceCoverage` and `ChemicalTraitEvidence`.

Recommended figures: Source coverage heatmap, occurrence evidence heatmap, compound class bar chart, trait matrix heatmap, and a focused pathway-context table.

Key caveats: Natural product occurrence in another organism supports hypothesis generation, not direct occurrence in the student's sample. KEGG pathway membership is pathway context, not pathway activity. Tentative GC-MS annotation requires cautious wording.

Minimum viable project: 10 to 20 tentative compounds, one plant system or comparison, source coverage report, occurrence or class grouping dictionary, two figures, and one limitation paragraph.

Advanced project: Compare treatment or tissue groups using a medium-confidence trait matrix and add a targeted follow-up list for standards or MS/MS confirmation.

Example final product: “Source-backed natural product and pathway-context profile of tentative volatile annotations from treated and untreated plant tissue.”

15.2 Track 2: Environmental Exposure and Toxicology

Good research questions:

- Which detected or queried chemicals have hazard, property, or regulatory signals that justify follow-up review?
- Do sample groups differ in safety-relevant trait profiles or physicochemical properties?
- Which chemicals combine strong source coverage with properties relevant to volatility, hydrophobicity, persistence, or exposure screening?

Bad or overbroad questions:

- Are these samples toxic?
- Which chemical caused the biological effect?
- Is a compound risky because a hazard annotation exists?

Recommended uafR tables and outputs:

- `ChemicalHazards`, `SafetyProfile`, `PubChemProperties`, and `PubChemSafety`.
- `ChemicalMeasurementSummary`, `ChemicalTraits`, and `DerivedGroups`.
- `SourceCoverage` and `ValidationSummary`.

Recommended figures: Hazard-term summary, property scatterplot, source coverage heatmap, confidence-weighted trait matrix, and table of chemicals prioritized for follow-up.

Key caveats: Hazard screening is not risk assessment. Risk requires exposure, concentration, dose, route, organism, and context. Database safety annotations may come from different formulations or use cases.

Minimum viable project: One chemical set, a source coverage and hazard screening table, one property figure, one evidence-limited prioritization list, and a risk-assessment caveat.

Advanced project: Add sensitivity analysis showing how conclusions change when low-confidence safety or exposure traits are removed.

Example final product: “Evidence-screening profile for tentative chemicals with safety-relevant annotations in an environmental sample set.”

15.3 Track 3: Food, Flavor, Aroma, and Sensory Chemistry

Good research questions:

- Which tentative volatile compounds have source-backed flavor, odor, or sensory descriptors relevant to a food or plant system?
- Do treatment groups differ in sensory-relevant trait profiles or physicochemical properties?
- Which compounds should be prioritized for standards or sensory validation?

Bad or overbroad questions:

- Which sample tastes best?
- Did this compound cause a sensory perception based only on a database descriptor?
- Are all compounds with flavor annotations relevant at the measured abundance?

Recommended uafR tables and outputs:

- FEMAProfile, ChemicalTerms, and ChemicalTraits.
- PubChemProperties, ChemicalMeasurementSummary, and ChemicalTraitMatrix.
- ChemicalTraitEvidence.

Recommended figures: Sensory descriptor heatmap, abundance versus sensory evidence plot, trait count by treatment, property summary, and source coverage table for sensory evidence.

Key caveats: Sensory descriptors do not prove perceptibility. Concentration, odor threshold, matrix effects, mixture effects, and human or instrument validation matter.

Minimum viable project: 8 to 15 volatile compounds, sensory descriptor evidence map, one heatmap, one abundance or property summary, and a cautious sensory interpretation.

Advanced project: Add literature-derived odor thresholds where available and classify compounds as strong evidence, context evidence, or follow-up candidates.

Example final product: “Source-backed sensory descriptor profile for tentative volatile annotations in a food aroma comparison.”

15.4 Track 4: Biomedical, Pharmacology, and Public Health

Good research questions:

- Which query chemicals have biomedical terms, pharmacologic actions, assay annotations, or target evidence in public databases?
- Which compounds have evidence strong enough for background discussion versus only future-work prioritization?
- How does bioactivity source coverage differ across a chemical set?

Bad or overbroad questions:

- Which compounds cure disease?
- What mechanism explains the biological effect based only on PubChem bioassay records?
- Are compounds clinically relevant because they have MeSH terms?

Recommended uafR tables and outputs:

- `MeSHProfile`, `PubChemBioactivity`, and `PubChemBioAssayDetails`.
- `ChemicalBioassays`, `ChemicalBioactivities`, `ChemicalTargets`, and `ChemicalPotencies`.
- `ChemicalTraitEvidence` and `LiteratureProfile`.

Recommended figures: Bioactivity evidence summary, target evidence table, biomedical trait matrix, assay-source coverage plot, and evidence ladder figure.

Key caveats: Bioassay or pharmacologic annotation does not prove mechanism, efficacy, safety, clinical relevance, or activity in the student's system. Assay context, potency, target type, and evidence source matter.

Minimum viable project: Small chemical set, biomedical evidence map, one source coverage figure, one table of supported and unsupported claims, and a literature-context paragraph.

Advanced project: Compare assay-level details for a few priority compounds while keeping broad compound interpretation conservative.

Example final product: "Biomedical evidence map and claim-boundary analysis for a focused set of tentative chemical annotations."

15.5 Track 5: Methods, Data Science, and Tool Evaluation

Good research questions:

- How do `chemicalTraitMatrix()` settings change grouping or similarity conclusions?
- Which database sources contribute the most usable evidence for a particular chemical class?
- How robust are project conclusions when low-confidence traits, sparse columns, or weakly covered chemicals are removed?

Bad or overbroad questions:

- What can uafR do?
- Which setting gives the best-looking heatmap?
- Can all output tables be combined into one master table without a research question?

Recommended uafR tables and outputs:

- `ValidationSummary`, `TableQuality`, `SourceDiagnostics`, and `SourceCoverage`.
- `ChemicalTraitMatrix`, `ChemicalTraitSimilarity`, and `ChemicalTraitReport`.
- `DataDictionary` and saved RDS result objects.

Recommended figures: Matrix sensitivity comparison, source coverage summary, sparsity plot, trait overlap heatmap, and confidence distribution.

Key caveats: Methods projects still need a scientific or training question. A large output inventory is not a result unless it changes how evidence is interpreted or workflows are run.

Minimum viable project: One chemical set, at least two matrix settings, source coverage summary, one sensitivity figure, and a recommendation memo.

Advanced project: Build a reusable project-specific audit function or visualization function and test it against a second chemical set.

Example final product: “Sensitivity of source-backed chemical grouping to trait matrix profile and confidence thresholds in uafR.”

15.6 Track 6: Chemistry Education and Communication

Good research questions:

- Which chemical traits are most useful for teaching new students about evidence limits?
- Can a small compound set teach identity ambiguity, source coverage, and cautious interpretation?
- Which visualization best communicates tentative annotation and database evidence to non-specialists?

Bad or overbroad questions:

- How can chemistry be made easy?
- Can uncertainty be removed from the training materials?
- Which graphic looks most engaging without preserving source evidence?

Recommended uafR tables and outputs:

- `ChemicalTraitReport`, `ChemicalTraitEvidence`, and `DataDictionary`.
- `SourceCoverage`, `ValidationSummary`, `ChemicalTerms`, and selected identity tables.

Recommended products: Teaching dataset, annotated figure set, evidence interpretation guide, worksheet example packet, and short mini-workshop.

Key caveats: Communication products must preserve scientific caution. Simplification should not erase tentative identity, missingness, source limitations, or the distinction between source annotation and sample evidence.

Minimum viable project: One teaching compound set, evidence map, two annotated figures, a short guide, and an assessment question set.

Advanced project: Pilot the material with a peer group and revise based on documented confusion points.

Example final product: “A teaching module for interpreting tentative GC-MS annotations and public database evidence with uafR.”

15.7 Capstone Options

Students can finish with one of several products:

- Manuscript-style research report suitable for internal dsDNA Core review.

- Poster-ready figure set with captions and methods.
- Reusable analysis pipeline for a larger lab or collaborator project.
- Database evidence atlas for a chemical class.
- Methods comparison or sensitivity analysis for uafR workflows.
- Teaching module for future interns.

15.8 Track Selection Worksheet

Prompt	Response
Chosen track	
Why this track fits the student's interest	
Primary uafR tables	
Most important figure	
Most important limitation	
Minimum viable product	
Advanced extension	
Likely final product	

16 Program Completion Standard

The program is complete when the student leaves behind a research package that is understandable, auditable, and useful for the next step. A strong final package does not need to prove a dramatic biological result. It must preserve the evidence chain from chemical list to uafR output, from output to matrix or figure, and from figure to cautious interpretation.

16.1 Required Final Package

Required item	What it must contain	Acceptance question
Reproducible R project folder	README, scripts, data folders, outputs, figures, logs, and docs.	Can a reviewer find the starting point and script order?
Cleaned chemical query list	Original names, cleaned query names, identifiers where available, inclusion rules, exclusions, and ambiguity notes.	Could another student rebuild the query list?
Saved uafR result object	RDS file or equivalent saved object from the final enrichment workflow.	Is the enriched evidence preserved without rerunning live services?
Validation summaries	<code>ValidationSummary</code> , <code>TableQuality</code> , <code>SourceDiagnostics</code> , and notes from <code>validateCategorateResult()</code> .	Was interpretation delayed until validation was reviewed?
Source coverage report	Coverage by chemical and source family, failed or empty sources, and missingness consequences.	Are source gaps visible in interpretation?
Trait table or matrix	Final <code>ChemicalTraitMatrix</code> or grouped chemical table with settings documented.	Are matrix settings and confidence choices reproducible?
Grouping dictionary	Source table, source field, extraction rule, allowed values, confidence rule, and caveat for each variable.	Can variables be rebuilt from source evidence?

Required item	What it must contain	Acceptance question
Publication-style figures or tables	At least three scripted figures or tables with captions naming source objects and limitations.	Do figures answer clear questions without overclaiming?
Manuscript-style report	Title, abstract, introduction, methods, results, discussion, limitations, reproducibility statement, and references.	Could the report support internal review or a draft publication section?
Presentation or poster	Clear research question, workflow, central result, evidence, limitation, and next step.	Can the student explain the project to a scientific audience?
Research log	Dated entries for weekdays and weekend work, including commands, outputs, decisions, errors, and evidence.	Can the decision history be reconstructed?
Session information	R version, package versions, uafR source or bundle version, and platform details.	Can package state be checked later?
Final package audit	Completed audit worksheet and clean-session rerun status.	Has the package been tested from a fresh starting point?
Continuation note	First next step, unresolved issues, recommended follow-up, and archive location.	Can a future student continue without interviewing the author?

16.2 Claims That Require Explicit Boundaries

Every final package must include a short statement of what the project does not show. This statement protects the student and the program from overclaiming. It should be adapted to the project, but it must address all relevant boundaries:

- Tentative GC-MS matches are not confirmed identities unless standards or equivalent confirmatory evidence were used.
- Database records do not prove that a compound is present in the student's sample.
- Source annotations do not automatically become analysis variables until they are cleaned and documented.
- KEGG pathway context does not demonstrate pathway activity, flux, or regulation.
- Hazard annotations support screening, not risk assessment without exposure and dose context.
- Natural product occurrence in another organism does not prove occurrence in the student's organism.
- Bioactivity or pharmacology annotations do not prove mechanism, efficacy, clinical relevance, or safety.
- Sparse matrices may support exploration but not strong grouping claims without sensitivity checks.

16.3 Final Acceptance Review

Final Review Questions

1. What is the research question, and where is it stated in the project?
2. Which script creates the final saved uafR result object?
3. Which validation and source coverage outputs were reviewed before interpretation?
4. Which variables appear in the final grouping dictionary, and where did they come from?
5. Which three figures or tables carry the main results?
6. Which claim is strongest, and which evidence supports it?
7. Which claim is most uncertain, and how is that uncertainty written?
8. Which chemicals or sources were excluded, unmatched, sparse, or weakly supported?
9. Can the final figures be regenerated from a clean R session?
10. What should the next student do first?

16.4 Minimum Passing Package

The minimum passing package is intentionally smaller than the advanced package but not less rigorous. It includes a reproducible folder, 8 to 15 chemicals or a program-approved case dataset, saved uafR output, validation and source coverage notes, one final matrix or grouped table, two to three figures or tables, a short manuscript-style report, a presentation, research log, and continuation note. It must still preserve tentative identity language, missingness, source limitations, and reproducibility.

16.5 High-Impact Package

A high-impact package adds a stronger literature bridge, sensitivity analysis, external peer review response, advanced figure set, and a follow-up design that could support standards-based confirmation, targeted quantification, new sample collection, or a manuscript section. High-impact packages are still cautious: they are stronger because their evidence trail is better, not because their claims are louder.

A Student Worksheets

Worksheets are working documents, not decorations. Each one exists to preserve a decision, expose uncertainty, or make a final product easier to review. Students may complete them on paper, in a spreadsheet, or in a project Markdown file, but the final project folder must contain a completed version or equivalent evidence.

A.1 Worksheet 1: Project Brief

Purpose

Convert a broad interest into a feasible chemical research question with a product that can be completed in 10 weeks.

When to use: Week 1 scope gate and any later major project revision.

Instructions: Write in complete sentences. The question must name the chemical set, system or comparison, evidence type, and expected final product.

Required fields: Project title, research system, primary question, comparison or grouping variable, starting chemical list source, evidence types, expected product, main risk, and backup plan.

Common mistakes: Questions that ask whether chemicals are important without defining importance; questions that require confirmed identity when only tentative annotations are available; missing risk notes.

Prompt	Student response
Working project title	
Research system or domain	
Primary research question	
Comparison or grouping variable	
Starting chemical list source	
Evidence types needed	
Expected final product	
Main risk or limitation	
Backup question	

Completion standard: A reviewer can tell what will be analyzed, what evidence is needed, and what claim will remain out of scope.

A.2 Worksheet 2: Chemical Intake Table

Purpose

Preserve the original chemical list while creating a clean, auditable query list.

When to use: Week 1 through Week 3, before enrichment and any time identifiers change.

Instructions: Never overwrite submitted names. Add cleaned query names, identifiers, ambiguity flags, and inclusion decisions in separate columns.

Required fields: Original name, cleaned query name, source, identifier if available, ambiguity type, inclusion decision, confidence, project relevance, and notes.

Common mistakes: Deleting difficult chemicals without explanation; mixing salts, mixtures, and parent compounds without a rule; treating synonyms as independent compounds.

Original name	Clean query	Source	Identifier	Ambiguity	Include?	Notes
---------------	-------------	--------	------------	-----------	----------	-------

Original name	Clean query	Source	Identifier	Ambiguity	Include?	Notes
---------------	-------------	--------	------------	-----------	----------	-------

Completion standard: The cleaned list can be rerun by another student using the same inclusion rules.

A.3 Worksheet 3: Database Evidence Map

Purpose

Show which evidence sources are useful for each chemical and what each source can support.

When to use: Week 3 source audit and Week 5 enrichment review.

Instructions: Mark coverage using yes, no, weak, or not relevant. Add notes for source gaps that affect interpretation.

Required fields: Chemical, identity evidence, property evidence, hazard evidence, occurrence evidence, pathway or reaction evidence, sensory evidence, biomedical evidence, strongest source, and limitation.

Common mistakes: Treating every database field as equally reliable; hiding empty sources; assuming a source annotation applies to the student's sample.

Chemical	Identity	Property	Hazard	Occur.	Pathway	Sensory	Best source/limit
----------	----------	----------	--------	--------	---------	---------	----------------------

Completion standard: Every major source gap has an interpretation note.

A.4 Worksheet 4: Source Coverage Report

Purpose

Summarize what was returned by `categorate()` before analysis begins.

When to use: Week 5 validation and any rerun after changing query chemicals.

Instructions: Use `SourceCoverage`, `TableQuality`, and `SourceDiagnostics`. Distinguish empty sources from failed requests.

Required fields: Result object name, date, query size, source family, returned rows, missing chemicals, diagnostics, and interpretation consequence.

Common mistakes: Interpreting rich chemicals more strongly without noting uneven coverage; rerunning live queries repeatedly without preserving settings.

Source family/table	Coverage status	Diagnostics	Interpretation consequence
---------------------	-----------------	-------------	----------------------------

Completion standard: The report explains whether missingness is expected, concerning, or project-limiting.

A.5 Worksheet 5: GC-MS Input Column Audit

Purpose

Confirm that GC-MS hit tables have the required columns and usable values before running the core workflow.

When to use: Week 4 before `spreadOut()` and again after input repair.

Instructions: Compare the file to the required fields used by the workflow. Record renaming and type coercion in a script.

Required fields: File name, row count, required columns, missing columns, duplicate keys, suspicious values, repair decision, and processed output location.

Common mistakes: Editing the raw export manually; ignoring duplicate peak keys; assuming column names are obvious without checking.

Check	Pass/fail	Evidence	Action needed
<code>Component.RT</code> present and numeric			
<code>Component.Area</code> present and numeric			
<code>Base.Peak.MZ</code> present and numeric			
<code>File.Name</code> present			
<code>Compound.Name</code> present			
<code>Match.Factor</code> present and numeric			
Duplicate peak keys checked			

Completion standard: The processed input can be produced from raw input by running a documented script.

A.6 Worksheet 6: Query Settings Log

Purpose

Preserve the exact settings used for enrichment so results can be reproduced and interpreted.

When to use: Every `categorate()` run, especially live database runs.

Instructions: Record code-level settings and service behavior. Save the result object immediately.

Required fields: Date, compounds, `detail`, cache setting, cache directory, throttle, assay detail limit, trait profile, confidence threshold, max traits, and result path.

Common mistakes: Running large live queries without a smoke test; forgetting whether cached or fresh results were used; losing the RDS object.

Setting	Value
Run date and operator	
Compound list file	
<code>detail</code> setting	
<code>cache</code> and cache location	
<code>throttle</code>	
<code>assay_detail_limit</code>	
Trait matrix profile and confidence	
Saved RDS path	
Warnings or failed sources	

Completion standard: Another student can rerun the same query or explain why it should not be rerun.

A.7 Worksheet 7: Trait Evidence Audit

Purpose

Trace important traits or matrix columns back to source evidence before using them in figures or claims.

When to use: Week 6 and every time a final claim depends on a trait.

Instructions: Use `chemicalTraitEvidence()` when possible. If evidence must be inspected manually, record the source table and field.

Required fields: Trait key, chemical, source table, source field, evidence text or value, confidence, extraction rule, source URL or identifier, and decision.

Common mistakes: Treating trait names as self-evident; failing to record extraction rules; using low-confidence traits without caveat.

Trait	Chemical	Source table	Field	Evidence/value	Conf.	Decision
-------	----------	--------------	-------	----------------	-------	----------

Completion standard: Every final trait can be defended with source evidence or removed.

A.8 Worksheet 8: Trait Grouping Dictionary

Purpose

Define analysis-ready variables from source-backed fields.

When to use: Week 6 variable gate and later matrix revisions.

Instructions: Each row is a variable definition. Include the allowed values and the rule used to extract them.

Required fields: Variable, source table, source field, extraction rule, allowed values, confidence rule, caveat, and final use.

Common mistakes: Using full sentences as factor levels; merging unrelated traits; creating variables without source fields.

Variable	Source table	Field	Extraction rule	Allowed values	Caveat
----------	--------------	-------	-----------------	----------------	--------

Completion standard: The dictionary is precise enough that another student could rebuild the variables.

A.9 Worksheet 9: Matrix Sensitivity Log

Purpose

Document how matrix settings affect dimensions, sparsity, and interpretation.

When to use: Week 6 sparsity challenge and any final matrix selection.

Instructions: Change one setting at a time. Preserve dimensions and nonzero counts.

Required fields: Profile, mode, confidence threshold, max traits, row count, column count, sparsity, changed conclusion, and final decision.

Common mistakes: Selecting the prettiest heatmap; changing chemicals and settings at the same time; hiding unstable conclusions.

Profile/mode	Threshold	Rows	Cols.	Sparse?	Interpretation change
--------------	-----------	------	-------	---------	-----------------------

Completion standard: The selected matrix settings are justified by evidence quality and interpretability.

A.10 Worksheet 10: Figure Planning Worksheet

Purpose

Plan a figure before plotting so the graphic answers a scientific question.

When to use: Week 7 before each major figure.

Instructions: Write the result sentence first. Then choose the source object and figure type.

Required fields: Figure question, source table or matrix, rows, columns, grouping variable, expected pattern, missingness plan, export format, and draft caption.

Common mistakes: Creating a figure only because the data are available; missing source table in caption; no plan for missing values.

Prompt	Student response
Result sentence the figure should support	
Source table, matrix, or script	
Rows or observations	
Columns or variables	
Grouping or color variable	
Missing-value treatment	
Expected pattern	
Draft caption	

Completion standard: The figure plan names the evidence object and the limitation before plotting begins.

A.11 Worksheet 11: Figure Caption Checklist

Purpose

Make captions scientifically useful and reviewer-ready.

When to use: Week 7 figure clinic and final report preparation.

Instructions: Check each caption against the figure and underlying data.

Required fields: Figure number, source object, sample or chemical set, variables, missingness note, statistical or visual summary, and claim boundary.

Common mistakes: Captions that only restate the title; captions that imply causation or confirmation; unlabeled missing values.

Caption element	Present?	Revision note
Source object or script named		
Chemical set or sample group defined		
Variables and units described		
Missing or sparse values explained		
Claim strength matches evidence		
Limitation included when needed		

Completion standard: A reader can understand what the figure shows and what it does not show without guessing.

A.12 Worksheet 12: Literature and Evidence Synthesis

Purpose

Connect uafR outputs to literature without inflating claims.

When to use: Week 8 and report discussion drafting.

Instructions: Summarize each paper by system, method, chemical evidence, main claim, and limitation. Link it to a specific project result.

Required fields: Chemical or trait, project result, citation, system, method, evidence type, support level, conflict or limitation, and final use.

Common mistakes: Citing a paper only because it mentions the same chemical; ignoring method differences; treating contextual support as direct proof.

Chem./trait	Project result	Citation	System	Method	Support/limit
-------------	----------------	----------	--------	--------	---------------

Completion standard: Each retained source has a clear role in background, interpretation, methods, conflict, or future work.

A.13 Worksheet 13: Claim-Evidence-Limitation Memo

Purpose

Draft defensible interpretation by pairing every claim with evidence and limitation.

When to use: Week 8 interpretation gate and Week 9 report drafting.

Instructions: One row per claim. Include the strongest evidence and the most important caveat.

Required fields: Claim, result object, source evidence, literature support, limitation, reviewer challenge, and revised wording.

Common mistakes: Writing claims before checking evidence; omitting limitations; claiming pathway activity, clinical relevance, or sample occurrence from database records.

Draft claim	Evidence	Limitation	Revised wording
-------------	----------	------------	-----------------

Completion standard: Every final claim has a source-backed version and a clear boundary.

A.14 Worksheet 14: Peer Review Form

Purpose

Structure peer feedback around evidence, reproducibility, and claim strength.

When to use: Weeks 7 through 10, especially draft report and presentation review.

Instructions: Review the artifact first, then write specific comments that point to a figure, table, paragraph, or script.

Required fields: Research question, strongest result, weakest evidence, reproducibility issue, figure revision, methods gap, first revision priority, and reviewer name/date.

Common mistakes: Generic praise; comments without file locations; asking for stronger claims when evidence is weak.

Review question	Reviewer notes
-----------------	----------------

What is the research question?

What is the strongest result?

Where is the evidence weakest?

Which script or method detail needs attention?

Which figure needs revision?

Which claim should be weakened or defended?

What should the author revise first?

Completion standard: The author receives at least three actionable revisions tied to specific locations.

A.15 Worksheet 15: Daily Research Log

Purpose

Create a dated record of work, decisions, errors, and evidence.

When to use: Every active calendar day. Weekend entries may be shorter but still required.

Instructions: Complete the log before ending work. Include commands, files, outputs, and decisions.

Required fields: Date, objective, script or file changed, command run, output produced, evidence inspected, problem, decision, next step, and question for review.

Common mistakes: Entries that say only that work was done; missing file names; undocumented interpretation changes.

Prompt	Student response
Date and day number	
Today's objective	
Command/script/file changed	
Output produced	
Evidence inspected	
Problem encountered	
Decision made	
Question for review	
Next first step	

Completion standard: A reviewer can reconstruct what changed that day and why.

A.16 Worksheet 16: Weekly Progress Review

Purpose

Summarize the week's product, quality gate status, and next-week risks.

When to use: End of every week before moving to the next milestone.

Instructions: Attach links or paths to completed artifacts. State what passed, what failed, and what changed after review.

Required fields: Weekly milestone, deliverables, quality gates, evidence reviewed, unresolved problems, mentor/peer feedback, and next-week plan.

Common mistakes: Moving forward with failed gates; failing to record feedback; treating unfinished tasks as complete.

Prompt	Response
Week and milestone	
Primary deliverable location	
Quality gates passed	
Quality gates not yet passed	
Evidence or outputs reviewed	
Feedback received	
Revision plan	

Completion standard: The next week's work starts with a known priority, not a vague continuation.

A.17 Worksheet 17: Evidence Court

Purpose

Test one important claim by tracing it from statement to evidence.

When to use: Week 5 weekend, Week 8 interpretation, and final report review.

Instructions: Select a claim that matters. Fill every field before deciding whether to keep it.

Required fields: Draft claim, figure/table, uafR source table, evidence row or trait key, source database, confidence, literature support, revised claim, and reviewer decision.

Common mistakes: Choosing easy claims only; leaving confidence blank; keeping claims that only have narrative support.

Prompt	Student response
Draft claim	
Figure or table supporting claim	
uafR source table or matrix	
Evidence row, trait key, or field	
Source database	
Confidence or limitation	
Literature support or conflict	
Rewritten claim	
Reviewer decision	

Completion standard: The claim is kept, weakened, moved to future work, or removed with evidence.

A.18 Worksheet 18: Final Handoff

Purpose

Prepare the project so another person can continue it.

When to use: Week 10 final package audit.

Instructions: List file locations, not just yes/no status. Include unresolved issues and next recommended step.

Required fields: README status, raw data location, processed data, scripts, saved uafR result, validation outputs, matrices, figures, report, session information, continuation note, and non-claims.

Common mistakes: Saying files exist without naming locations; leaving private-data access unexplained; omitting what should not be claimed.

Handoff item	Location/status	Reviewer note
Project README is current		
Raw data preserved or access documented		
Processed data are scripted		
Saved uafR result object is present		
Validation summaries are saved		

Handoff item	Location/status	Reviewer note
Trait matrix and grouping dictionary are saved		
Figures regenerate from scripts		
Final report and presentation are saved		
Session information is saved		
Continuation note is complete		
Claims not to make are listed		

Completion standard: A new student can identify the project status and first next step in less than 15 minutes.

A.19 Worksheet 19: Presentation Planning

Purpose

Build a clear presentation around one central result and its evidence.

When to use: Week 10 before slide or poster production.

Instructions: Choose one central result. Everything else should support context, method, evidence, limitation, or next step.

Required fields: Audience, central result, slide sequence, main figure, evidence source, limitation, anticipated questions, and closing next step.

Common mistakes: Too many results; unexplained methods; no limitation slide; figures that are unreadable at presentation size.

Prompt	Response
Audience and time limit	
One-sentence central result	
Opening context	
Main figure or table	
Evidence source behind result	
Key limitation	
Likely audience questions	
Closing next step	

Completion standard: The presentation can be understood by a scientifically literate audience that has not seen the project folder.

A.20 Worksheet 20: Final Package Audit

Purpose

Confirm that the final package meets the program completion standard.

When to use: Final day and any archive or transfer event.

Instructions: Run the reproducibility check before marking the audit complete. Use file paths or project-relative locations.

Required fields: Project folder, report, presentation, scripts, data, saved objects, validation, traits, figures, research log, session information, continuation note, and archive location.

Common mistakes: Submitting only the final PDF; missing saved result objects; figures that cannot be regenerated; no statement of unsupported claims.

Requirement	Pass?	Evidence/location
Reproducible R project folder		
Cleaned query list and inclusion rules		
Saved uafR result object		
Validation summaries and source coverage		
Trait table or matrix		
Grouping dictionary		
At least three final figures or tables		

Requirement	Pass?	Evidence/location
Manuscript-style report		
Presentation or poster		
Research log		
Session information		
Continuation note		
Explicit claims not to make		

Completion standard: The package is complete, auditable, and ready for program archive or controlled transfer.

B Assessment Rubrics

The rubrics are designed to reward research quality rather than speed. A student may produce a modest project and still meet a high standard if the work is reproducible, cautious, source-backed, and useful for continuation. Suggested weights can be adjusted for course credit, stipend programs, or noncredit research training.

B.1 Performance Levels

Level	Meaning
Exemplary	Ready for dsDNA Core archive or external sharing after normal proofreading. Evidence is traceable and decisions are clear.
Proficient	Meets the program standard with minor revisions. The main work is reproducible and scientifically cautious.
Developing	Useful progress, but one or more quality gates need revision before the project can move forward.
Incomplete	Missing evidence, unclear methods, unsupported claims, or outputs that cannot be rerun.

B.2 Rubric 1: Weekly Deliverables

Criterion	Wt.	Exemplary	Prof./dev.	Incomplete
Milestone product	25	Complete artifact addresses the week's milestone and is stored in the project folder.	Product is present but needs revision, clarification, or missing file links.	Product is absent or unrelated to the milestone.
Quality gate	25	Gate is passed with evidence and review notes.	Gate is partly passed but unresolved issues remain.	Gate status is unknown or failed without plan.

Criterion	Wt.	Exemplary	Prof./dev.	Incomplete
Documentation	20	Research log and README show what changed and why.	Documentation exists but misses commands, outputs, or decisions.	Work cannot be reconstructed.
Evidence discipline	20	Claims cite data, source tables, figures, or literature.	Some claims are supported; some need traceability.	Claims are unsupported or overstate evidence.
Responsiveness	10	Feedback is incorporated and documented.	Feedback is acknowledged but not fully resolved.	Feedback is ignored or lost.

B.3 Rubric 2: Research Log

Criterion	Wt.	Exemplary	Prof./dev.	Incomplete
Daily coverage	20	Every active day has an entry, including weekend asynchronous work.	Most days are covered; a few entries are thin.	Large gaps or no weekend record.
Command and file traceability	25	Commands, scripts, files, outputs, and object names are recorded.	File names or commands are sometimes missing.	Entries say only what was attempted.
Decision history	25	Changes in scope, identifiers, filters, variables, and claims are dated and justified.	Decisions appear but rationale is incomplete.	Important decisions are invisible.
Error learning	15	Errors include context, diagnosis, fix, and remaining risk.	Errors are recorded but not fully explained.	Errors are hidden or only mentioned vaguely.
Scientific caution	15	Log separates tentative matches, database evidence, and final claims.	Some caution appears but language sometimes overstates.	Log treats database output as proof.

B.4 Rubric 3: Code and Reproducibility

Criterion	Wt.	Exemplary	Prof./dev.	Incomplete
Project structure	15	Root folder, README, data, scripts, outputs, logs, and docs are clear.	Structure is usable but some folders or names are confusing.	Files are scattered or unexplained.
Script order	20	Scripts are numbered or otherwise clearly ordered and rerun from the project root.	Most scripts run but order or dependencies need clarification.	Workflow depends on manual steps or hidden state.
Path safety	15	Relative paths are used and no local machine paths are required.	A few local assumptions remain but are easy to fix.	Analysis only runs on one machine.
Data preservation	15	Raw data are preserved and processed outputs are scripted.	Raw data mostly preserved; some manual cleaning unclear.	Raw data were edited or cannot be found.
Session state	15	Package versions and session information are saved.	Session info is present but incomplete.	No version record.
Reproducibility check	20	Clean-session test runs or failures are documented and fixed.	Test partially runs with known issues.	No rerun attempt.

B.5 Rubric 4: Chemical Evidence Traceability

Criterion	Wt.	Exemplary	Prof./dev.	Incomplete
Identifier handling	20	Original names, cleaned names, identifiers, ambiguity, and exclusions are documented.	Most identifiers are clear but some ambiguity remains.	Chemical list cannot be audited.
Tentative identity caution	20	GC-MS hits are described as tentative unless confirmed by standards.	Mostly cautious with occasional imprecise wording.	Tentative hits are presented as confirmed.
Source provenance	25	Important values trace to source table, field, URL or identifier, and evidence row.	Source names are present but row-level trace is incomplete.	Source provenance is missing.

Criterion	Wt.	Exemplary	Prof./dev.	Incomplete
Missingness	15	Empty sources and sparse coverage are visible in interpretation.	Missingness is noted but not connected to claims.	Missingness is hidden.
Claim strength	20	Claims match the evidence ladder and include limitations.	Claims are plausible but need stronger caveats.	Claims exceed available evidence.

B.6 Rubric 5: Trait Matrix and Grouping Dictionary

Criterion	Wt.	Exemplary	Prof./dev.	Incomplete
Variable definition	25	Each variable has source table, field, extraction rule, allowed values, confidence rule, and caveat.	Variables are mostly undefined but some rules are vague.	Variables are unexplained.
Discrete extraction	20	Narrative text is converted into clean, reproducible values or excluded.	Some values are clean; some text remains too broad.	Full sentences are used as factors without rules.
Matrix settings	20	Profile, mode, threshold, and maximum traits are recorded.	Settings are partly recorded.	Settings cannot be reconstructed.
Sensitivity	15	Sparsity and confidence choices are tested.	Sensitivity is mentioned but not quantified.	No sensitivity check.
Interpretability	20	Matrix supports a specific research question and avoids column overload.	Matrix is useful but could be narrowed.	Matrix is too noisy to interpret.

B.7 Rubric 6: Figures and Captions

Criterion	Wt.	Exemplary	Prof./dev.	Incomplete
Purpose	20	Each figure answers a stated question or supports a result sentence.	Figure is relevant but purpose could be sharper.	Figure is decorative or unfocused.

Criterion	Wt.	Exemplary	Prof./dev.	Incomplete
Reproducibility	20	Figures regenerate from scripts with documented source objects.	Most figures are scripted; some manual steps remain.	Figures cannot be regenerated.
Readability	20	Labels, legends, colors, and sizing are publication-ready.	Understandable but needs formatting revision.	Hard to read or misleading.
Missingness and uncertainty	20	Missing values, sparse coverage, and tentative status are visible or captioned.	Some caveats present but incomplete.	Missingness and uncertainty are hidden.
Caption quality	20	Caption names source, variables, pattern, and limitation.	Caption describes figure but lacks source or caveat.	Caption overclaims or only repeats title.

B.8 Rubric 7: Literature Integration

Criterion	Wt.	Exemplary	Prof./dev.	Incomplete
Search strategy	15	Search terms include chemicals, identifiers, system, methods, and traits.	Search terms are reasonable but incomplete.	Search process not documented.
Source quality	20	Papers are relevant and roles are clear.	Some papers are useful; some are broad background only.	Sources are weak or unrelated.
Evidence connection	25	Literature is tied to specific figures, traits, or claims.	Connection is present but sometimes generic.	Literature is pasted into discussion without purpose.
Limitations	20	Differences in system, method, dose, organism, and evidence type are explained.	Some limitations are included.	Literature is treated as direct proof.
Synthesis	20	Paragraphs use claim-evidence-limitation structure.	Interpretation is plausible but needs tighter evidence.	Interpretation overclaims or lacks synthesis.

B.9 Rubric 8: Final Report

Area	Wt.	Exemplary	Prof./dev.	Incomplete
Abstract and introduction	15	Question, system, evidence type, and significance are clear.	Mostly clear but broad or missing caveat.	Vague or disconnected from analysis.
Methods	20	Data sources, uafR functions, settings, filters, validation, and versions are reproducible.	Core workflow present but settings or validation missing.	Methods cannot be rerun.
Results	20	Results are tied to figures/tables and avoid unsupported interpretation.	Results relevant but some claims need restraint.	Results are descriptive or unsupported.
Discussion	20	Interpretation, uncertainty, literature, and next steps are balanced.	Discussion is useful but overstates some points.	Discussion ignores limitations.
Figures/tables	15	At least three publication-style products are clear and scripted.	Products are useful but need polish.	Products are missing or not traceable.
Reproducibility statement	10	Project folder, scripts, objects, and session info are named.	Statement present but incomplete.	No reproducibility statement.

B.10 Rubric 9: Presentation

Criterion	Wt.	Exemplary	Prof./dev.	Incomplete
Opening	15	Audience understands the problem, system, and question quickly.	Opening is understandable but too broad.	Purpose is unclear.
Workflow explanation	20	Data source, uafR workflow, validation, and analysis are explained without clutter.	Workflow mostly clear but missing settings or validation.	Workflow is confusing or absent.
Central result	25	One main figure or table anchors the presentation.	Result is relevant but not clearly central.	Presentation lists outputs without story.

Criterion		Wt.	Exemplary	Prof./dev.	Incomplete
Evidence and limitation		25	Student can answer what supports the result and what it cannot prove.	Some evidence and caveats present.	Overclaims or cannot name source evidence.
Delivery and questions		15	Clear, professional, and responsive to critique.	Understandable with minor issues.	Hard to follow or unable to answer basic questions.

B.11 Rubric 10: Final Package and Handoff

Component		Wt.	Exemplary	Prof./dev.	Incomplete
Package completeness		25	All required files, saved objects, figures, reports, logs, and notes are present.	Nearly complete with minor missing pieces.	Major products missing.
Rerun readiness		25	Clean-session rerun succeeds or failures are documented with fixes.	Partial rerun with known issues.	No meaningful rerun path.
Evidence trail		20	Final claims trace to data, uafR outputs, source rows, and citations.	Main claims mostly traceable.	Claims cannot be audited.
Continuation note		15	Next steps, unresolved issues, and claims not to make are explicit.	Note is present but thin.	No continuation guidance.
Archive quality		15	Package is organized for controlled DSU storage and future use.	Usable but needs cleanup.	Archive would confuse future users.

B.12 Recommended Overall Program Weight

Component	Weight	Evidence
Weekly deliverables and research log	25%	Weekly products, daily logs, peer review, revision history.
Code and reproducible project package	20%	Folder structure, scripts, saved outputs, clean-session check.
Chemical evidence and trait matrix quality	20%	Source coverage, evidence audit, grouping dictionary, matrix sensitivity.
Final report and figures	20%	Manuscript-style report, figures, tables, limitations, citations.
Presentation and handoff	15%	Final talk or poster, package audit, continuation note.

C Program Implementation Guide

C.1 Program Philosophy

This program treats undergraduate interns as early-career researchers. Students are not expected to arrive knowing R, chemistry, statistics, mass spectrometry, or manuscript writing. They are expected to learn how to make their work traceable, honest, reproducible, and useful. The curriculum is intentionally full-day: research skill develops through repeated cycles of trying, inspecting, documenting, revising, and explaining.

C.2 Pre-Program Setup Checklist

Before Week 1

- Prepare the current private uafR student bundle or another approved installation route.
- Confirm that the bundle includes installation instructions, preflight checks, acceptance tests, training scripts, and this manual.
- Prepare a case dataset and short chemical list for students without immediate research data.
- Decide where students will store project folders and which locations may contain private or unpublished data.
- Prepare printed or digital copies of the Project Brief, Chemical Intake Table, Database Evidence Map, Daily Research Log, and Weekly Progress Review worksheets.
- Schedule weekly review checkpoints and final presentation time.
- Identify which public database exercises can use saved examples if network access or live services fail.
- Create a list of program contacts for R help, chemistry interpretation, data privacy, and presentation review.

C.3 Daily Check-In Script

The daily check-in is short and artifact-centered. It can be used at the start or end of a workday.

1. What did the student produce since the last check-in?
2. Where is the file, table, figure, worksheet, or code object saved?
3. What evidence supports the main thing the student thinks they learned?
4. What is uncertain, missing, broken, or overclaimed?
5. What will be produced by the next check-in?

C.4 Weekly Mentor Review Script

Weekly review should inspect the product ladder artifact rather than reviewing the whole project from scratch.

1. Confirm the week's primary deliverable and file location.
2. Ask the student to explain the research question in one minute.
3. Ask the student to show the source evidence behind one important claim.
4. Ask the student to show one reproducibility check or rerun result.
5. Ask the student to name the strongest limitation.
6. Decide whether the next week can proceed, proceed with a reduced scope, or pause for revision.

C.5 Decision-Gate Review Questions

Gate	Review questions	Evidence required
Scope gate	What chemical set is being studied? What claim is out of scope?	Project brief, risk note, chemical intake draft.
Reproducibility gate	Can the project be opened and rerun from its root?	README, setup script, session info, clean folder structure.
Evidence gate	Which sources returned useful data and which did not?	Validation summary, source coverage, diagnostics, saved result object.
Variable gate	How did source evidence become analysis-ready variables?	Grouping dictionary, trait evidence audit, matrix settings.
Interpretation gate	Which claims survive evidence review?	Claim-evidence-limitation memo, figures, literature table.
Handoff gate	Can someone continue the project without the student present?	Final audit, continuation note, rerun status, archive path.

C.6 Expected Student Answers by Program Stage

Stage	Strong answer	Answer needing revision
Question design	Names system, chemical set, comparison, evidence type, and limitation.	"I want to see what chemicals are important."
Identifier cleanup	Preserves original names and documents cleaned query names and ambiguity.	Original names are overwritten or difficult compounds disappear.
GC-MS workflow	Describes match factor, retention time, base peak, area, and tentative identity caveats.	"The software identified the compound."
Database enrichment	Starts with validation and source coverage.	Jumps directly to the most exciting pathway or bioactivity term.
Trait design	Names source table, field, extraction rule, allowed values, confidence, and caveat.	Uses broad sentences as categories.
Visualization	Explains which question the figure answers and which values are missing.	Shows a figure because it looks interesting.
Literature	Uses papers to refine interpretation and limitations.	Searches only for papers that support a desired conclusion.
Final report	Methods, results, and limitations are reproducible and source-backed.	Report reads like a story without an evidence trail.

C.7 Common Coaching Moments

Pattern	Useful program response
Project is too broad	Reduce to one chemical set, one comparison, and one main evidence family.
Student is stuck on installation	Use the private bundle path, run preflight checks, and switch to case-data reading while installation is repaired.
Student has weak R background	Keep chemical set small, require script comments that explain intent, and prioritize clean reruns over advanced analysis.
Student has no chemistry background	Use identifier ambiguity drills and claim-boundary examples before interpreting traits.
Student has strong background and needs more challenge	Add sensitivity analysis, reviewer response, literature conflict, or a reusable audit/plotting function.
Live service fails	Use saved result objects or offline examples and document that live access was unavailable.
Matrix is too wide	Return to the grouping dictionary and remove variables that lack project relevance or source support.
Writing overclaims	Rewrite the claim using evidence type, source, and limitation.

C.8 When To Slow Down

The program should slow down when a student cannot explain the question, cannot find the files they produced, ignores source coverage, interprets before validation, uses absolute paths, treats database text as proof, or cannot explain what a figure shows. Slowing down means reducing scope and improving evidence quality, not abandoning the project.

C.9 When To Accelerate

The program can accelerate when the student can rerun the workflow from a clean session, source coverage and validation are documented, the grouping dictionary is clear, figures have source-backed captions, and the student can explain what evidence would change their interpretation. Acceleration should add depth: sensitivity checks, literature conflict analysis, stronger figures, or reusable functions.

C.10 Running With Real Data, Case Data, or No Data

Mode	Best use	Required adjustment
Real student data	Strongest ownership and publication potential.	Add data privacy notes, raw-data preservation, and realistic scope limits.
Program case data	Best for teaching a consistent workflow.	Make clear that scientific claims are about the case exercise, not new discovery.
No available chemical data	Best for students entering early or from communication tracks.	Use a curated chemical list, database evidence atlas, methods evaluation, or teaching module.
Mixed mode	Best when real data are messy.	Train on case data first, then apply only reviewed steps to real data.

C.11 Protecting Unpublished and Private Data

Private and unpublished data require deliberate handling. Student projects should not contain tokens, passwords, private collaborator files, raw data with restricted identifiers, or uncontrolled copies of unpublished sample metadata. Public-facing products should use generalized descriptions unless sharing has been approved. Reproducibility can be preserved with documented access rules, checksums where appropriate, case-data substitutes, and clear separation of public scripts from private data.

C.12 Database Etiquette

Live database access is deliberately conservative:

- Start with one or two compounds before scaling.
- Use caching for repeated work.
- Use throttling for broad queries.
- Save RDS result objects so students do not repeat identical requests.
- Use offline saved examples when teaching in rooms with unstable internet.
- Treat service failures as workflow realities, not as student failures.

C.13 Evaluating Final Packages

Final review uses three questions:

1. Can the package be understood from the README, report, and continuation note?
2. Can the figures and tables be traced to scripts, saved objects, source evidence, and literature?
3. Are claims scientifically restrained by tentative identity, source coverage, missingness, and evidence type?

Packages that pass these questions can be archived even if they contain negative or inconclusive findings. Inconclusive projects are valuable when they document why the evidence was insufficient and what should be done next.

C.14 Archiving Projects

The final archive should include the final project folder, report, presentation, saved uafR result object, validation outputs, source coverage report, trait matrix, grouping dictionary, figures, research log, session information, final package audit, and continuation note. Archive names should include student initials or project code, year, and short title. Restricted data should remain in controlled storage with a note explaining access requirements.

C.15 Recommended Grading Weight

Component	Weight	Notes
Weekly deliverables and research log	25%	Completion, quality, daily documentation, and responsiveness to feedback.
Reproducible project package	20%	File structure, scripts, logs, clean-session check, and saved objects.
Chemical evidence and trait design	20%	Source coverage, evidence traceability, grouping dictionary, and matrix quality.
Final report and figures	20%	Scientific logic, methods, results, figures, interpretation, and limitations.
Presentation and handoff	15%	Clear communication, final audit, continuation note, and archive readiness.

D dsDNA Core Review Notes and Answer Guidance

D.1 Facilitating Without Doing the Project for the Student

In dsDNA Core sessions, evidence stays at the center before answers. The strongest teaching question is often, “Which table, row, script, or citation supports that?” Students learn to diagnose problems by moving from output to source, not by guessing.

D.2 Daily Check-In Structure

This five-minute structure works at the start or end of each day:

1. What did you produce yesterday?
2. What evidence supports the main thing you think you learned?
3. What is uncertain, broken, missing, or confusing?
4. What will you produce by the end of today?
5. What decision do you need from the program lead or dsDNA Core mentor?

D.3 Expected Answers by Program Stage

Stage	Strong student answer	Needs coaching
Question design	Names the system, comparison, chemical evidence, and limitation.	Says only that they want to see what chemicals are important.
Database interpretation	Distinguishes identity, source annotation, occurrence, pathway, hazard, and assay evidence.	Treats any database hit as equivalent support.
GC-MS workflow	Explains match factor, retention time, base peak, area, and tentative identity caveats.	Says the software identified the compound without caveat.
Enrichment	Starts with validation, source coverage, and provenance.	Jumps directly to the most exciting trait or pathway.

Stage	Strong student answer	Needs coaching
Trait matrix	Explains source fields, confidence, matrix keys, and why weak traits were removed.	Uses every available column because more data seem better.
Figure interpretation	Ties each figure to a question and a source table.	Makes decorative plots without a decision or claim.
Literature integration	Uses papers to contextualize results and limitations.	Searches only for papers that support the desired conclusion.
Final report	Methods can be rerun and claims are traceable.	Reads like a narrative without reproducible evidence.

D.4 Common Program Responses

Student pattern	Program response
Overly broad project	The work is limited to one chemical list, one comparison, and one main evidence family.
Too much time cleaning messy data	The student switches to a program-reviewed case dataset for training, then returns to the real data later.
Interpreting before validation	A validation summary screenshot or table is reviewed before biological meaning is discussed.
Matrix is too wide	The matrix is narrowed to medium-confidence traits tied to the research question.
Weak writing	The student writes one claim, one evidence sentence, and one limitation sentence for each figure.
No progress after errors	The workflow moves to a small reproducible example with the exact error, input, expected output, and actual output.

D.5 Review Notes for Real-World Exercises

D.5.1 Unmatched Chemical

A good answer does not delete unmatched chemicals silently. It should preserve them in a missingness table and identify the next action: synonym search, identifier correction, manual review, or follow-up validation.

D.5.2 Hazard Code Trap

A good answer separates hazard screening from risk interpretation. Students should state that risk requires exposure, dose, route, organism, and context.

D.5.3 Pathway Does Not Mean Presence

A good answer rewrites “this pathway is active” as “this compound has KEGG pathway context consistent with...” unless independent pathway activity evidence exists.

D.5.4 Natural Product Occurrence

A good answer separates direct taxonomic evidence from broad natural product context. Students should not imply occurrence in their organism unless the source supports it.

D.5.5 Matrix Sparsity

A good answer favors a smaller matrix if it has clearer interpretation, better confidence, and sufficient coverage across compounds. Students should explain what was removed and why.

D.6 When To Slow the Program Down

The program slows down when:

- Students cannot explain their research question without reading notes.
- Source coverage is weak and students are ignoring it.
- Scripts contain absolute paths or manual steps that cannot be repeated.
- Figures are being made before variables are defined.
- Literature is being used as decoration rather than evidence.

D.7 When To Accelerate

The program accelerates when:

- The student can rerun their workflow from a clean R session.
- The grouping dictionary is reviewed and matrix outputs are stable.
- Figures have clear captions and limitations.
- The student can explain what would change their interpretation.

D.8 Final Review Questions

Before the final package is accepted, the review asks:

1. Which result is strongest, and what evidence supports it?
2. Which result is most uncertain, and why?

3. Which chemicals were excluded, unmatched, or weakly supported?
4. Which database source was most useful, and which was least useful?
5. What should the next student do first?
6. What claim should not be made from this project?

E Installation and Setup Appendix

E.1 Recommended Student Installation

The recommended student installation path is a private uafR student bundle distributed by the DSU dsDNA Core program. This keeps the package repository private and gives every student the same RStudio-based setup.

The student bundle includes:

- `START_HERE.md`, the first checklist to open.
- `uafR_QUICK_REFERENCE.md`, the compact command reference.
- `CHECKSUMS.csv`, the generated file size and MD5 checksum table.
- `verify_bundle_integrity.R`, the checksum verification script.
- A local uafR source package archive in `packages/`.
- `preflight_check.R`, the machine readiness check.
- `install_uafR_from_bundle.R`, the student installer.
- `update_uafR_from_bundle.R`, the reinstall/update script for newer bundles.
- `verify_uafR_install.R`, the post-install verification script.
- `run_student_acceptance_test.R`, the offline readiness test.
- The current training manual PDF.
- Training scripts and data templates used during the internship.

The installer still needs internet access for CRAN and Bioconductor dependencies. The uafR package itself is installed from the local bundle archive.

E.2 Install From the Student Bundle

The installation workflow is:

1. Install current R and RStudio.
2. Unzip the whole bundle folder.
3. Open RStudio.

4. Open `START_HERE.md` from the unzipped bundle folder.
5. Source `verify_bundle_integrity.R`.
6. Source `preflight_check.R`.
7. If preflight has no failed checks, source `install_uafR_from_bundle.R`.
8. Source `run_student_acceptance_test.R`.
9. Keep `uafR_QUICK_REFERENCE.md` nearby and open `training/uafR_training_manual.pdf`.

The main setup commands are:

```
source("verify_bundle_integrity.R")
source("preflight_check.R")
source("install_uafR_from_bundle.R")
source("run_student_acceptance_test.R")
```

The installer prints five visible steps: checking the bundle and R library, installing CRAN dependencies, installing Bioconductor dependencies, installing uafR from the local archive, and verifying the installation. The installer writes `uafR_install_log.txt` in the bundle folder.

E.3 Create a New Student Bundle

From the repository root, the program build step is:

```
Rscript tools/build_student_bundle.R
```

This creates a versioned bundle folder and zip archive under `student_bundle/`. The zip archive can be copied to a USB drive, shared through an approved campus file-transfer method, or unpacked onto classroom machines.

E.4 Confirm Bundle Integrity

The bundle builder writes `CHECKSUMS.csv` with the file size and MD5 checksum for each file in the generated bundle. After a bundle is copied or unzipped, the integrity check confirms that the files still match the build record.

```
source("verify_bundle_integrity.R")
```

It should end with:

```
Bundle integrity result: PASS.
```

E.5 Confirm the Installation Without Live Web Queries

The verification script checks package loading, required dependencies, bundled datasets, and the saved standard categorate result. It does not query live web services.

```
source("verify_uafR_install.R")
```

E.6 Student Acceptance Test

The acceptance test confirms that the installed package is ready for the training workflow. It runs package verification, loads the core data objects, and runs the offline workflow from `training/scripts/04_core_workflow.R`. It should end with:

Acceptance test result: PASS.

```
source("run_student_acceptance_test.R")
```

E.7 Optional Live Smoke Test

After the offline verification passes, run a two-compound live test when internet access is available:

```
library(uafR)
data("library_data", package = "uafR")

quick_result <- categorate(
  compounds = c("aspirin", "caffeine"),
  chemical_library = library_data,
  input_format = "wide",
  detail = "research",
  cache = FALSE,
  throttle = 0.1,
  assay_detail_limit = 0
)

validateCategorateResult(quick_result)$Summary
quick_result$ChemicalTraitReport
```

The acceptance test can also include the live smoke test:

```
Sys.setenv(UAFR_STUDENT_LIVE = "1")
source("run_student_acceptance_test.R")
```

E.8 Troubleshooting

Problem	Likely cause	What to do
Bundle integrity check fails	One or more files changed, did not copy, or did not unzip correctly.	Delete the unzipped folder and unzip the original bundle again. If it still fails, get a fresh copy of the zip.
Installer cannot find uafR archive	The bundle was not fully unzipped or the <code>packages/</code> folder was moved.	Unzip the whole bundle again and keep the folder structure unchanged.
Preflight says the R library is not writable	R cannot install packages into the first library path.	Restart RStudio. If the problem remains, set a user library before installing.
Bioconductor dependency fails	Internet access, R version, or Bioconductor access is blocked.	Run <code>preflight_check.R</code> , update R if it is old, and try again on a network that can reach Bioconductor.
R asks about updating many packages	RStudio or Bioconductor is offering broad updates.	Choose the option that does not update unrelated packages.
<code>library(uafR)</code> fails right after install	RStudio has not refreshed the library.	Restart RStudio, then run <code>library(uafR)</code> .
No library detected	The chemical library argument was not supplied to <code>categorate()</code> .	Load <code>library_data</code> and pass <code>chemical_library = library_data</code> .
Long database query	Live PubChem, KEGG, LOTUS, or other services are responding slowly.	Start with one or two compounds, keep <code>assay_detail_limit = 0</code> , use caching, and scale up after the smoke test works.
Missing output tables	A live lookup failed or returned incomplete source data.	Run <code>validateCategorateResult()</code> and inspect <code>ValidationSummary</code> , <code>TableQuality</code> , and <code>SourceDiagnostics</code> .

F uafR Table and Script Guide

F.1 High-Value categorate Tables

Table	Research use
PubChemIdentity	Check CID matches, match status, and source URLs.
PubChemProperties	Compare formula, exact mass, molecular weight, and descriptor fields when available.
SourceCoverage	Identify which databases contributed evidence for each chemical.
ChemicalTerms	Inspect long-form extracted terms before committing to variables.
ChemicalTraits	Use cross-source discrete traits with confidence and evidence fields.
ChemicalTraitOntology	Use whitelisted ontology-style traits grouped by research domain.
ChemicalTraitMatrix	Analyze chemicals by binary, count, or confidence-weighted trait columns.
ChemicalTraitEvidence	Trace matrix columns back to source evidence.
ChemicalTraitReport	Review one compact row per compound for high-level interpretation.
ChemicalMeasurementSummary	Summarize extracted measured or computed properties.
ChemicalHazards	Inspect source-backed safety and hazard values.
ChemicalOccurrences	Review organism or natural product occurrence evidence.
ChemicalPathwayRoles	Interpret KEGG pathway context and biochemical roles.
KEGGReactionParticipants	Examine reaction participation and related biochemical context.
DataDictionary	Confirm expected table schemas and field meaning.
TableQuality	Audit missing columns, duplicate keys, and completeness.
ValidationSummary	Start every review here before interpreting output.

Validation Naming Note

Inside a saved `categorate()` result, the compact validation table is named `ValidationSummary`. A direct call to `validateCategorateResult(result)` returns a list whose corresponding validation table is named `Summary`, along with `TableQuality`, `SourceDiagnostics`, `Issues`, and `DataDictionary`. Both names are intentional; students should record which object they inspected.

F.2 Training Scripts

Script	Purpose
00_install_check.R	Confirms R, package, dependencies, and example data.
01_project_setup.R	Creates a reproducible student project folder.
04_core_workflow.R	Runs the core GC-MS example workflow.
05_categorate_research.R	Runs a small research enrichment smoke test.
06_trait_matrix_analysis.R	Builds selected trait matrices and evidence tables.
07_visualization.R	Creates example coverage and trait figures.
09_reproducibility_check.R	Audits a student project folder before final submission.
run_training_smoke_tests.R	Runs lightweight checks across the training scripts.

F.3 Student Bundle Scripts

Script	Purpose
<code>tools/build_student_bundle.R</code>	Builds the private student bundle folder and zip archive from the repository root.
<code>START_HERE.md</code>	Gives the first student checklist for setup, installation, acceptance testing, and opening the manual.
<code>uafR_QUICK_REFERENCE.md</code>	Gives students a compact command reference for setup, offline workflow, live smoke test, updates, and common problems.
<code>CHECKSUMS.csv</code>	Records generated bundle file sizes and MD5 checksums for copying and distribution checks.
<code>verify_bundle_integrity.R</code>	Confirms copied or unzipped bundle files still match <code>CHECKSUMS.csv</code> .
<code>preflight_check.R</code>	Checks R, RStudio, internet access, bundle contents, library write access, and required dependencies before installation.
<code>install_uafR_from_bundle.R</code>	Installs dependencies and the local uafR source archive from the unzipped student bundle.
<code>update_uafR_from_bundle.R</code>	Reinstalls uafR from a newer bundle and verifies the result.
<code>verify_uafR_install.R</code>	Confirms package loading, dependencies, bundled datasets, and saved standard categorate data without live web queries.
<code>run_student_acceptance_test.R</code>	Runs package verification, the offline training workflow, and an optional live database smoke test.
<code>examples/test_install.R</code>	Reruns the acceptance test from the bundle examples folder.

F.4 Choosing Detail Levels

- Use `detail = "standard"` for the legacy eight-table output.
- Use `detail = "research"` for most student projects. It gives rich tables while keeping the query smaller than full assay detail.
- Use `detail = "full"` only when the project needs broad assay details or deeper PubChem annotations.

F.5 Recommended Smoke Test Settings

```

categorate(
  compounds = c("aspirin", "caffeine"),
  chemical_library = library_data,

```

```
input_format = "wide",
detail = "research",
cache = TRUE,
throttle = 0.2,
assay_detail_limit = 0,
trait_matrix_profile = "core",
trait_matrix_min_confidence = "medium",
trait_matrix_max_traits = 100
)
```

Bibliography

- [1] J. J. Allaire, Yihui Xie, Christophe Dervieux, Jonathan McPherson, Javier Luraschi, Kevin Ushey, Aron Atkins, Hadley Wickham, Joe Cheng, Winston Chang, and Richard Iannone. *rmarkdown: Dynamic Documents for R*, 2024. R package.
- [2] Minoru Kanehisa, Yoko Sato, and Masayuki Kawashima. Kegg as a reference resource for gene and protein annotation. *Nucleic Acids Research*, 49(D1):D545–D551, 2021.
- [3] Kanehisa Laboratories. Kegg api, 2026. <https://www.genome.jp/kegg/rest/>.
- [4] Sunghwan Kim, Jie Chen, Tiejun Cheng, Asta Gindulyte, Jia He, Siqian He, Qingliang Li, Benjamin A. Shoemaker, Paul A. Thiessen, Bo Yu, Leonid Zaslavsky, Jian Zhang, and Evan E. Bolton. Pubchem 2019 update: improved access to chemical data. *Nucleic Acids Research*, 47(D1):D1102–D1109, 2019.
- [5] Sunghwan Kim, Paul A. Thiessen, Tiejun Cheng, Bo Yu, and Evan E. Bolton. An update on pug-rest: Restful interface for programmatic access to pubchem. *Nucleic Acids Research*, 46(W1):W563–W570, 2018.
- [6] National Center for Biotechnology Information. Pubchem pug-rest documentation, 2026. <https://pubchem.ncbi.nlm.nih.gov/docs/pug-rest>.
- [7] Adriano Rutz, Maria Sorokina, Jakub Galgonek, Daniel Mietchen, Egon Willighagen, Alexandre Gaudry, James G. Graham, Ralf Stephan, Roderic Page, Jiri Vondrasek, Christoph Steinbeck, Guido F. Pauli, Jean-Luc Wolfender, Jonathan Bisson, and Pierre-Marie Allard. Lotus, the natural products occurrence database. *eLife*, 11:e70780, 2022.
- [8] Lloyd W. Sumner, Alexander Amberg, David Barrett, Michael H. Beale, Richard Beger, Colin A. Daykin, Teresa W.-M. Fan, Oliver Fiehn, Royston Goodacre, Julian L. Griffin, Thomas Hankemeier, Nigel Hardy, James Harnly, Richard Higashi, Joachim Kopka, Andrew N. Lane, John C. Lindon, Philip Marriott, Andrew W. Nicholls, Mark D. Reily, Joshua J. Thaden, and Mark R. Viant. Proposed minimum reporting standards for chemical analysis. *Metabolomics*, 3:211–221, 2007.
- [9] Mark R. Viant, Timothy M. D. Ebbels, Richard D. Beger, Drew R. Ekman, David J. T. Epps, Henricke Kamp, Pim E. G. Leonards, George D. Loizou, James I. MacRae, Bennard van Ravenzwaay, Philippe Rocca-Serra, Reza M. Salek, Tilmann Walk, and Ralf J. M. Weber. Use cases, best practice and reporting standards for metabolomics in regulatory toxicology. *Nature Communications*, 10:3041, 2019.
- [10] Greg Wilson, Jennifer Bryan, Karen Cranston, Justin Kitzes, Lex Nederbragt, and Tracy K. Teal. Good enough practices in scientific computing. *PLOS Computational Biology*, 13(6):e1005510, 2017.